

HBF Technology, Workload Analysis and Roadmap

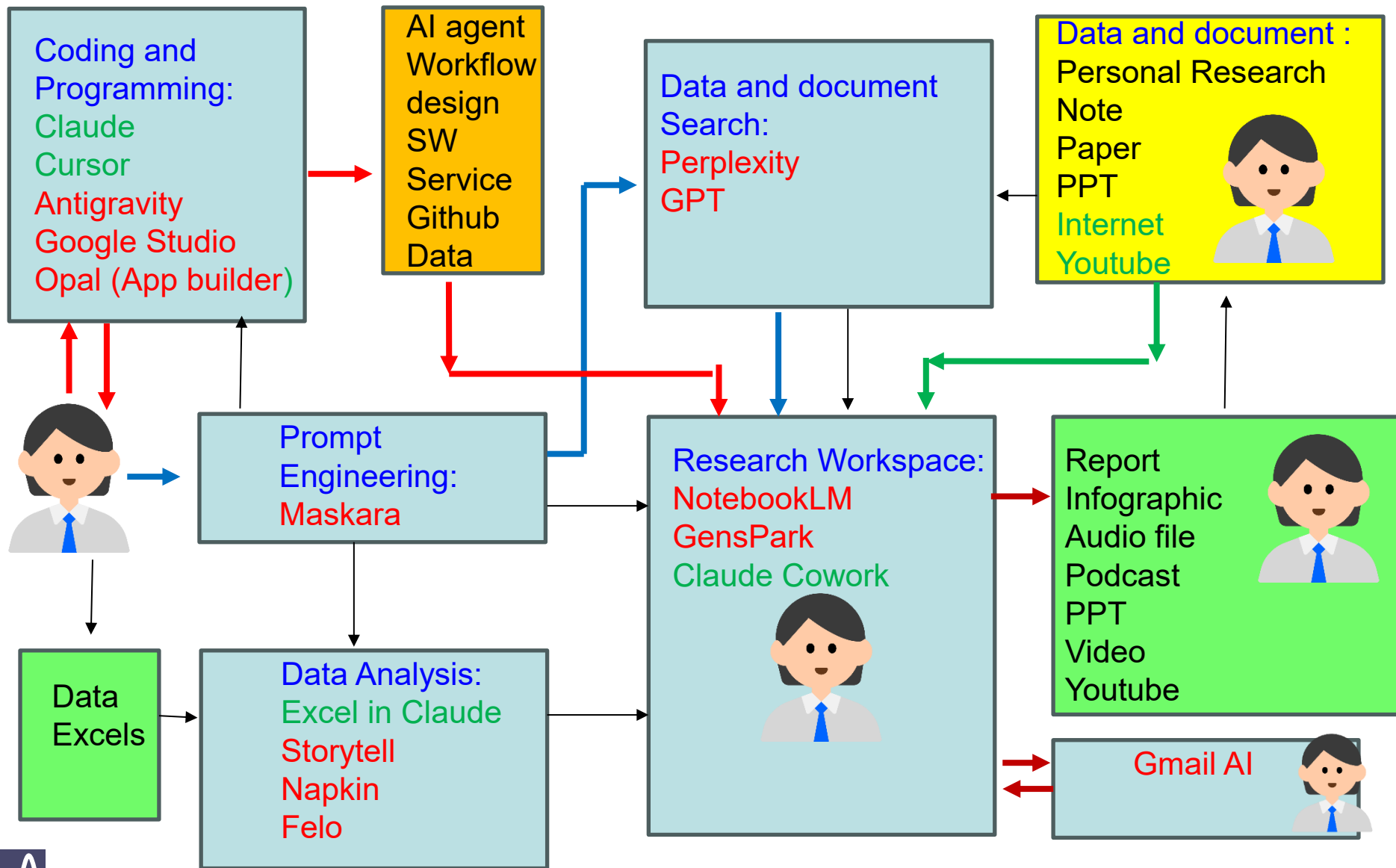
Prof. Jounggho Kim

TeraByte Interconnection and Package Laboratory
School of Electrical Engineering
KAIST

2026.2. 3

Part1: Beginning of Memory Hungry Century

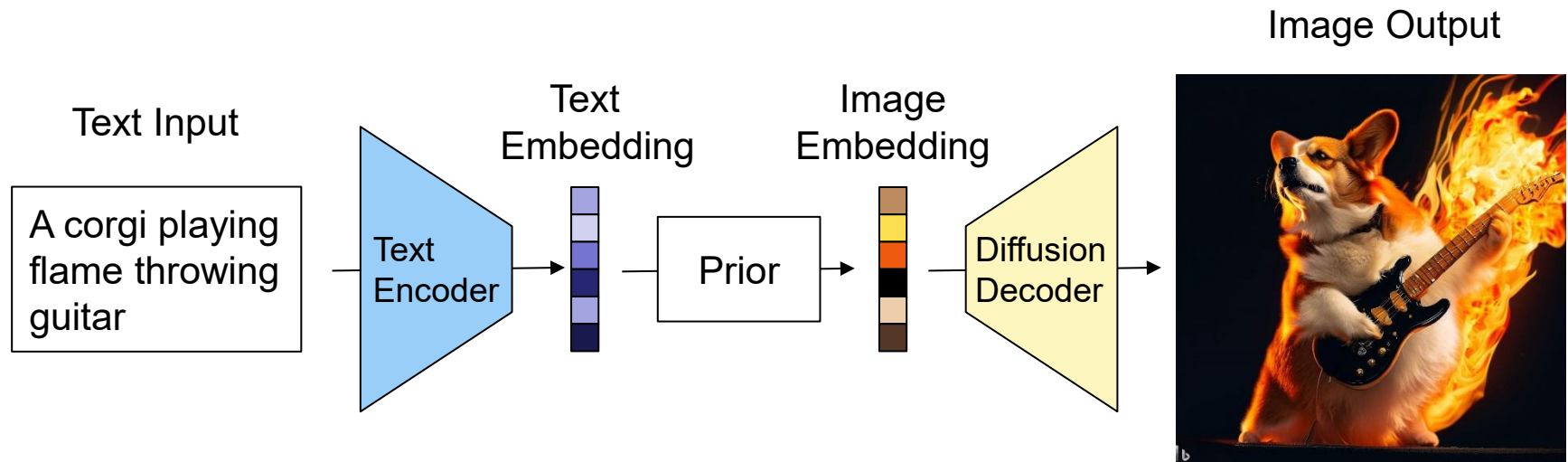
AI Workflow Design and Architecture



- 1) Reasoning and Thinking
- 2) Multi-modal
- 3) Mixture of Expert
- 4) Vibe coding
- 5) Personal AI
- 6) Real time training, RAG
- 7) Continuous training
- 8) Personnel training
- 9) MoE pre-training with specific data set, end-to-end training
- 10) Short term and long term memory effects
- 11) Voice interface
- 12) Agentic AI, A2A, MCP

- 1) Workspace Platform, Google NotebookLM
- 2) More emphasis on multi-modal data
- 3) Higher competition and frequency products/service
- 4) AI Echo-system
- 5) 3 computers, Training, Inference, Data Generation with GPU for automotive vehicles
- 6) Hybrid model(Generation, Discussion, Verification), hybrid computing
- 7) Synthetic data, Augmented data
- 8) Google dominance
- 9) Memory explosion

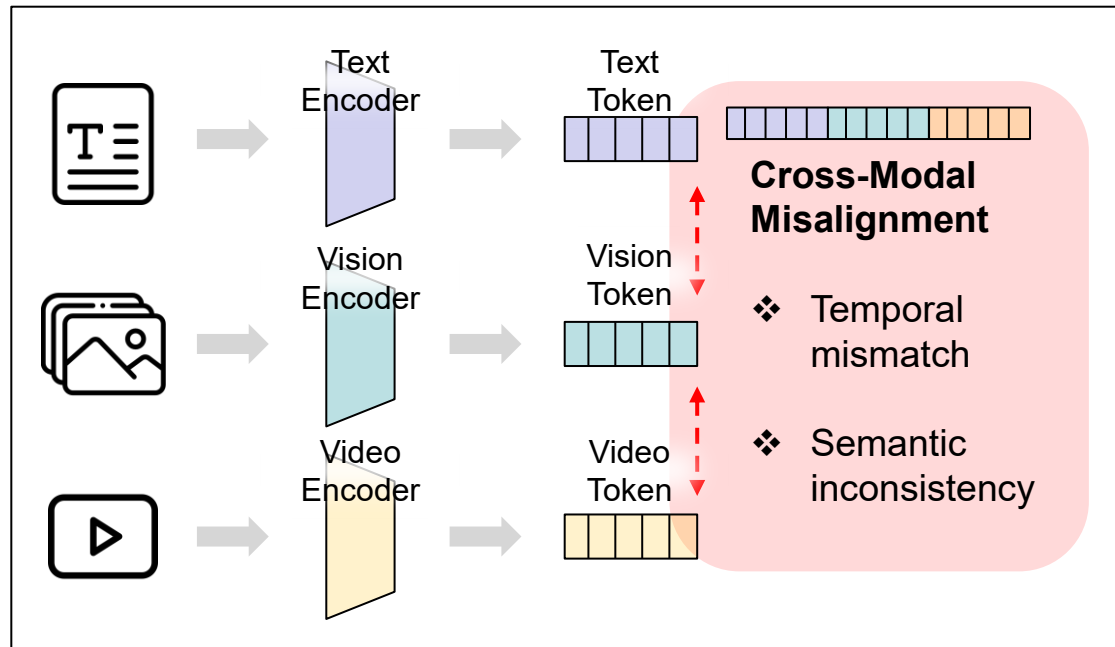
Principle of DALL·E 2: Encoder, Prior and Decoder



< Overview of the DALL·E 2 image-generation process >

- DALL·E 2 is a particular instance of a three-part model made of encoder, prior and decoder.
- The text embedding and image embedding represents the main features that are semantically meaningful: People, animals, objects, style, colors, background, etc.
- DALL·E 2 can generate a novel image that retains these characteristics while varying the non-essential features.

Problem Statement: The Limits of Disjointed Multimodal Processing



< 분리된 모달리티 처리 과정에서 발생하는 Cross-Modal Misalignment로 인한 구조적 문제 >

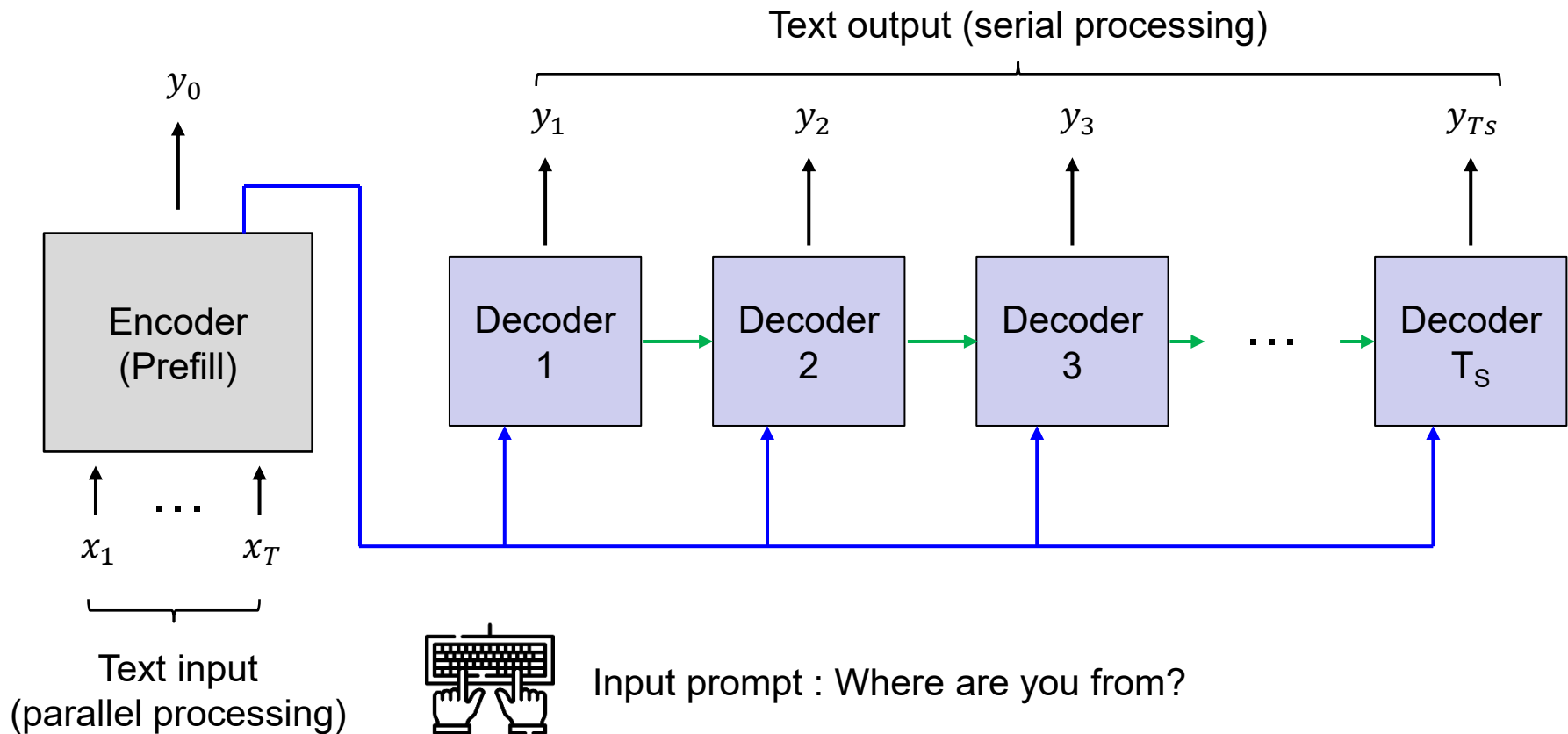
- Independent modality encoders: 각 모달리티가 분리된 인코더로 처리되며, 미세한 시간적/의미적 상관관계(cross-modal alignment)가 손실됨
- Limited context window: 긴 비디오/문서를 요약해 입력해야 하는 정보 병목(bottleneck)이 발생함
- Fragmented context understanding: 맥락 단절로 인해 부정확한 추론 및 hallucination 발생 가능성 증가

Basic LLM Transformer Model : Text Input / Text Output

→ KV cache (generated in prefill)

→ KV cache (generated in decode)

Output token : I am from South Korea.

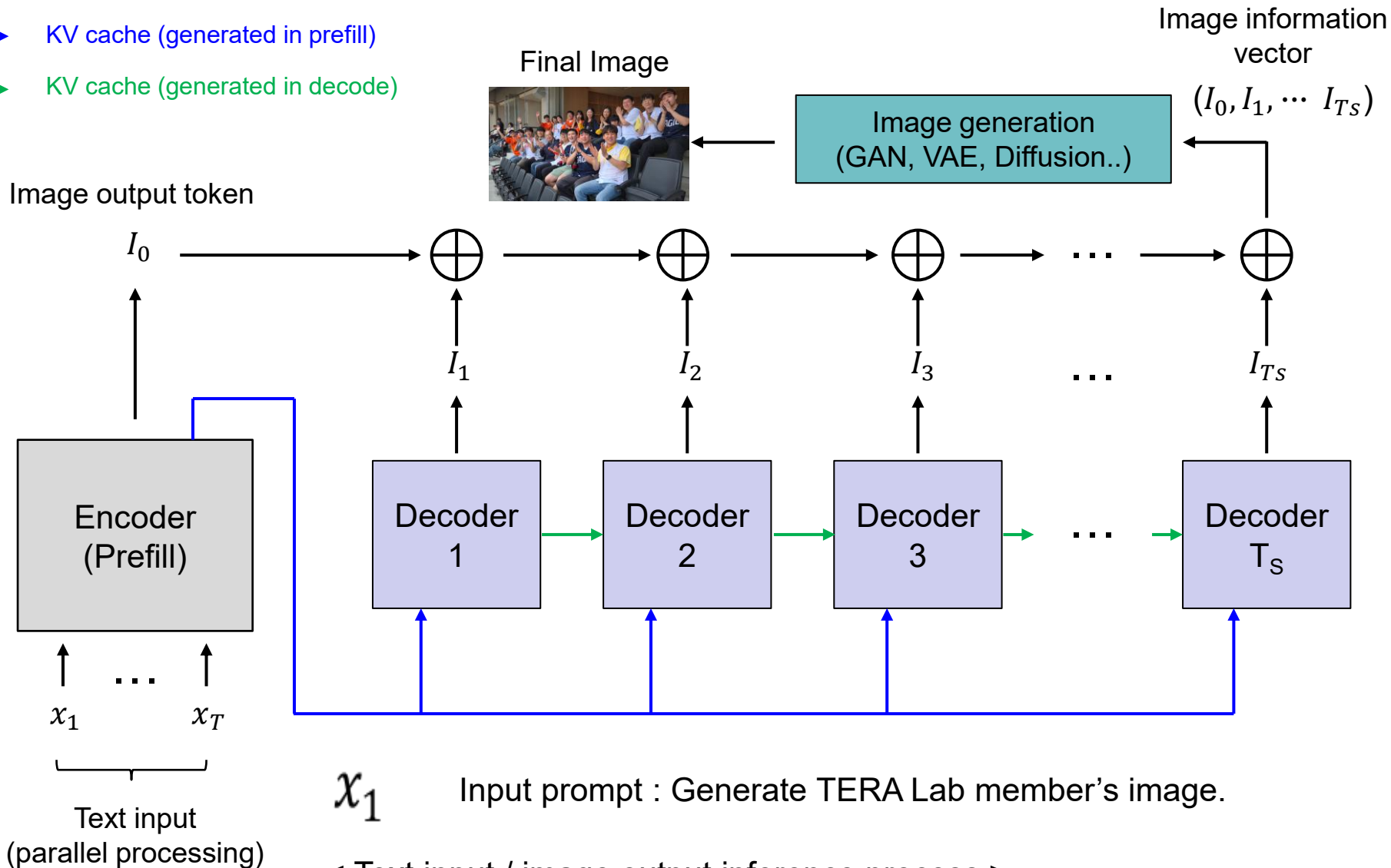


< Text input / text output inference process >

VLM Transformer Model [1/4] : Text Input / Image Output

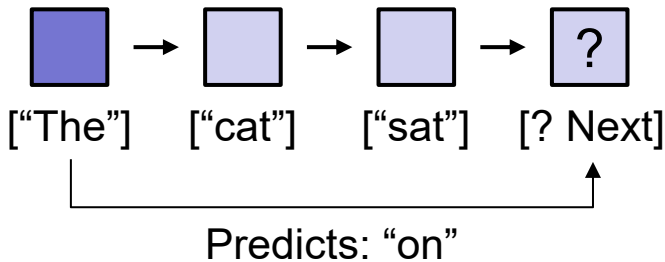
→ KV cache (generated in prefill)

→ KV cache (generated in decode)



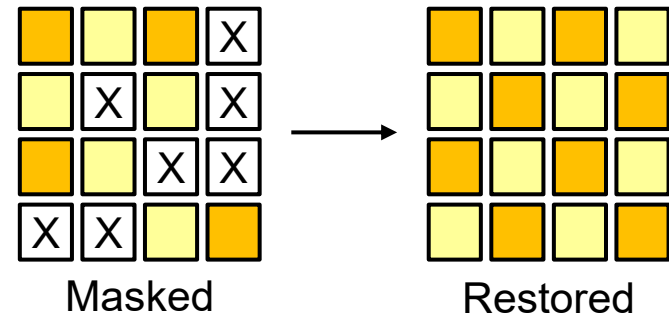
Unified Learning Objectives

1. Text Learning: Next Token Prediction (NTP)



- ✓ Rule: Guess the *next* word based on the past (Causal, Autoregressive)

2. Image Learning: Masked Token Prediction (MTP)



- ✓ Rule: Fill in the *missing* pieces using surroundings (Bidirectional, Denoising)

$$\text{Total Unified Loss} = \text{Loss}_{NTP} + \text{Loss}_{MTP}$$

< Dual-Objective Learning Strategy >

- Text Learning (NTP): 이전 문맥을 기반으로 다음 텍스트 토큰을 예측하여 언어의 순차적 구조를 학습.
- Image Learning (MTP): 마스킹된 이미지 latent tokens을 복원하는 denoising-style objective를 통해 공간적 구조를 학습.

Key Metrics in Inference [1/2] : Low Token Latency

Token Latency: The time for token generation [Unit : $s/token$]

Prefill

Generation

Text Input

Text Output



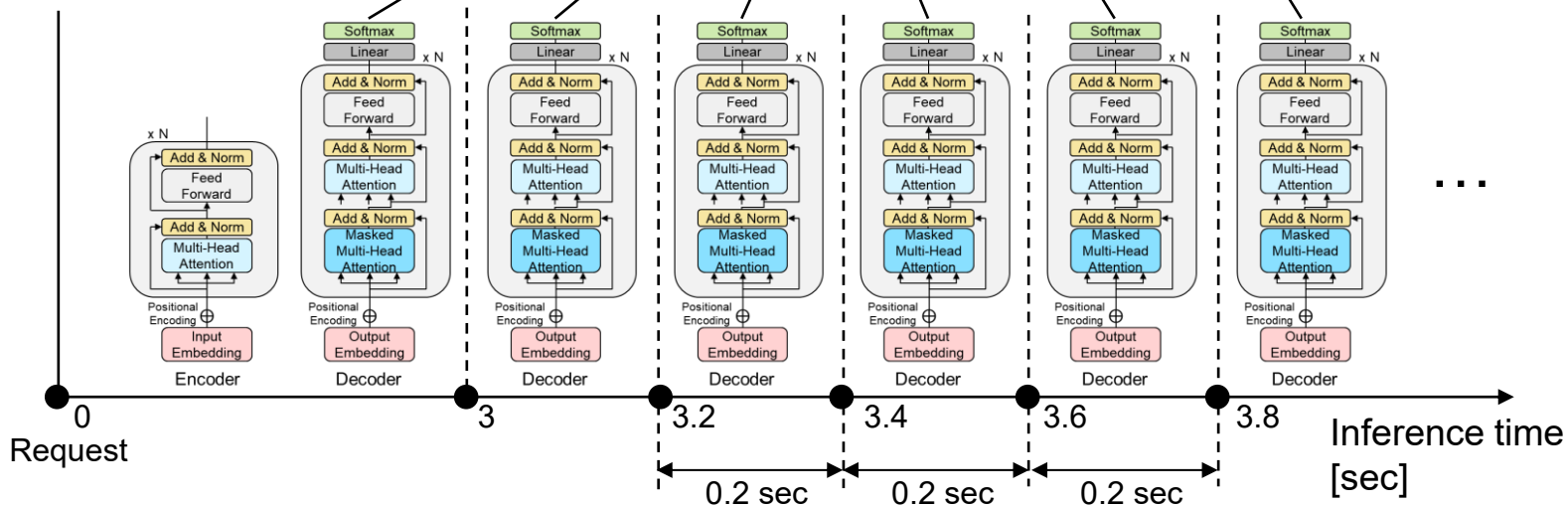
User

What is HBM?



LLM

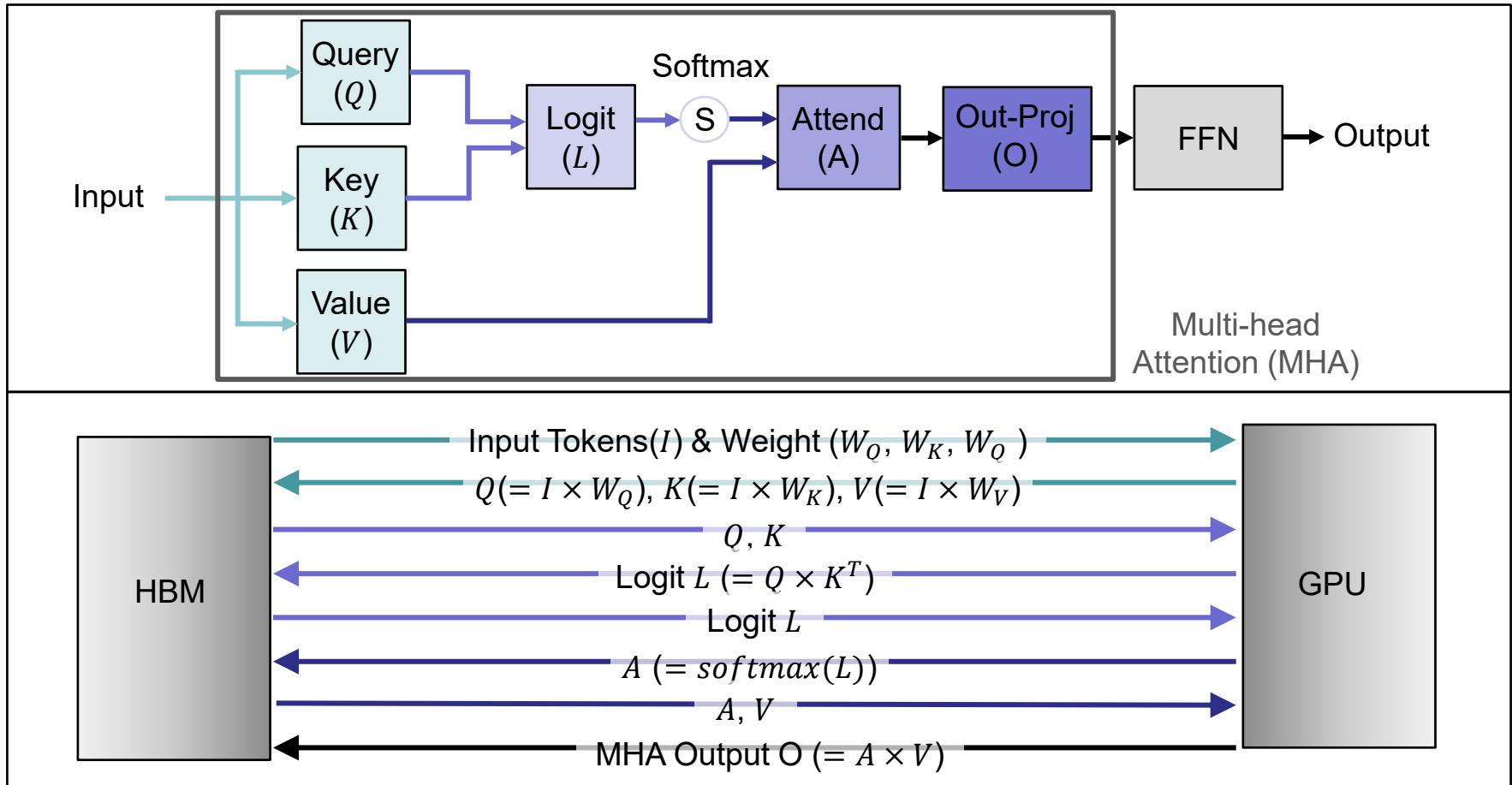
HBM stands for "High Bandwidth Memory," ...



Token Latency : 0.2 sec/token

- Lower token latency is important for user experience.

Massive Data Movement between GPU and HBM during Transformer-based AI Computing



< Structure of MHA and FFN in Transformer and Conceptual Data Movement between GPU and HBM >

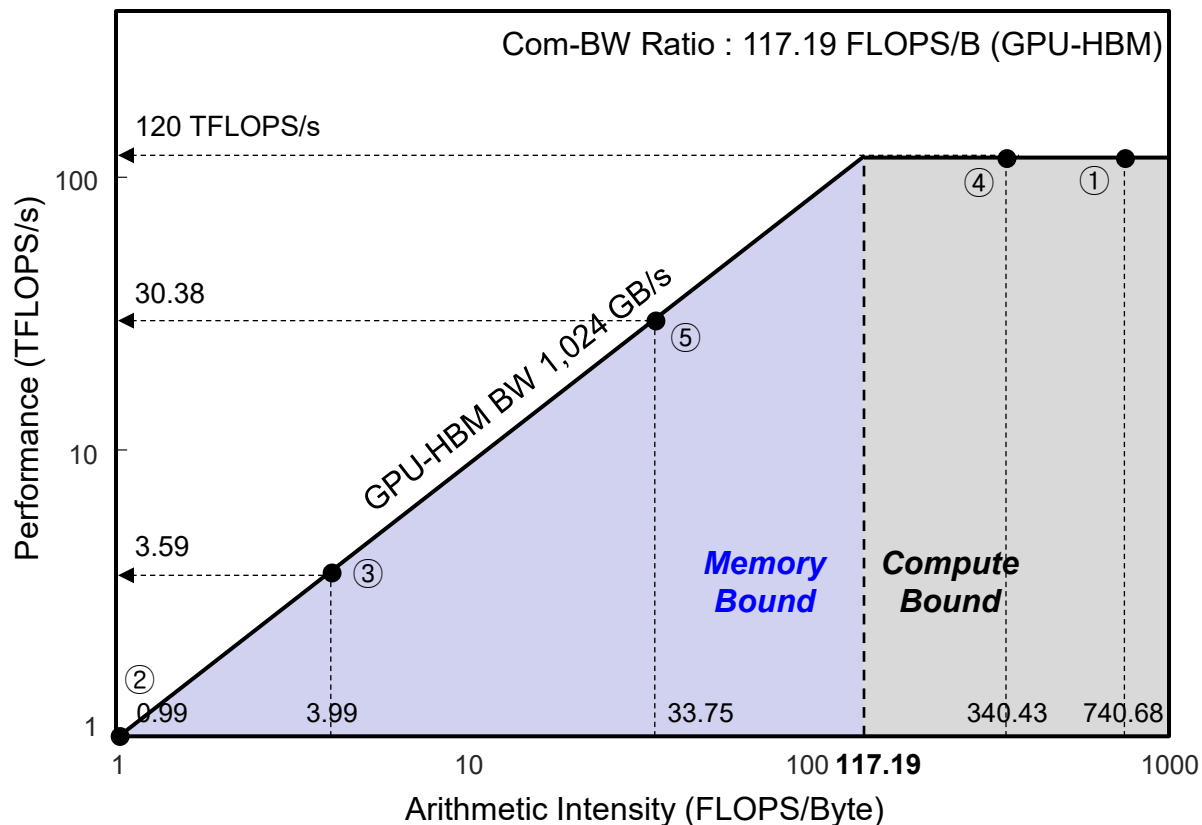
- Transformer model consists of an encoder and decoder, which are composed of two main components: multi-head attention (MHA) and feed forward network (FFN).
- During these processes, significant data movement occurs between GPU and HBM.

Performance Evaluation of GEMM Workloads with GPU-HBM Roofline Model

$$\text{Arithmetic Intensity} = \frac{M \cdot N \cdot K}{M \cdot K + N \cdot K + M \cdot N}$$

$$\text{Com-BW Ratio} = \frac{\text{Peak Performance}}{\text{Bandwidth}}$$

GEMM	Arithmetic Intensity [FLOPS/Byte]	Achieved Performance [TFLOPS/s]
①	740.68	120
②	0.99	0.89
③	3.99	3.59
④	340.43	120
⑤	33.75	30.38



< Roofline Model of Simulated GEMM Workloads >

- Assuming the GPU-HBM architecture meets the system's peak computation capabilities, the roofline model of GPU-HBM and GEMM workloads was plotted based on V100.
 - ✓ GEMM ①, ④ : Arithmetic Intensity > 117.19 FLOPS/B → **Compute-bound**
 - ✓ GEMM ②, ③, ⑤ : Arithmetic Intensity < 117.19 FLOPS/B → **Memory-bound**

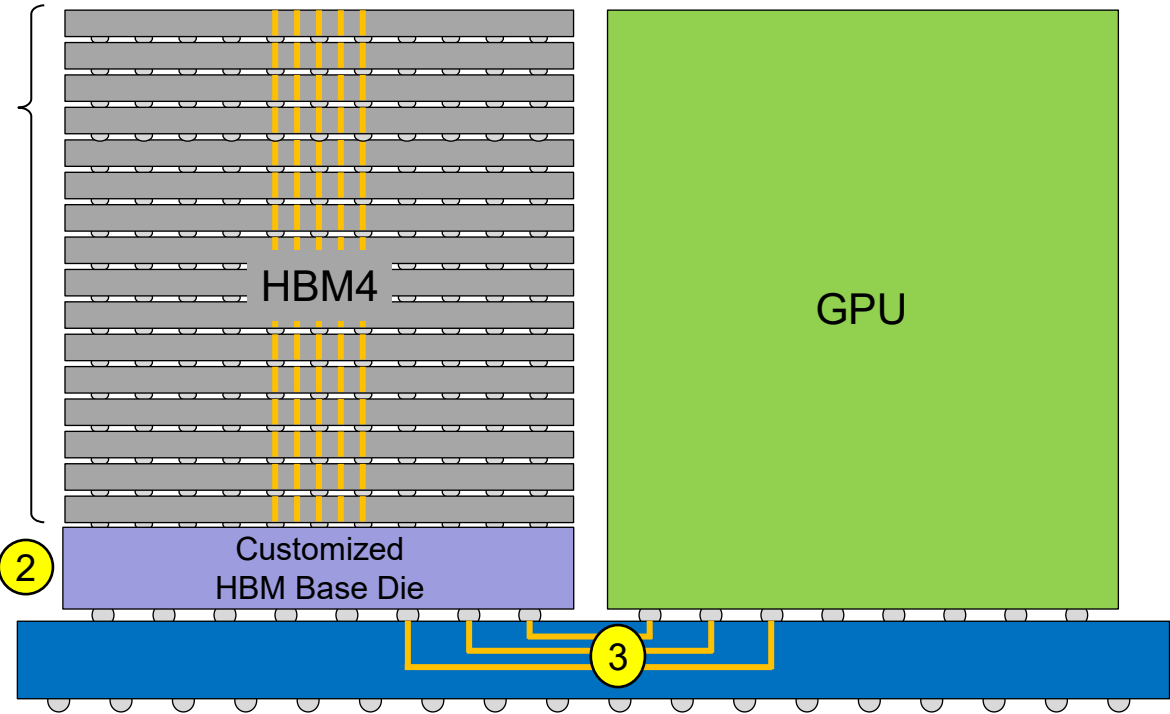
Part2: HBM Roadmap : HBM4-HBM8

TSV(실리콘 관통전극)

1 HBM 적층 수, 저장 용량 증가
8 ~ 12 단 → 12 ~ 16 단
16 ~ 24 GB → 36 ~ 48 GB

2 고객 맞춤형 (Customized)
Base Die 설계
일부 GPU 계산 기능이 이동

실리콘 인터포저 (기판)

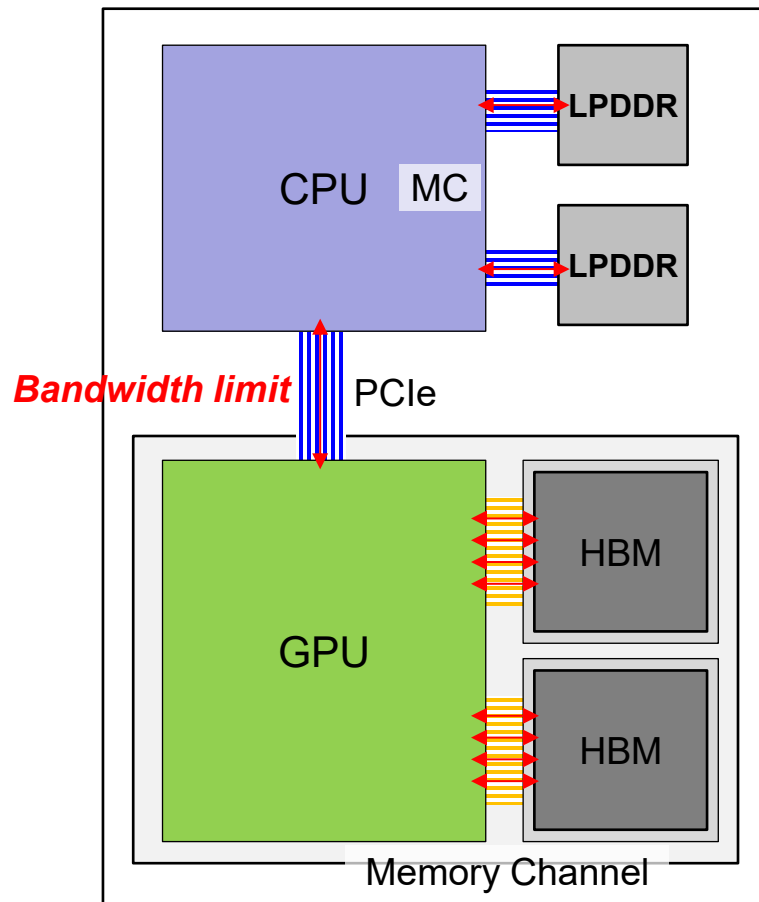


3 HBM3/3E 대비 2배 늘어난
HBM-GPU 연결선 (I/O)
2,048 개

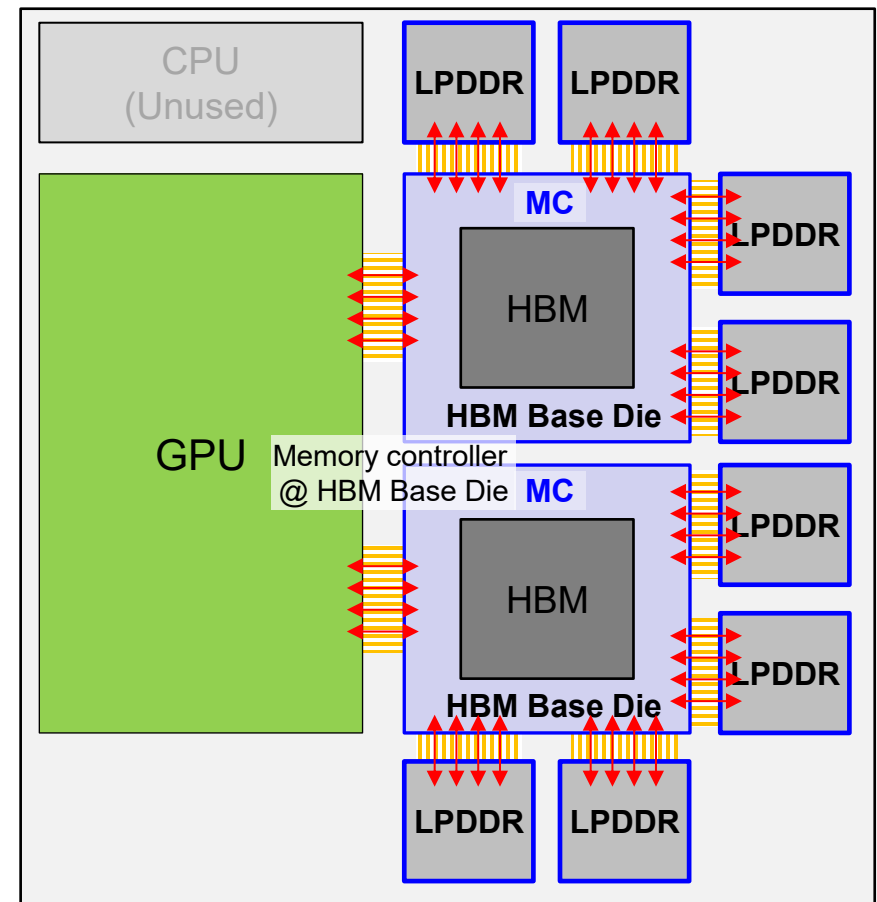
Custom HBM Base Die Design : LPDDR Memory Channel for High Capacity & Memory Bandwidth

HBM4
Architecture

↔ : Low-Bandwidth ↔↔↔ : High-Bandwidth

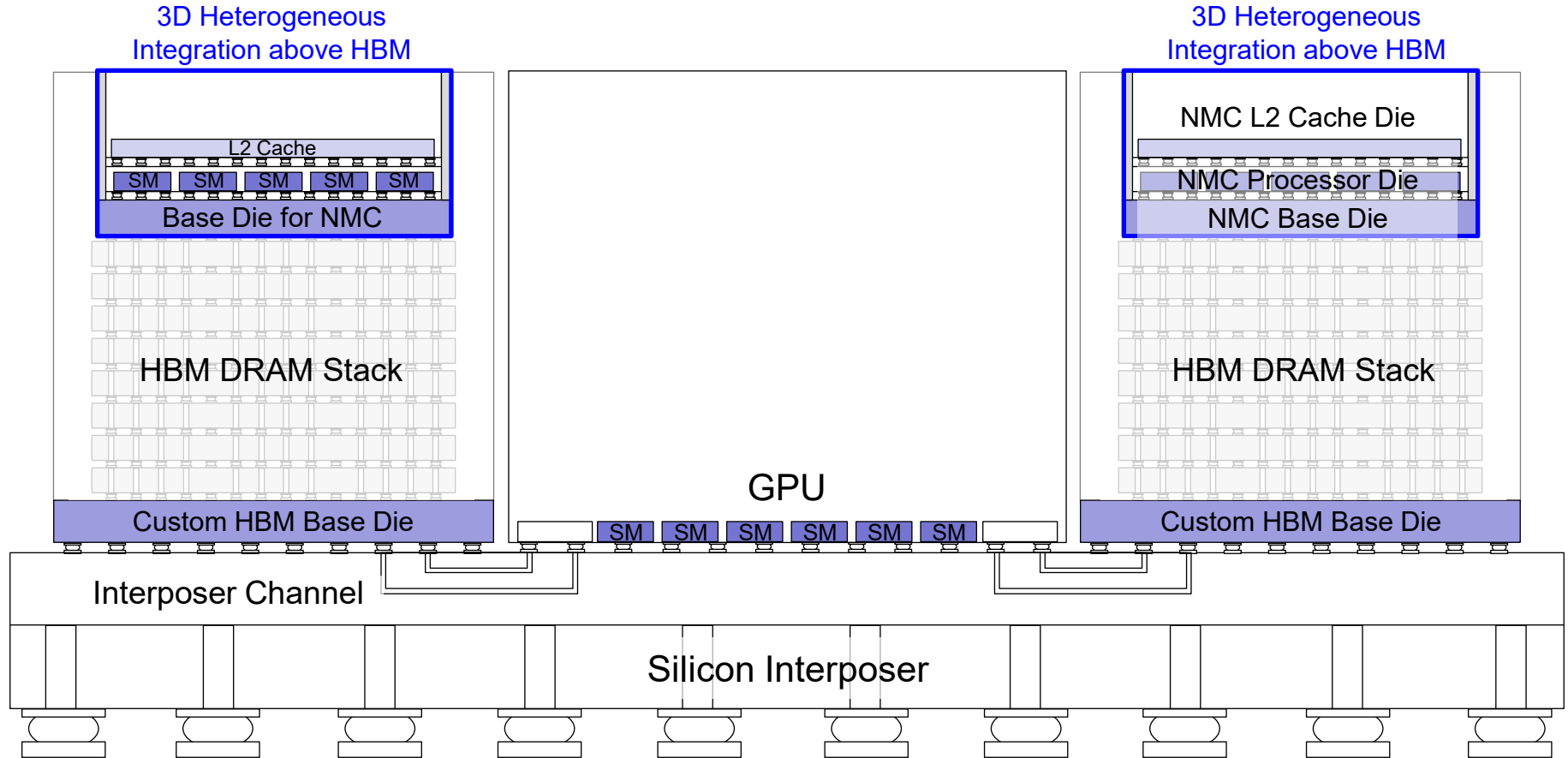


< Conventional CPU-GPU Architecture with HBM3 >



< HBM4 Architecture with LPDDR >

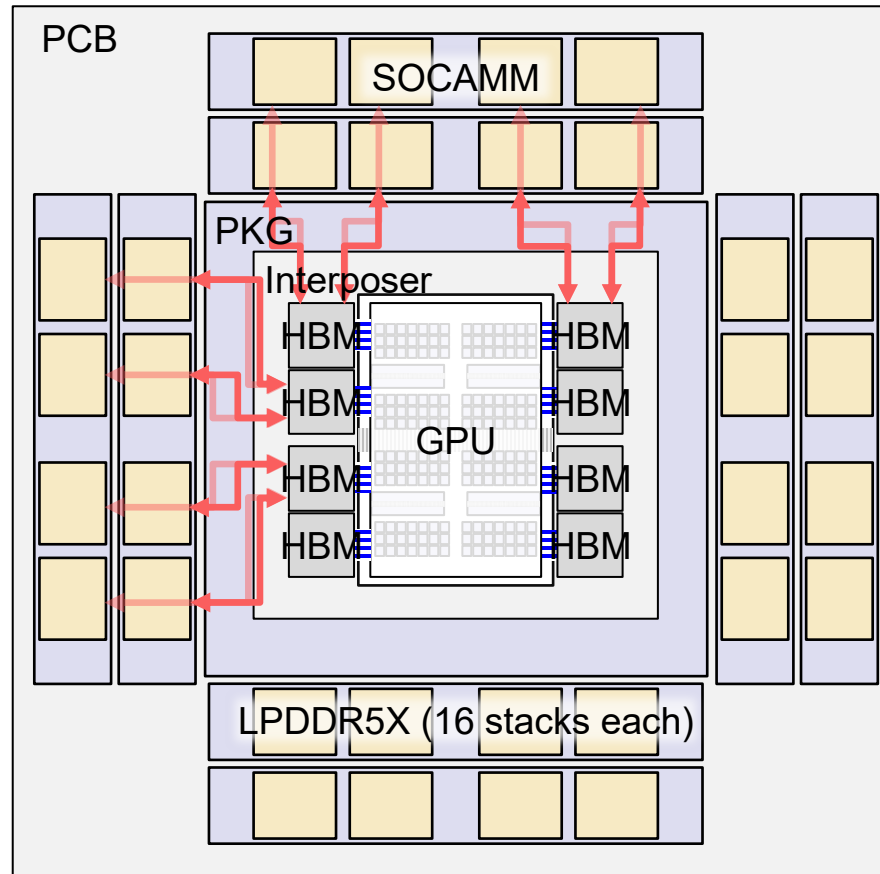
- The custom base die of HBM4 enables direct access to HBM and LPDDR, providing improved memory capacity without the CPU.



< 3D Integrated NMC-HBM Architecture above HBM DRAM >

- By integrating a NMC processor die and cache die above HBM, the proposed 3D NMC-HBM achieves high performance and energy efficient computing through dedicated TSV interconnection and power supply network.

 Interposer routing
 PCB routing



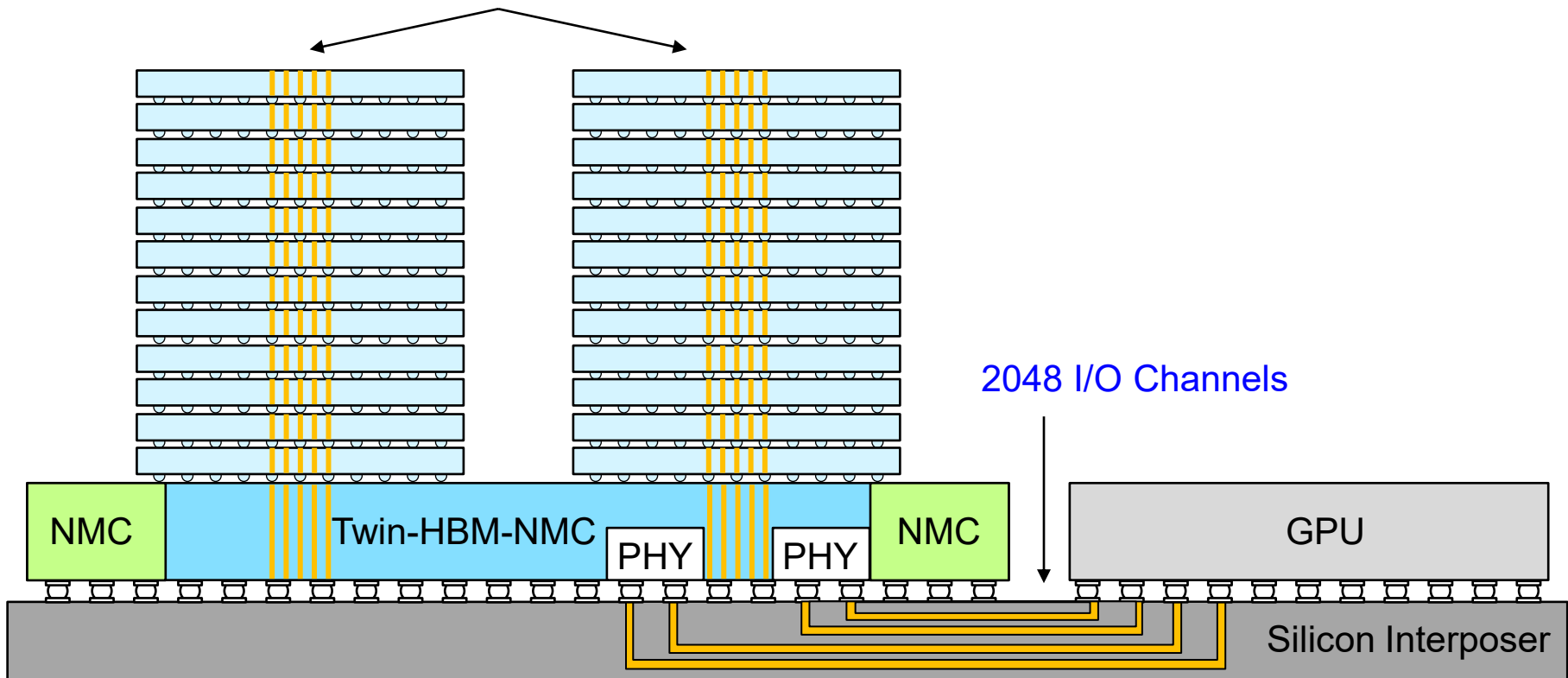
< Proposal of GPU-HBM-SOCAMM System: HBM-Centric Architecture with Off-Interposer SOCAMM Expansion >

- HBM은 인터포저 위에, SOCAMM은 인터포저 외부(PCB)에 배치된 구조로, 고대역폭과 대용량을 동시에 확보하는 아키텍처
- GPU-HBM 간은 고속 인터포저 연결, SOCAMM은 PCB 라우팅을 통해 추가 확장 가능

Proposal of Twin Tower High Bandwidth Memory with Near-Memory Computing (Twin-HBM-NMC) Architecture

HBM6
Architecture

Two DRAM Stacks

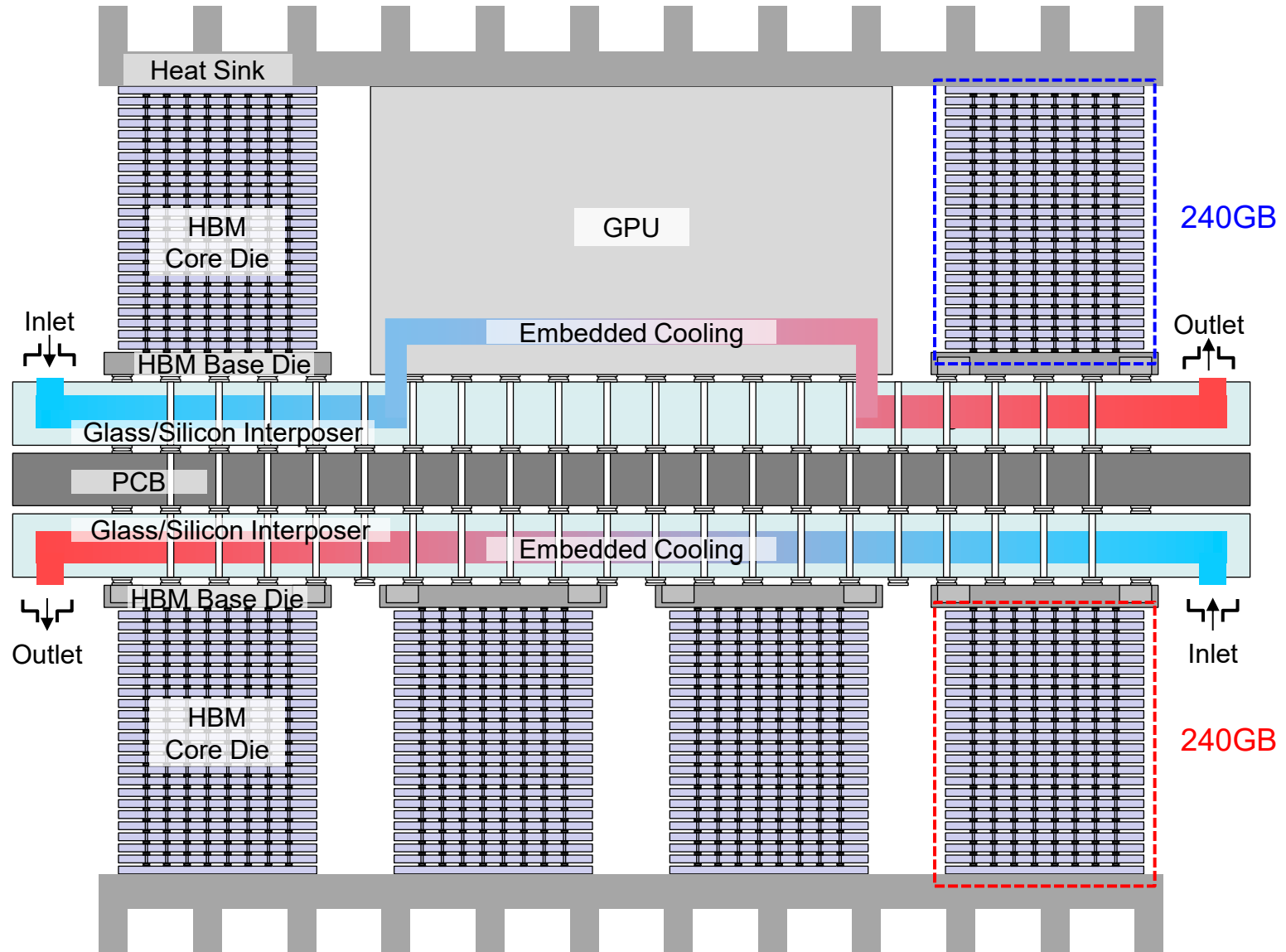


< Overview of Twin Tower HBM with NMC Architecture >

- In twin tower high bandwidth memory with near-memory-computing (Twin-HBM-NMC) architecture, two DRAM stacks are located on top of the large logic die.
- The logic die include NMC units and is connected to the GPU via 2048 interposer channels.

3D Memory Expansion Architecture with Embedded Cooling Structure for HBM8 with Double-Sided Interposer [1/3]: GPU-HBM-HBM

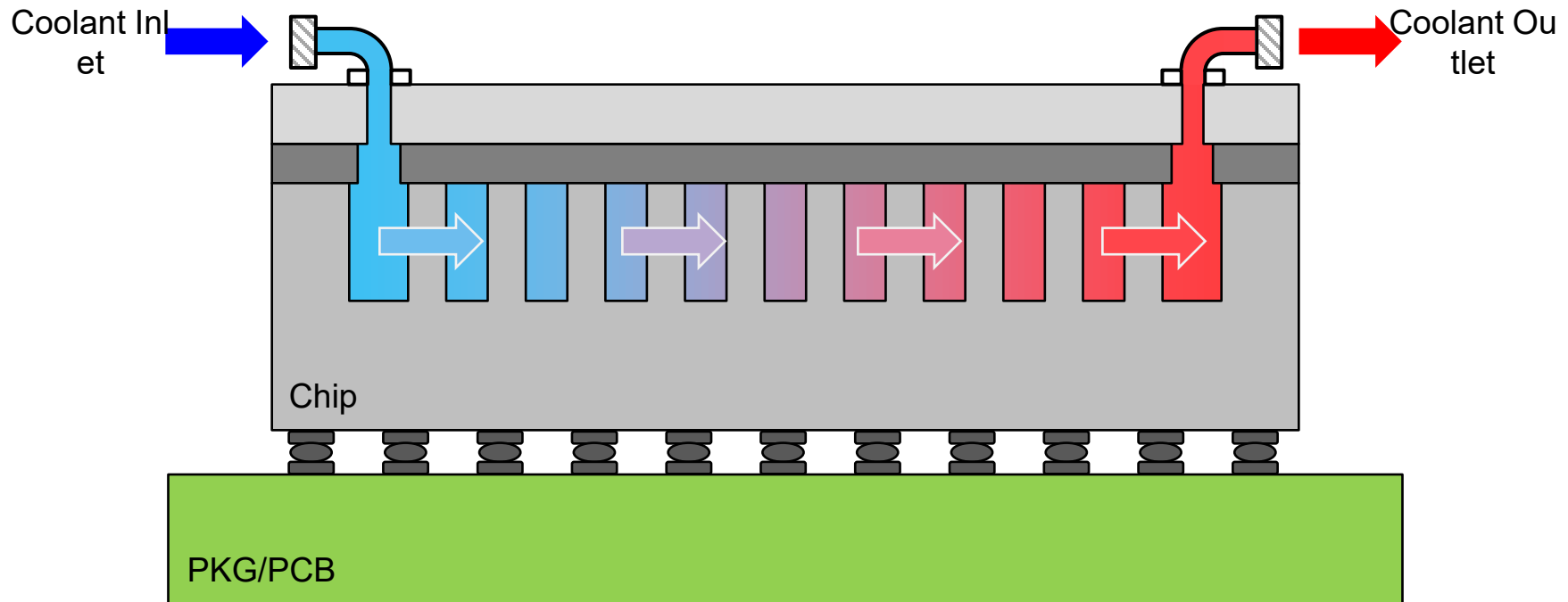
HBM8
Architecture



< 3D Memory Expansion Architecture with Embedded Cooling for HBM8 with Double-Sided Interposer using HBM >

Inter-Chip Cooling Method: Microfluidic Cooling

– Advancement of Microchannel Cooling

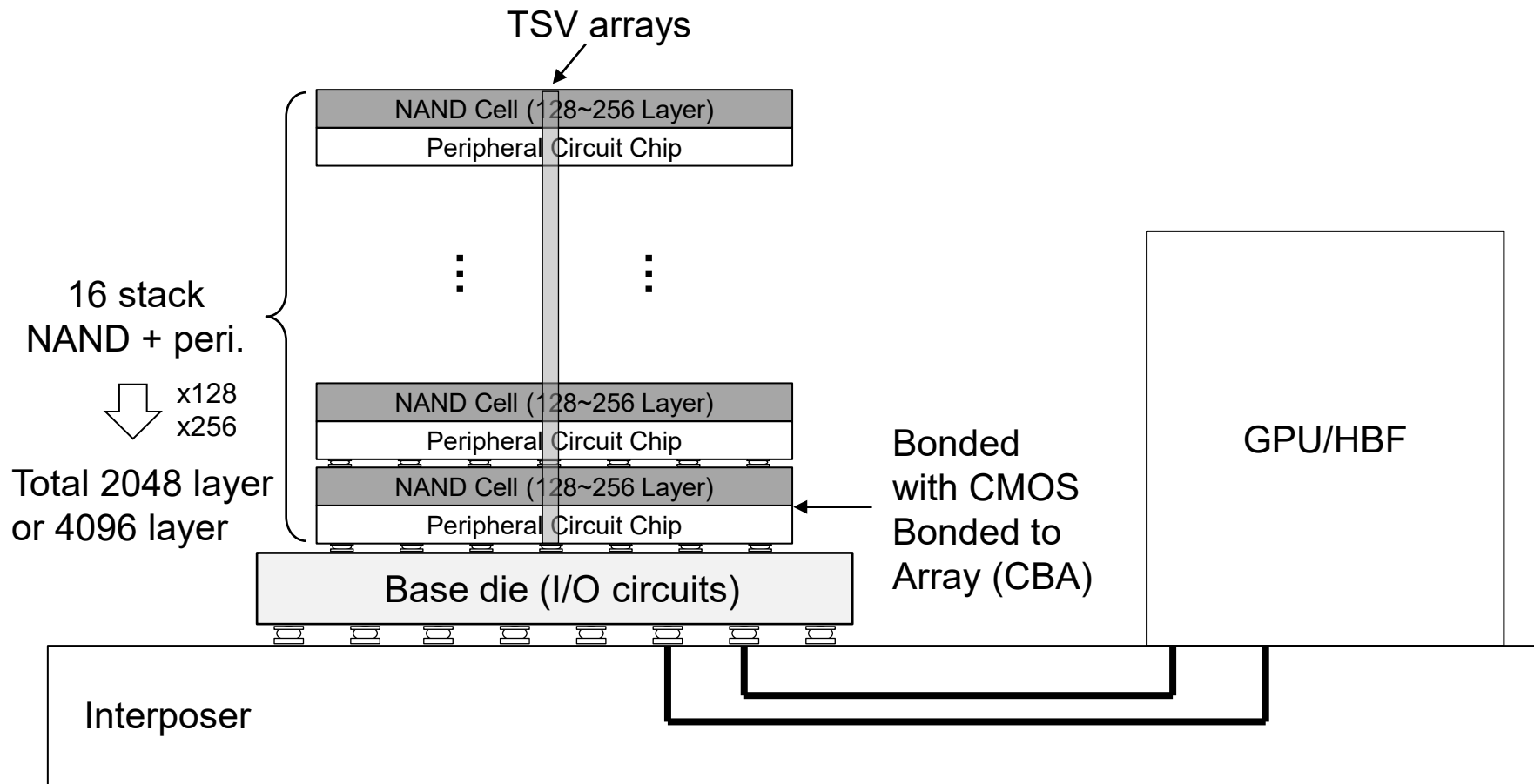


- Microfluidic cooling 방법은 microchannel cooling 방법의 발전형이며, Chip 내부에 microchannel 을 가공하여 direct cooling 하는 방법임.
- Chip 의 윗부분에서 coolant 를 inlet 하여, chip 내부에 직접 coolant 를 유동시킨 뒤, Chip의 열을 흡수한 coolant 를 다시 Chip 의 윗부분에서 outlet 하는 방식임.
- Chip 내부에서 direct cooling 하는 방법이기때문에, 다른 냉각 방법 대비 보다 높은 냉각 성능을 가짐.

Part3:

What is HBF and HBF Architectures

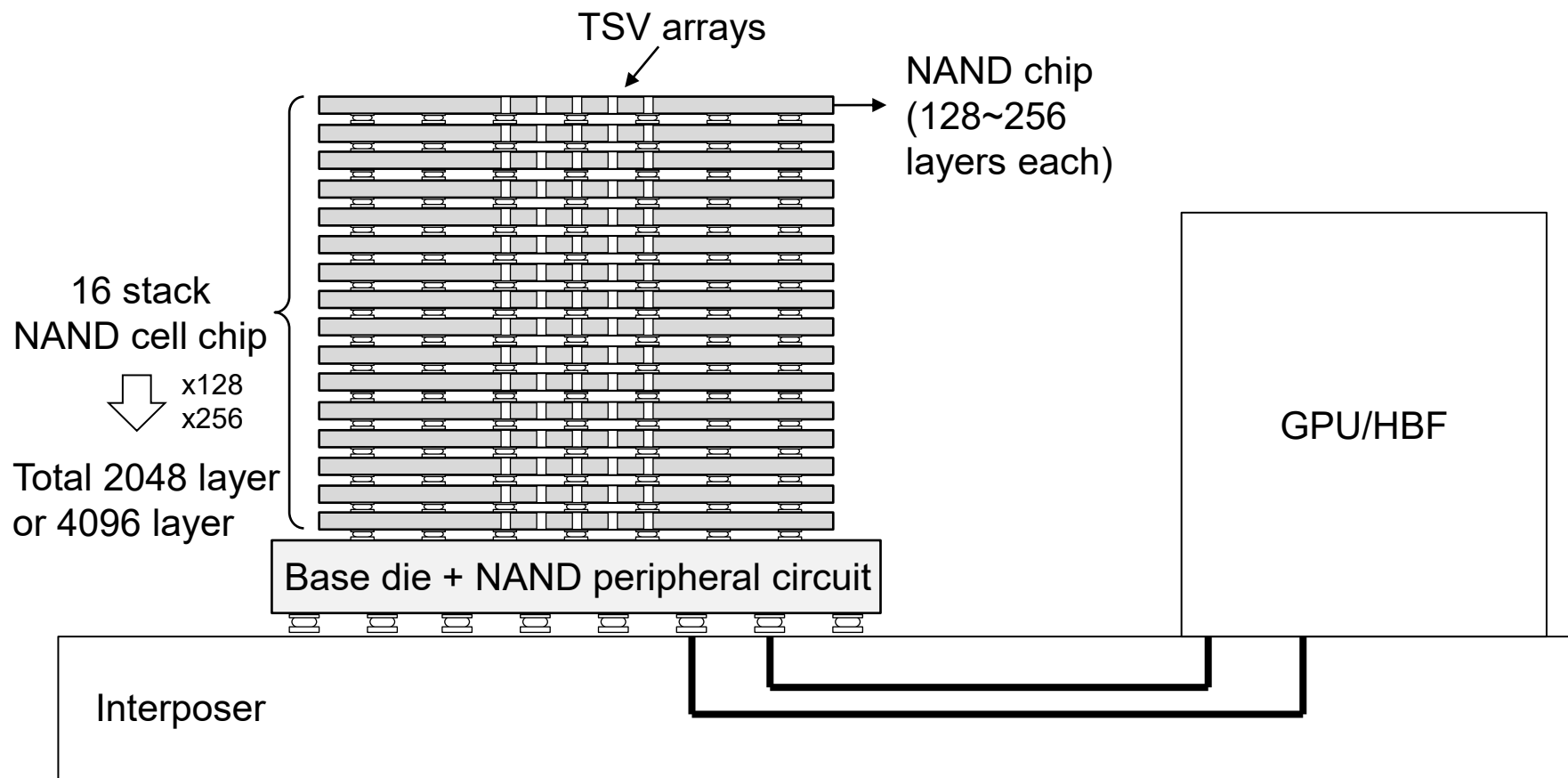
Type I. HBF



< HBF Type 2. Peripheral on every NAND die >

- Total 2048 (128 x 16) or 4096 (256 x 16) layers of NAND cell.
- Peripheral circuit on every die

Type II. HBF



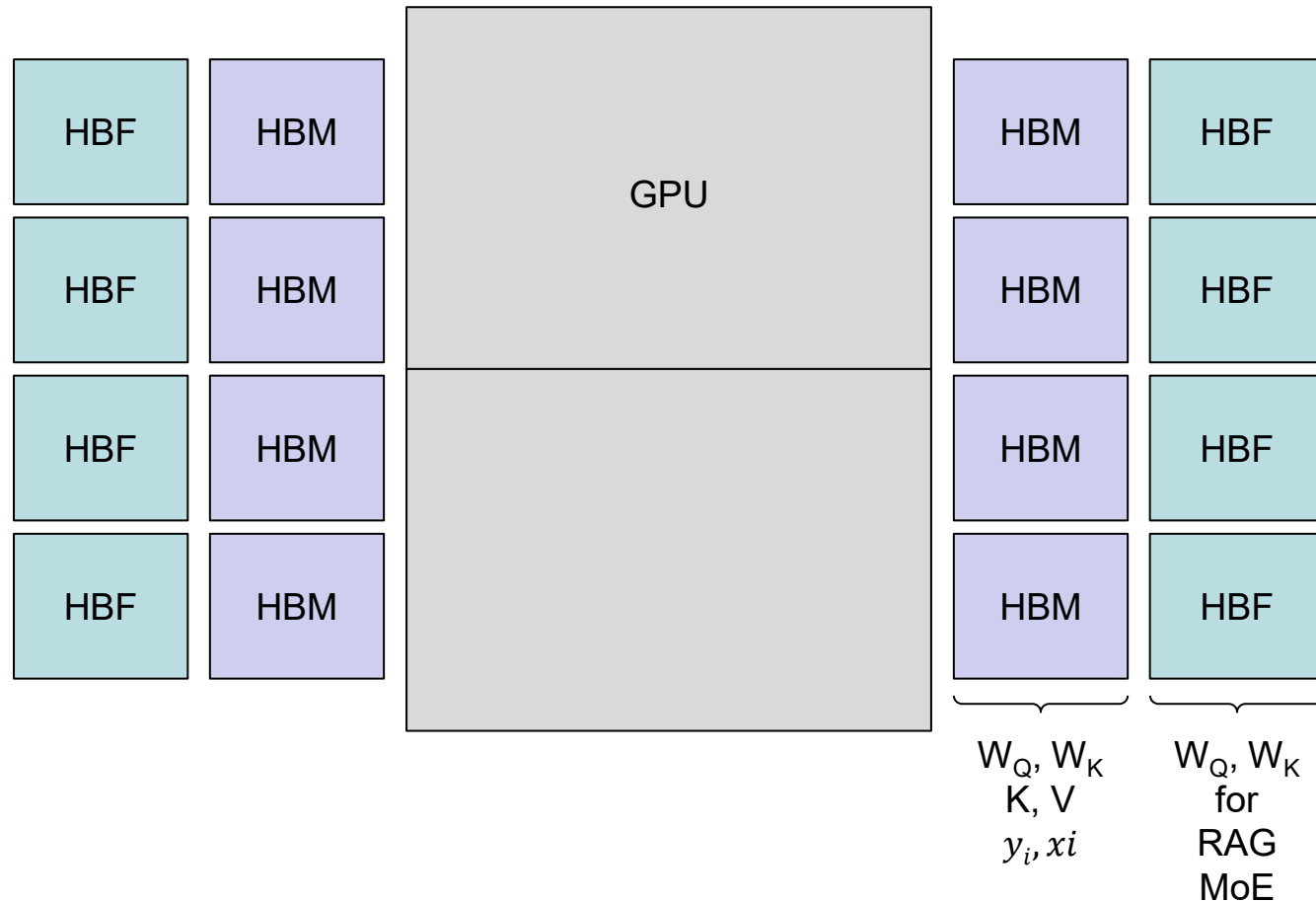
< HBF Type 1. Peripheral on base die >

- Total 2048 (128 x 16) or 4096 (256 x 16) layers of NAND cell
- Peripheral circuit only on base die

Advantages of HBF

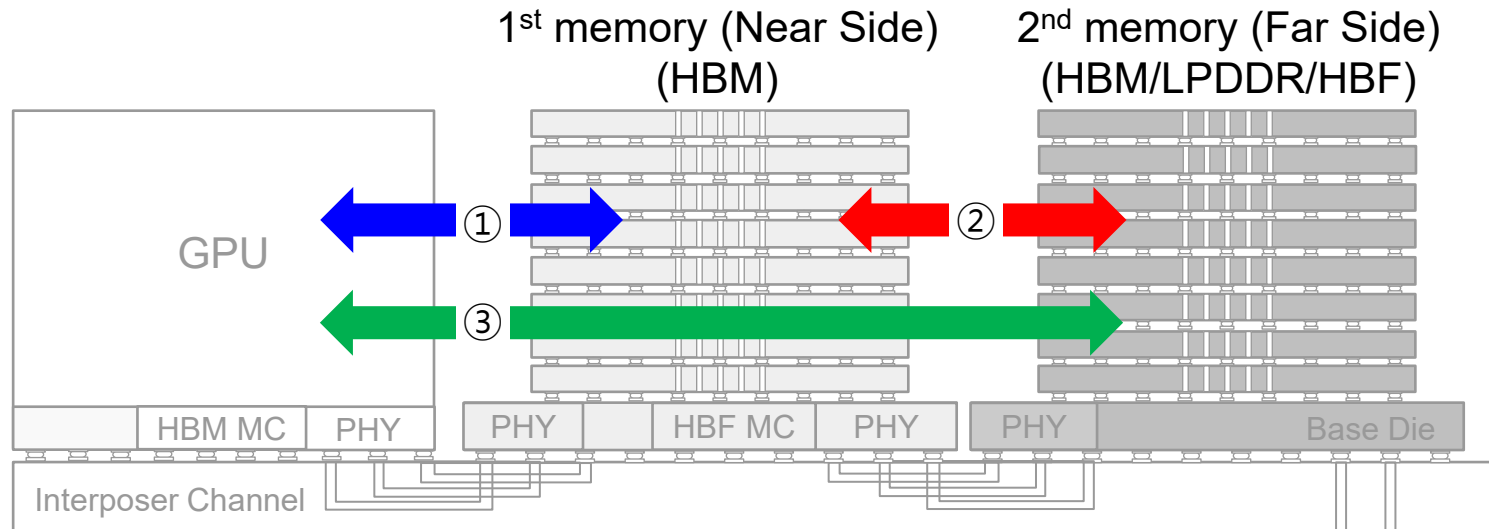
- ✓ Large Memory Capacity
- ✓ Higher Bandwidth
- ✓ Can be Combined with HBM
- ✓ Can be Used for KV Cache for inference in transformer model
- ✓ Can be used for RAG document saving storages
- ✓ Can be used mode parameter saving storage
- ✓ Big opportunity for NAND Flash companies in AI Era

Specialized Hybrid Architecture for RAG and MoE Workloads



< Workload-specific memory partitioning for RAG and MoE >

LLM Inference Latency Evaluation Setup Considering Memory Hierarchy

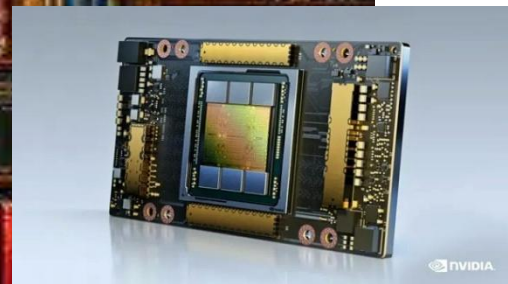
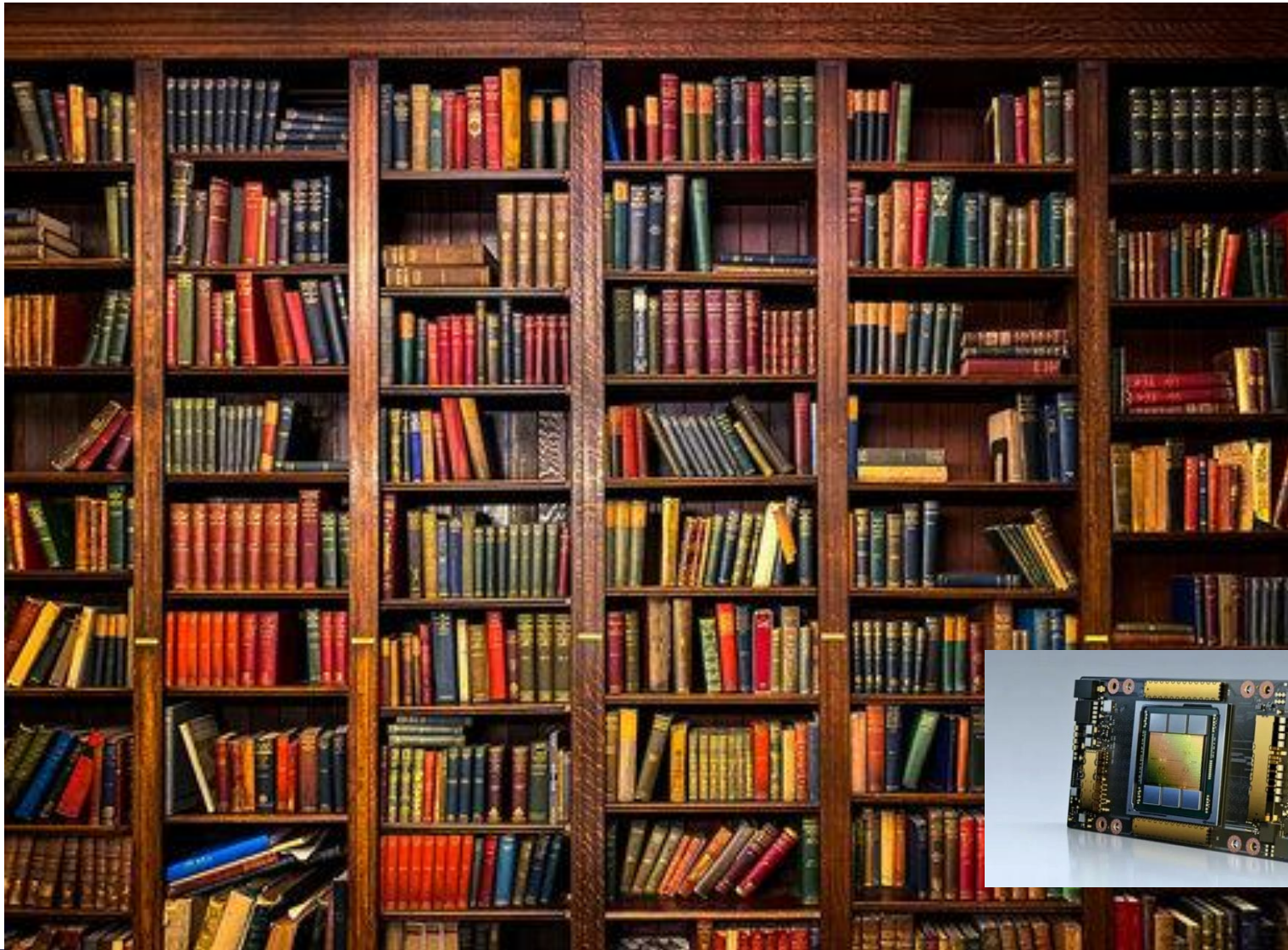


< Latency analysis setup with diverse next-generation memory for LLM Inference >

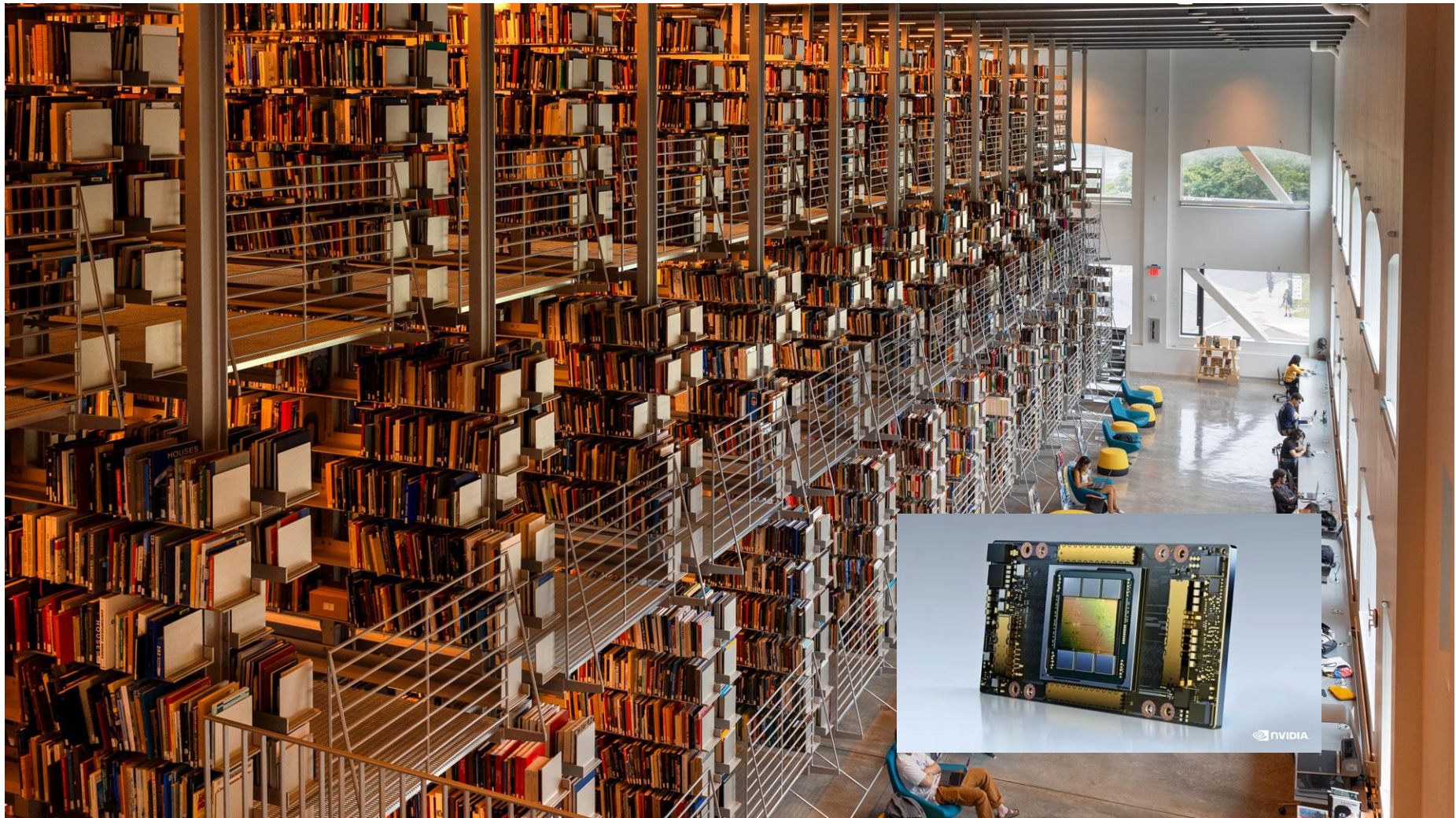
Memory Capacity	Multi-Stack HBM	HBM-LPDDR	HBM-HBF
1 st Memory (Near Side)	200 GB (HBM)	200 GB (HBM)	200 GB (HBM)
2 nd Memory (Far Side)	200 GB (HBM)	400 GB (LPDDR)	2 TB (HBF)
t_{access} (Read + Write)	Multi-Stack HBM	HBM-LPDDR	HBM-HBF
① GPU ↔ 1 st Memory	1 ns	1 ns	1 ns
② 1 st Memory ↔ 2 nd Memory	1 ns	9 ns	99 ns
③ GPU ↔ 2 nd Memory	1 ns	10 ns	100 ns

< Memory Capacity / Access(Read+Write) Latency Teardown of HBM-based Memory System >

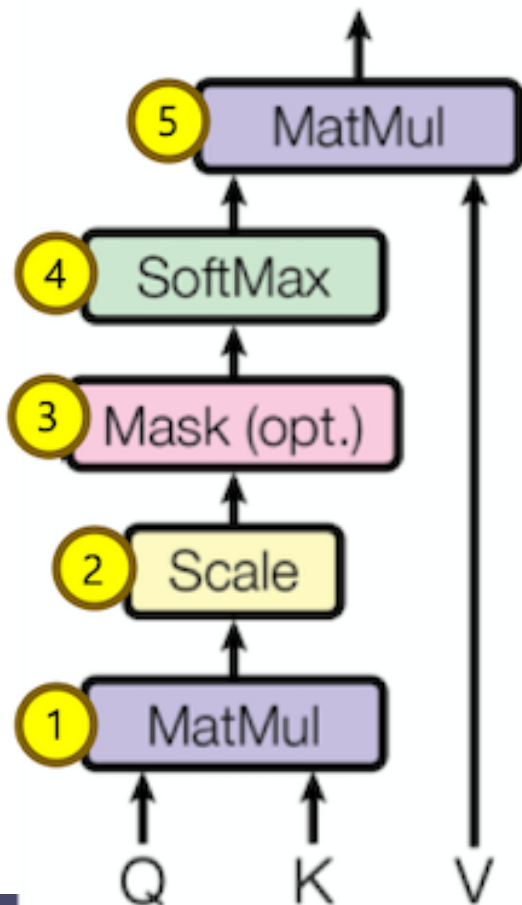
HBM: Bookshelf near GPU



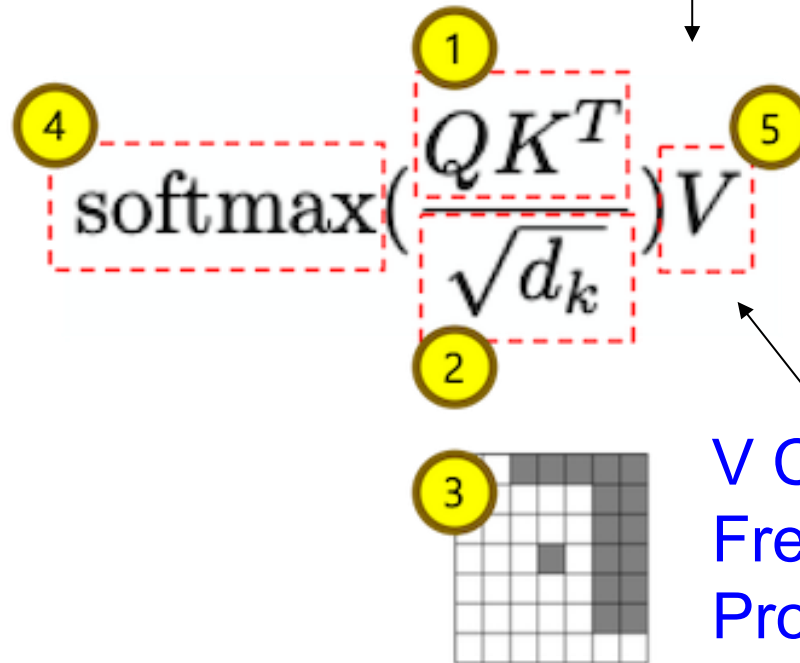
HBF: Library near GPU



K, V Cache in HBF for Transformer Model



K Cache in HBF
Frequent Read Process
(Token Length)
Large Size

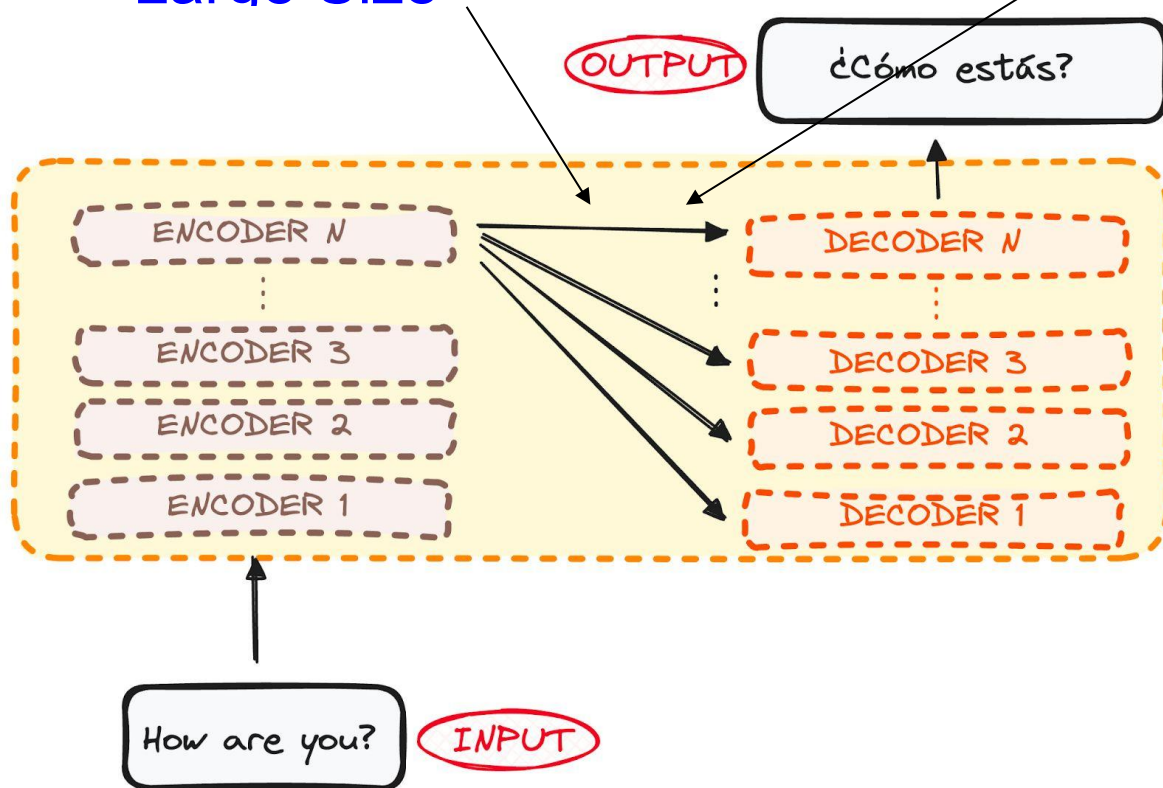


V Cache in HBF
Frequent Read Process (Token length)
Large Size

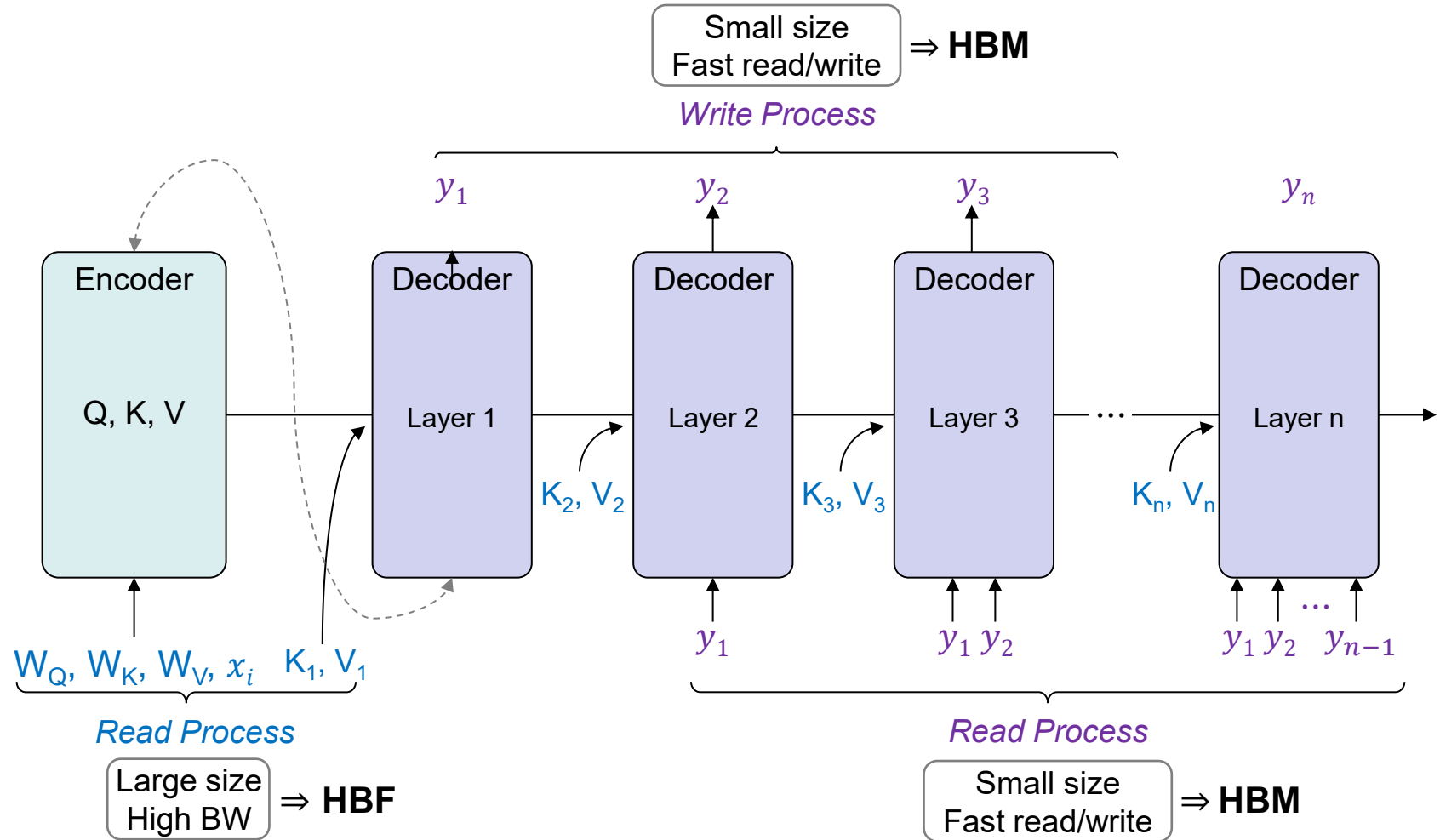
K, V Cache in HBF for Transformer Model

K Cache in HBF
Frequent Read Process
(Token Length)
Large Size

V Cache in HBF
Frequent Read Process
(Token length)
Large Size

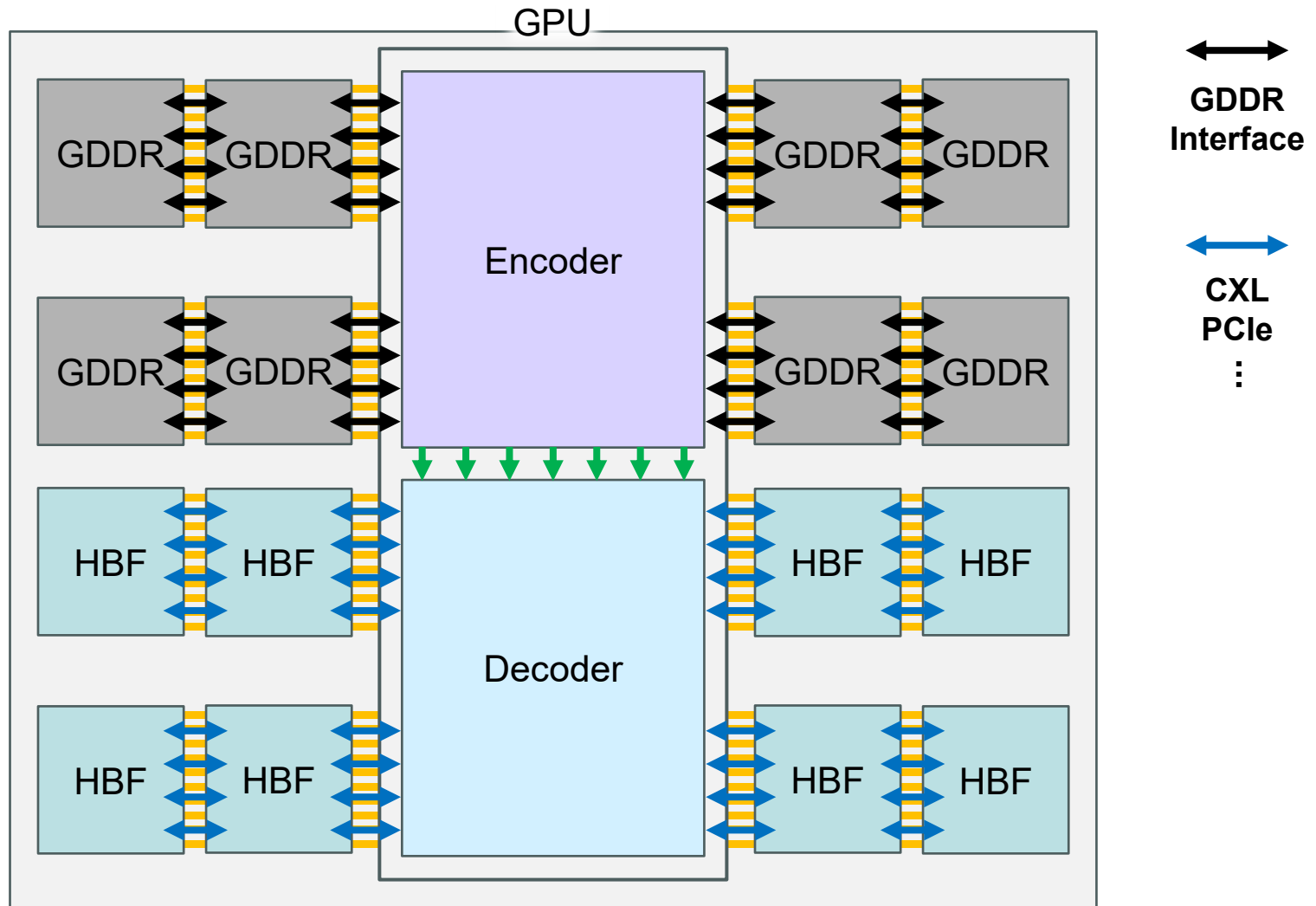


HBM-HBF Memory Allocation in Inference Pipeline



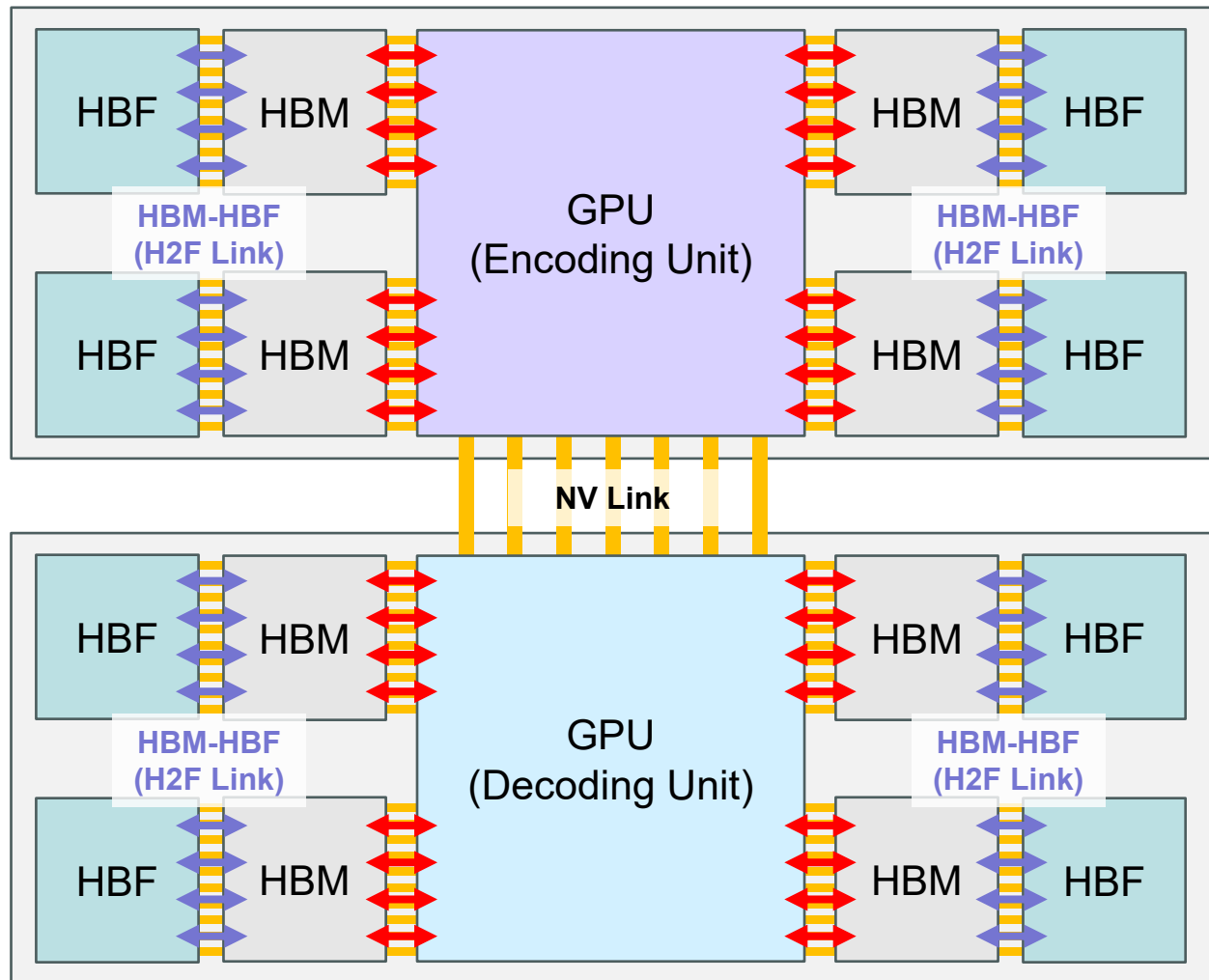
< Memory allocation strategy for encoding and decoding phases >

GDDR-HBF Single-GPU Architecture (small model, large context/token length)



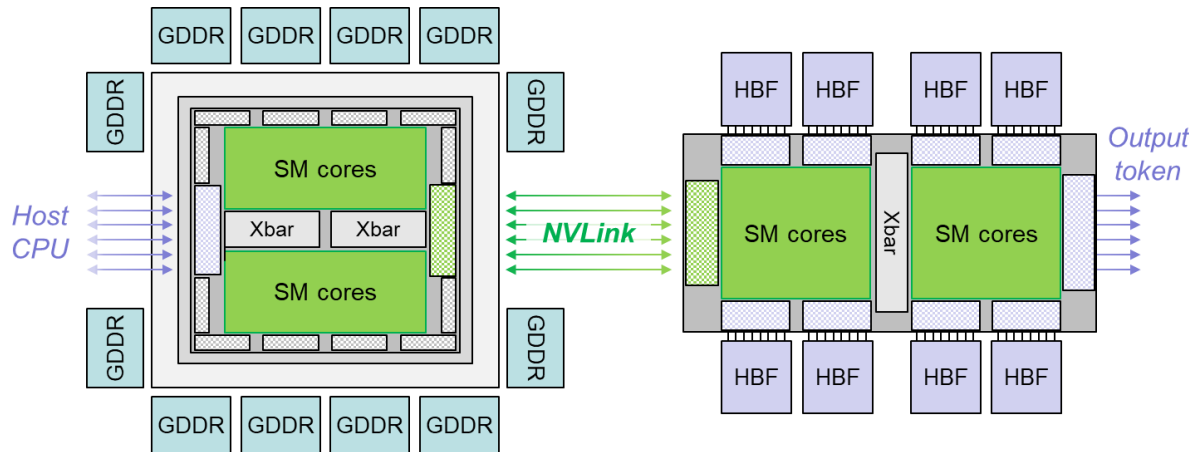
< GDDR-HBF Single-GPU Architecture: Top View >

HBM-HBF Dual-GPU Architecture (small model, large context/token length)



< HBM-HBF Dual-GPU Architecture: Top View >

Part4: Workloads allocations to HBF Roadmap

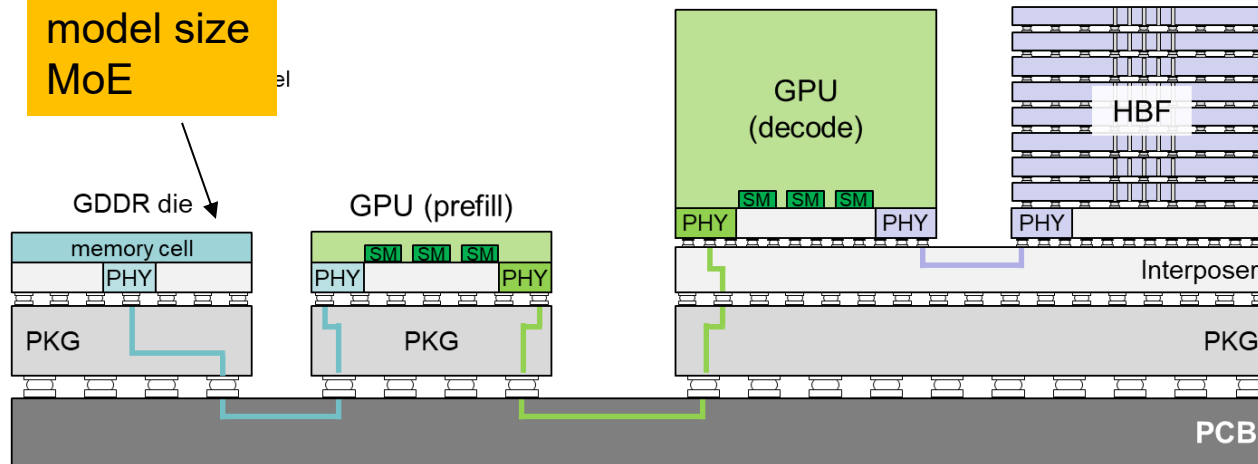


< Top-view of the GPU-GDDR and GPU-HBF hybrid architecture >

- one GDDR capacity : 4 GB
 - Total GDDR capacity : 48 GB
 - one HBF capacity : 512 GB
 - Total HBF capacity : 4 TB
-
- # of GDDR : 12
 - # of HBF : 8
 - Total # of memory : 20

Weight:
Small
model size
MoE

KV Cache:
Longer
context
length



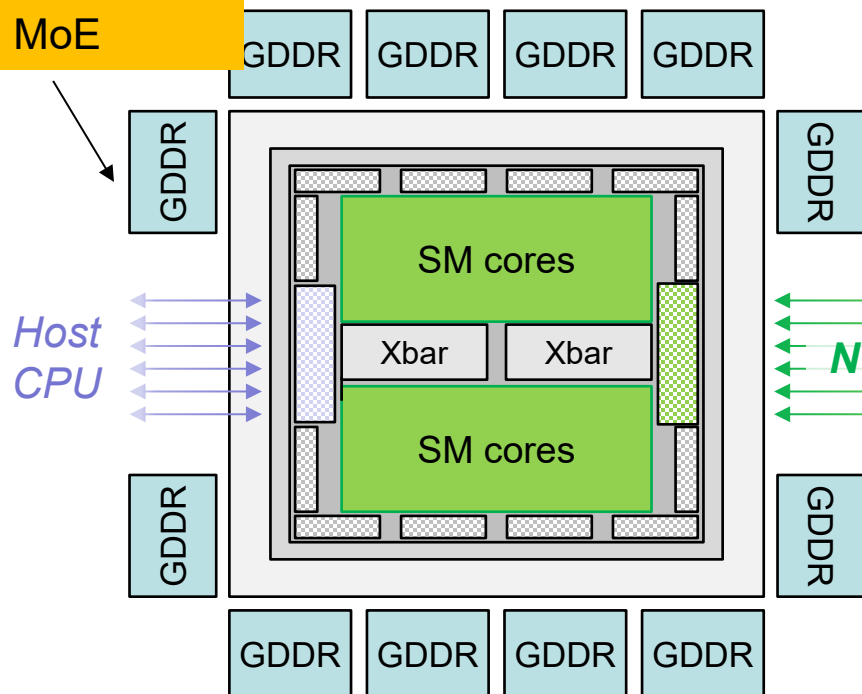
< Cross-sectional view of the GPU-GDDR and GPU-HBF hybrid architecture >

Proposal of Hybrid GDDR-HBF Memory Architecture with Multi-Agent Disaggregated Inference System

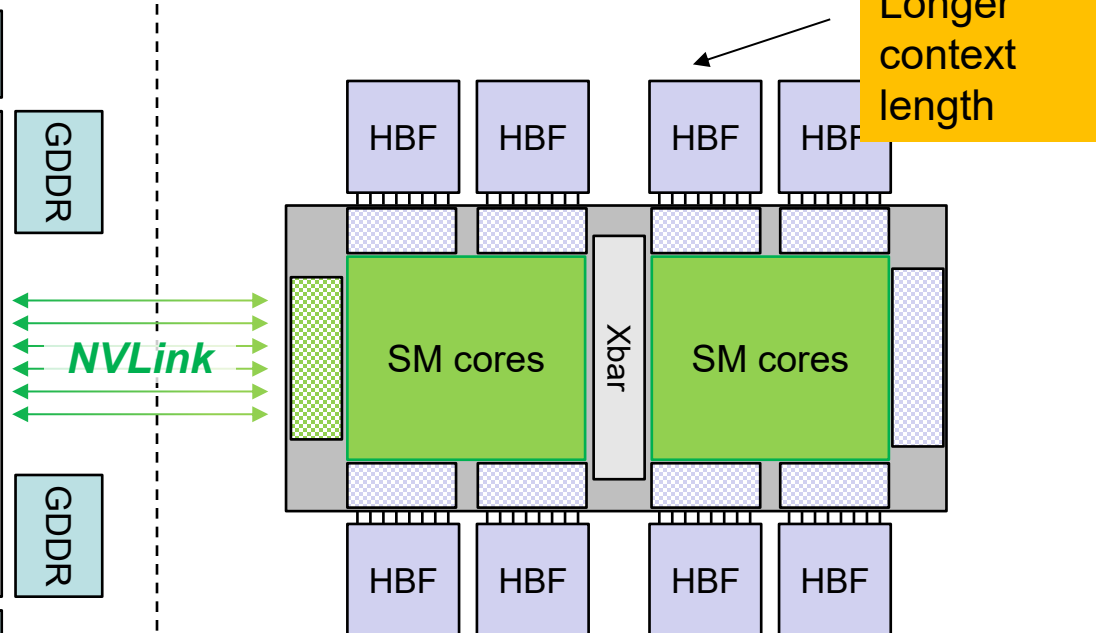
HBF1
Architecture

Weight:
Small
model size
MoE

Prefill GPU



Decode GPU

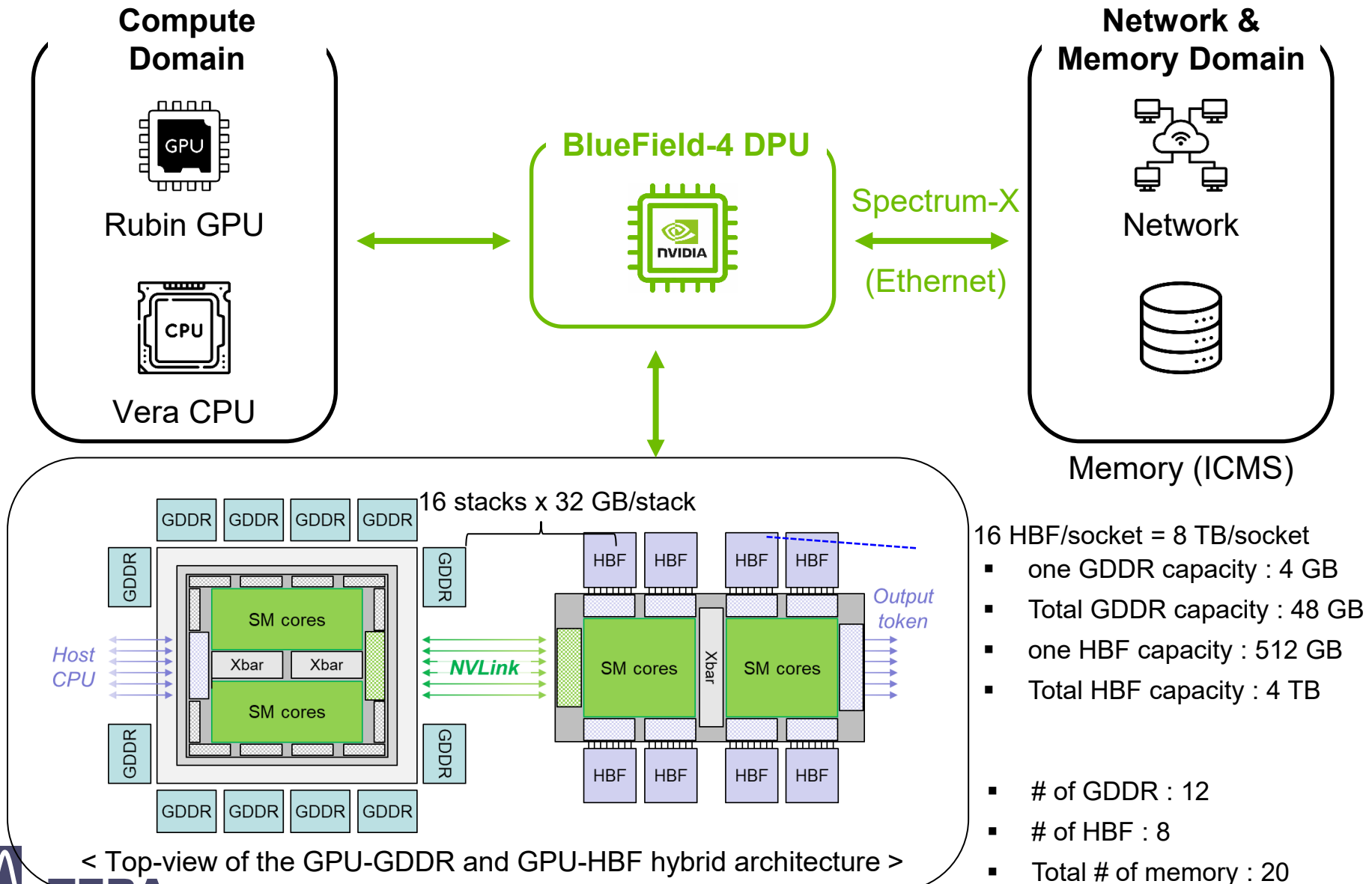


KV Cache:
Longer
context
length

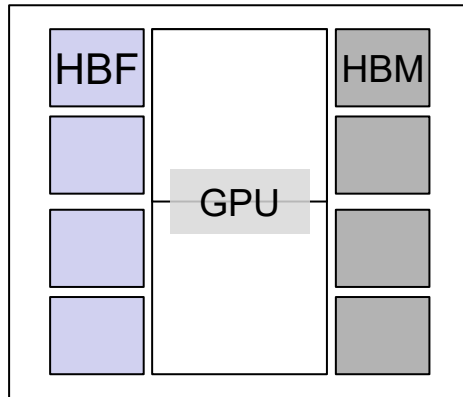
NVLink

< Overview of proposed hybrid-memory architecture with GDDR7 & HBF >

- Proposed architecture is composed of two GPUs specialized in prefill & decode, respectively.
 - Prefill GPU composed of GDDR7 which provides sufficient memory bandwidth and latency for prefill compared with HBM.
 - Decode GPU composed of HBF which provides large memory capacity.



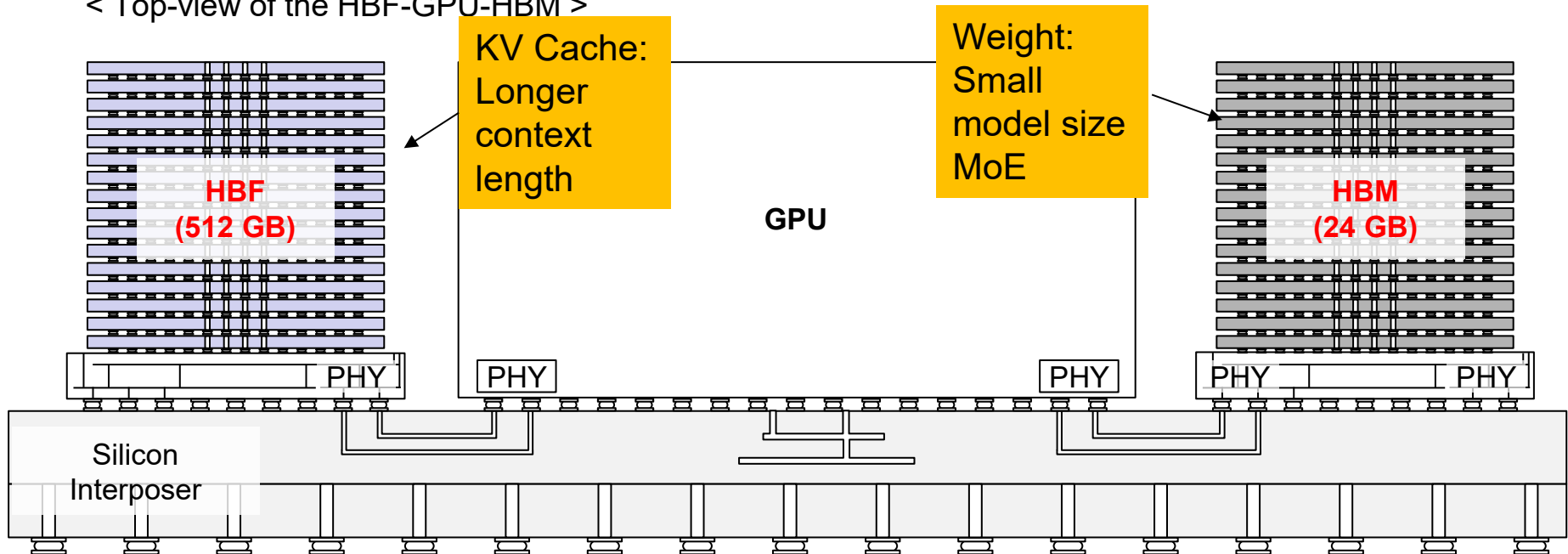
< Top-view of the GPU-GDDR and GPU-HBF hybrid architecture >



- one HBM capacity : 24 GB
- Total HBM capacity : 96 GB
- one HBF capacity : 512 GB
- Total HBF capacity : 2 TB

- # of HBM : 4
- # of HBF : 4
- Total # of memory : 8

< Top-view of the HBF-GPU-HBM >

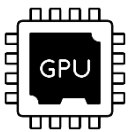


< Cross-sectional view of the HBF-GPU-HBM >

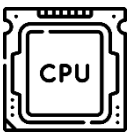
HBF for AI Accelerator - HBF-GPU-HBM Architecture

HBF2
Architecture

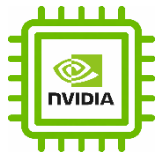
Compute Domain



Rubin GPU



BlueField-4 DPU



Spectrum-X
(Ethernet)

Network & Memory Domain



Network



Vera GPU

**HBF
(512 GB)**

KV Cache:
Longer
context
length

Weight:
Small
model size
MoE

GPU

**HBM
(24 GB)**

PHY

PHY

PHY

PHY

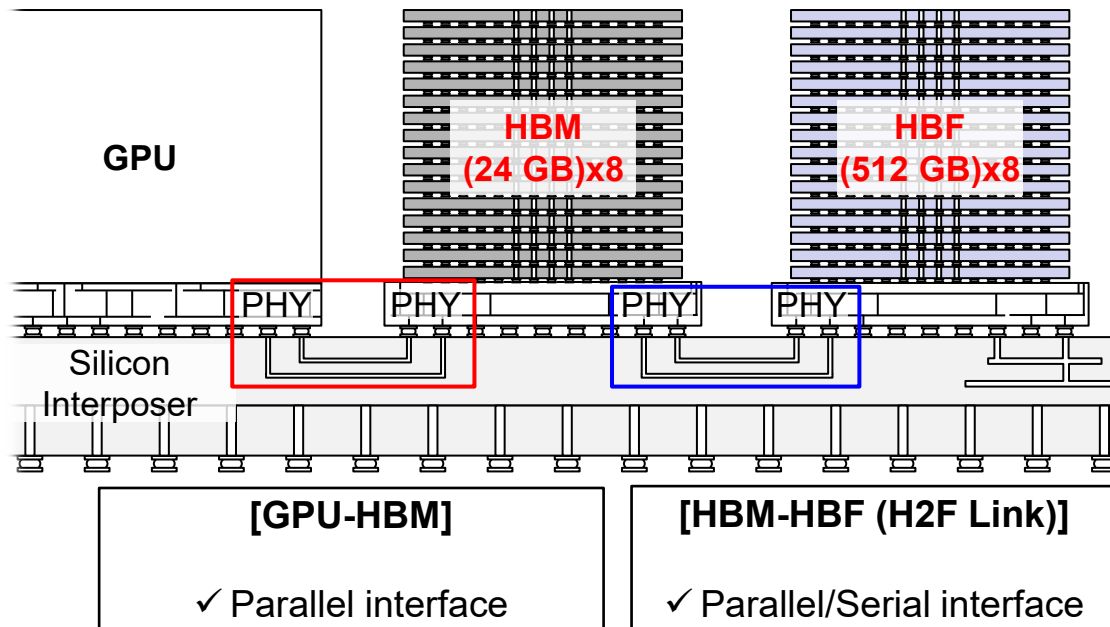
PHY

Silicon
Interposer

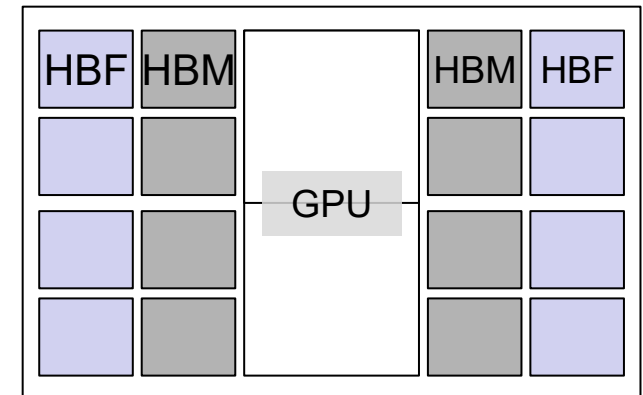
< Cross-sectional view of the HBF-GPU-HBM >

Weight:
Small
model size
MoE

KV Cache:
Longer
context
length

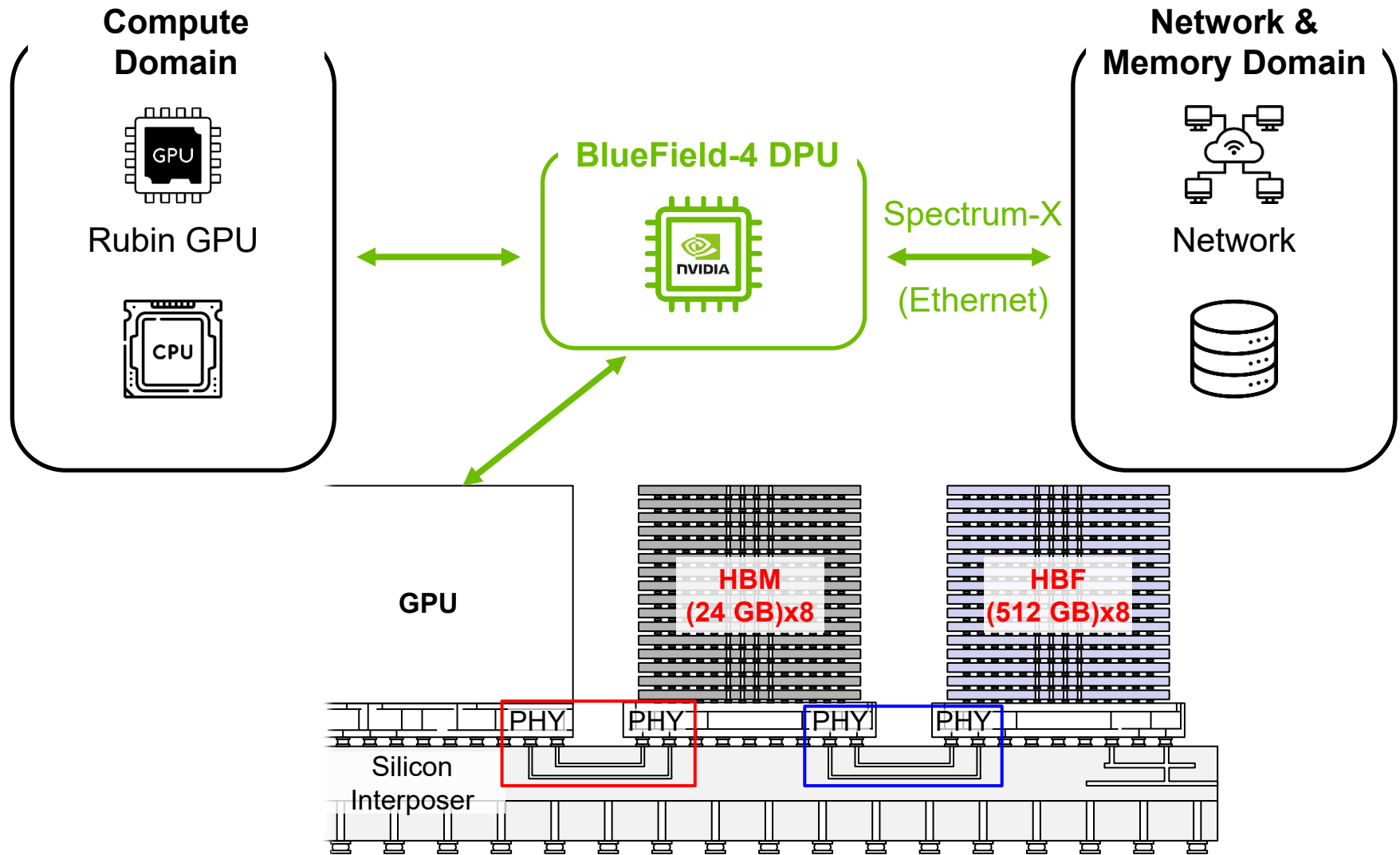


< Cross-sectional view of the GPU-HBM-HBF >

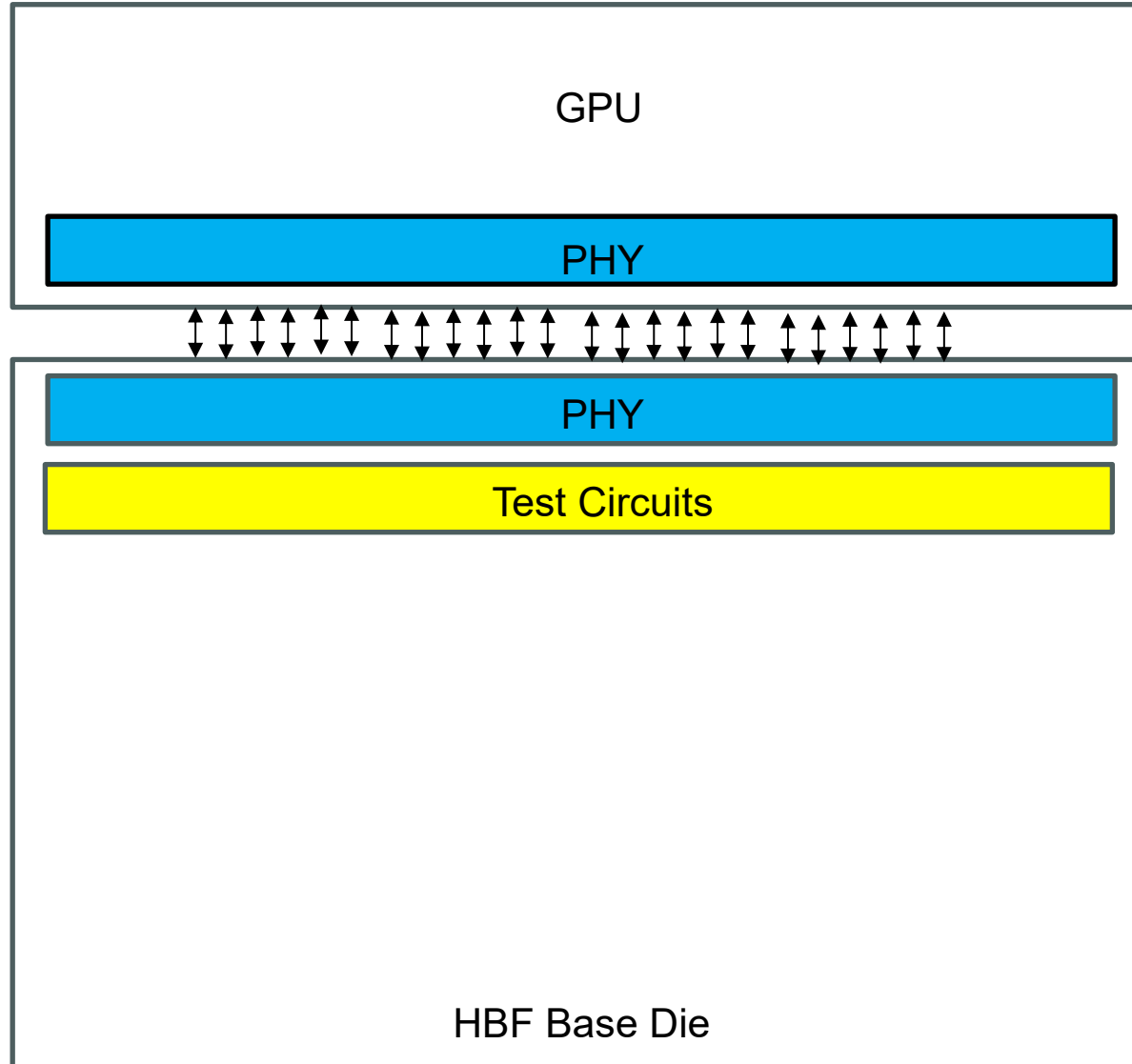


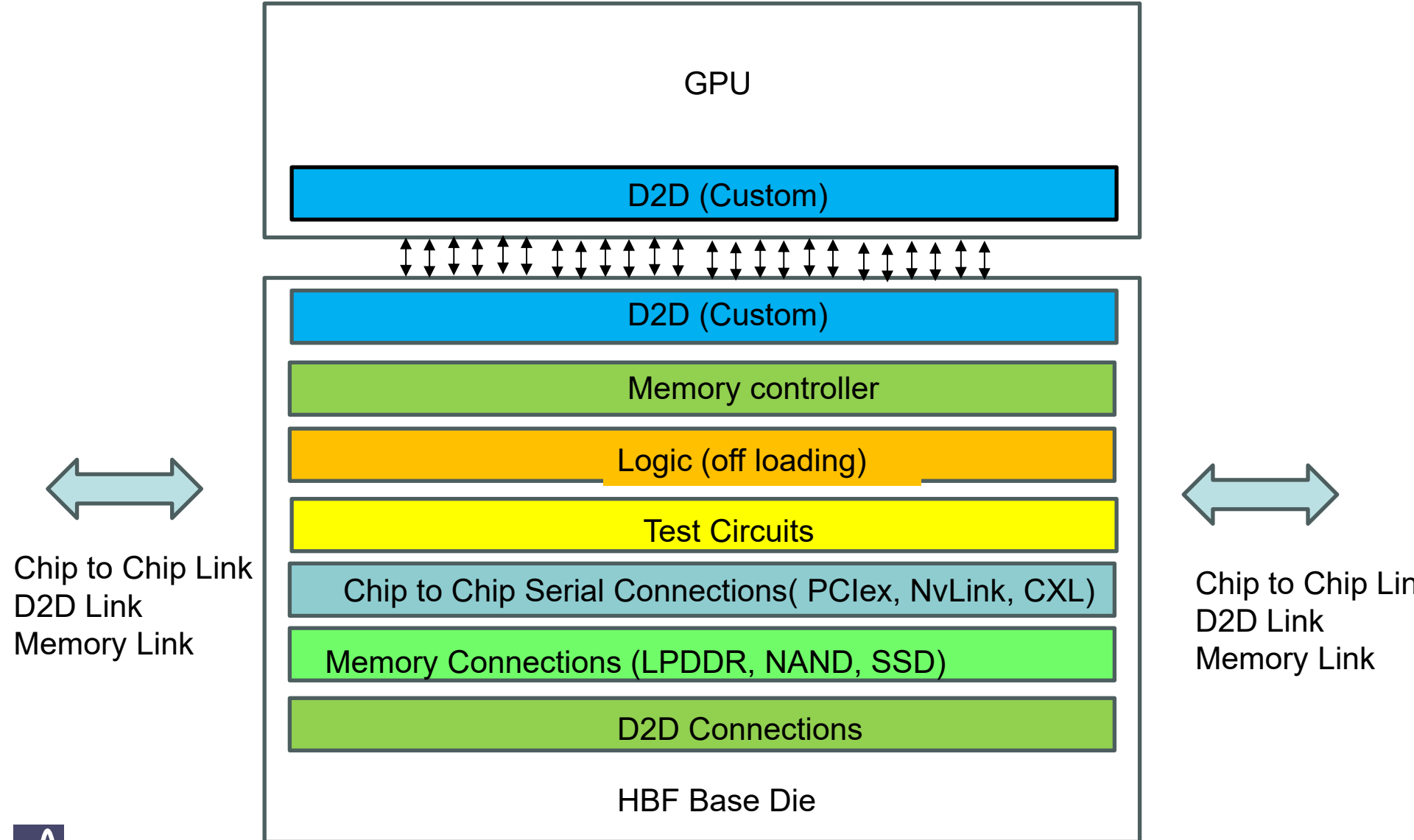
- ✓ HBM Capacity = 24 GBx8
- ✓ HBF Capacity = 512 GBx8

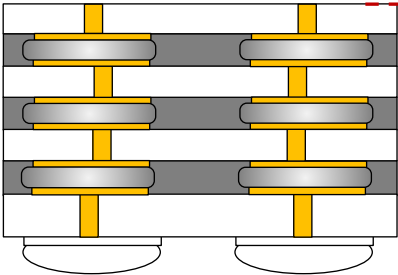
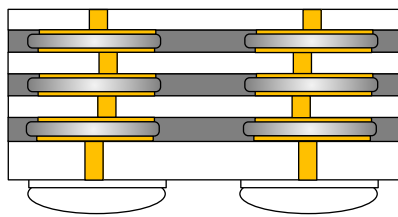
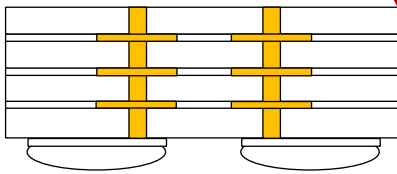
< Top-view of the GPU-HBM-HBF >



< Cross-sectional view of the GPU-HBM-HBF >





Bump/ Process	u-Bump (MUF)	u-Bump (*A-MUF)	Cu pad (Hybrid Bonding)
Structure			
Diameter	25 ~ 30 μm	20 ~ 15 μm	< 10 μm
Gap height	10~15 μm	5~10 μm	<5 μm
Stack Density	4~16Hi	4~24Hi	4~24Hi
Interconnection	MR / MUF	MR / A-MUF	Hybrid bonding
Risk	Target thickness not achieved	Feasible to overcome But, SI/ TI Risk ↑	Low technology maturity

Target for HBM4

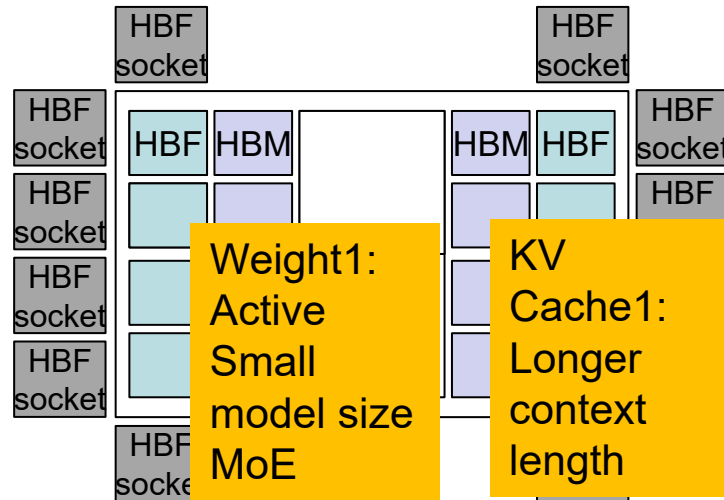
< Comparison of Bump Structures and Processes for High-Stack HBM/HBF Applications >

- The Advanced MUF process aims to achieve the target thickness required for post HBM4 while simultaneously improving the technological maturity of hybrid bonding.
- However, increased gap height and cumulative bump count compared to hybrid bonding lead to higher TI/ SI risks.

*Advanced MUF

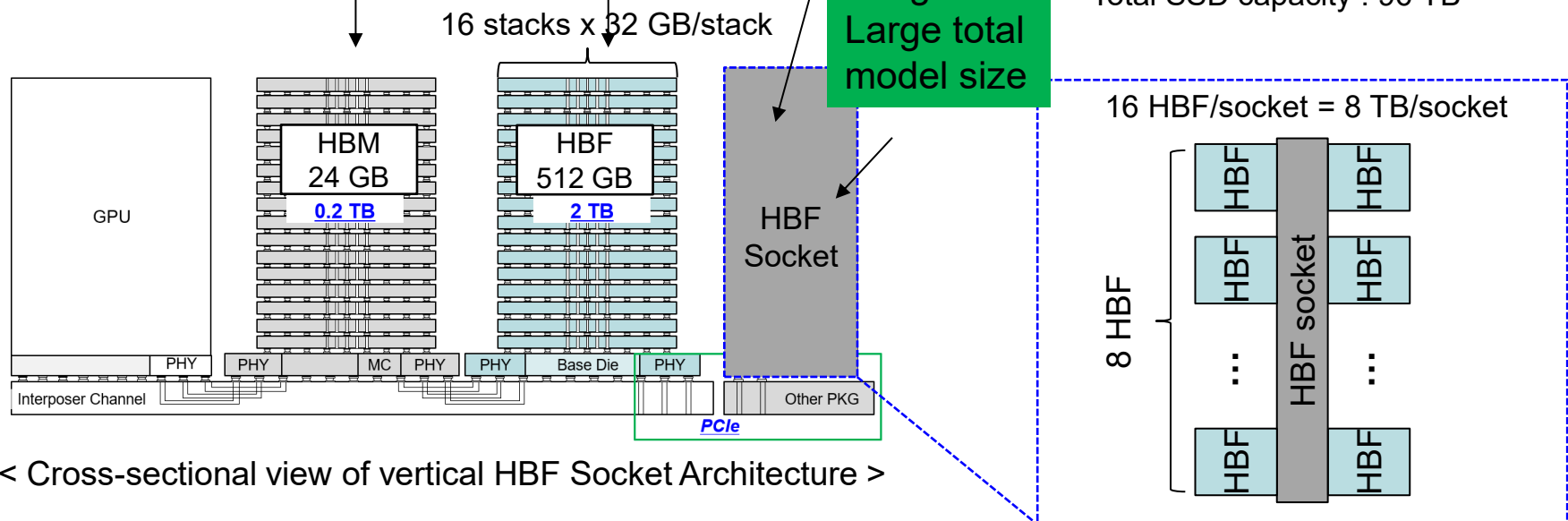
HBF for AI Accelerator – Vertical HBF Socket Architecture

**HBF4
Architecture**

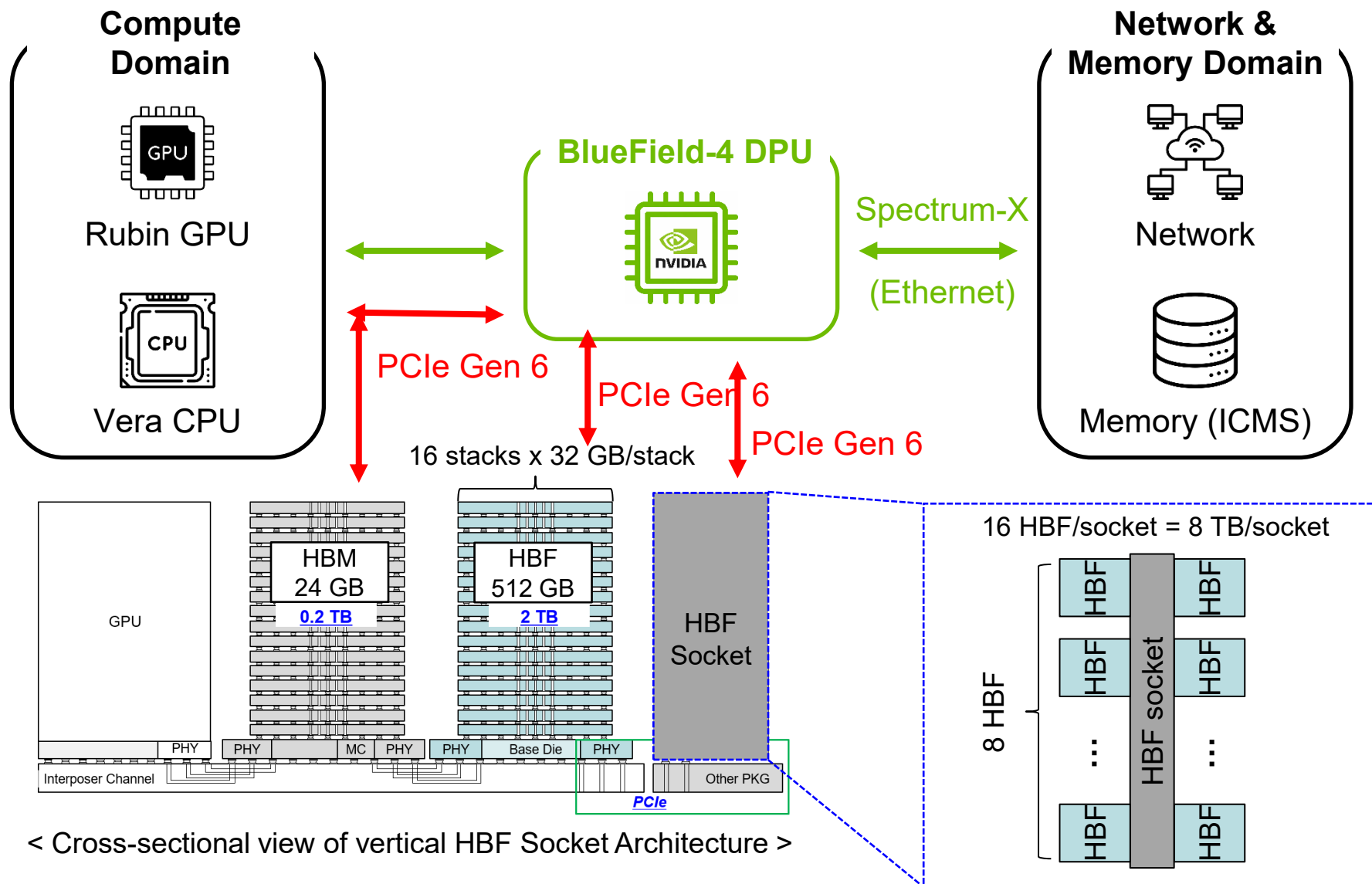


- # of HBM : 8
- # of HBF : 8
- # of socket : 12
- Total # of memory : 28
- one HBM capacity : 24 GB
- Total HBM capacity : 192 GB
- one HBF capacity : 512 GB
- Total HBF capacity : 4 TB
- One socket capacity : 8 TB
- Total SSD capacity : 96 TB

< Top-view of the vertical HBF Socket Architecture >

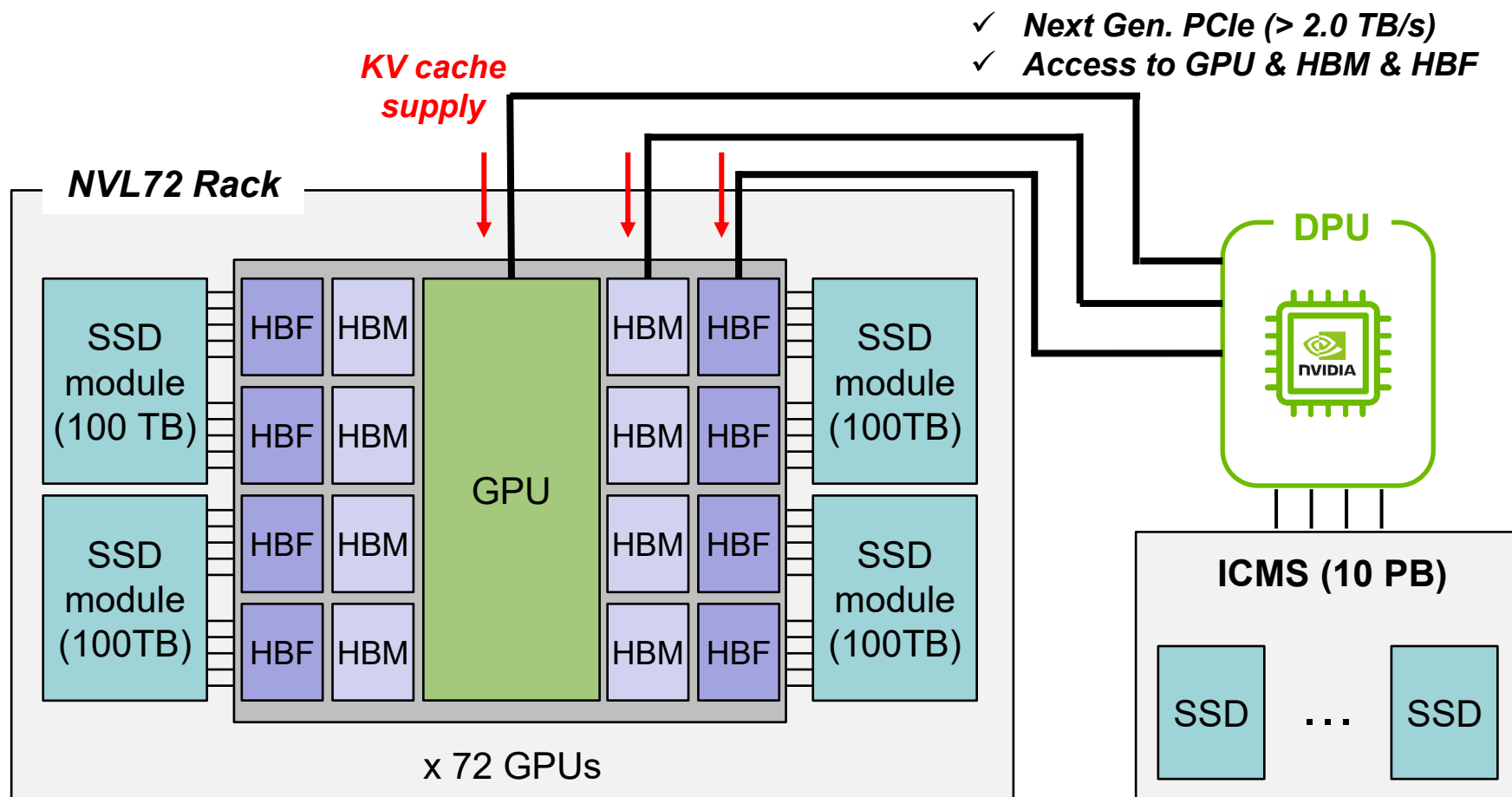


< Cross-sectional view of vertical HBF Socket Architecture >



< Cross-sectional view of vertical HBF Socket Architecture >

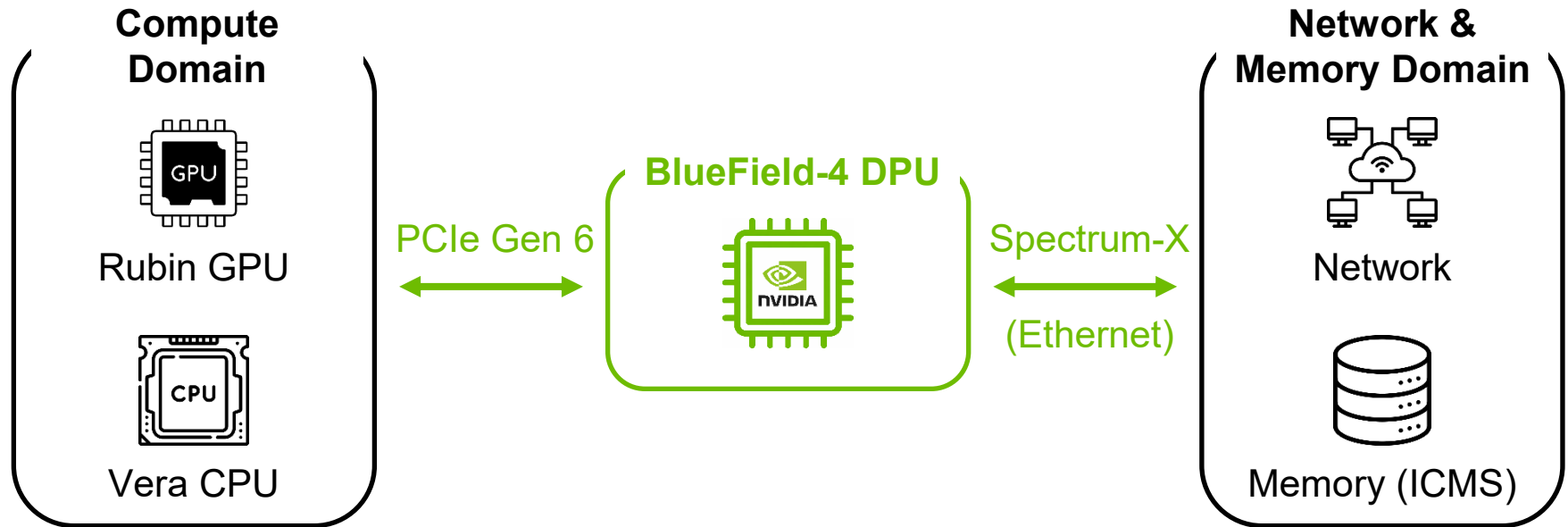
Future Architecture in Decode Stage : GPU-HBM-HBF-SSD + ICMS for Extreme Large KV Cache Storage



< GPU-HBM-HBF-SSD + ICMS architecture top-view >

- Inference Context Memory Storage (ICMS) systems are integrated in rack-scale system.
- Data Processing Unit (DPU) directly supplies KV caches from SSD cluster to each GPU's HBM and HBF.

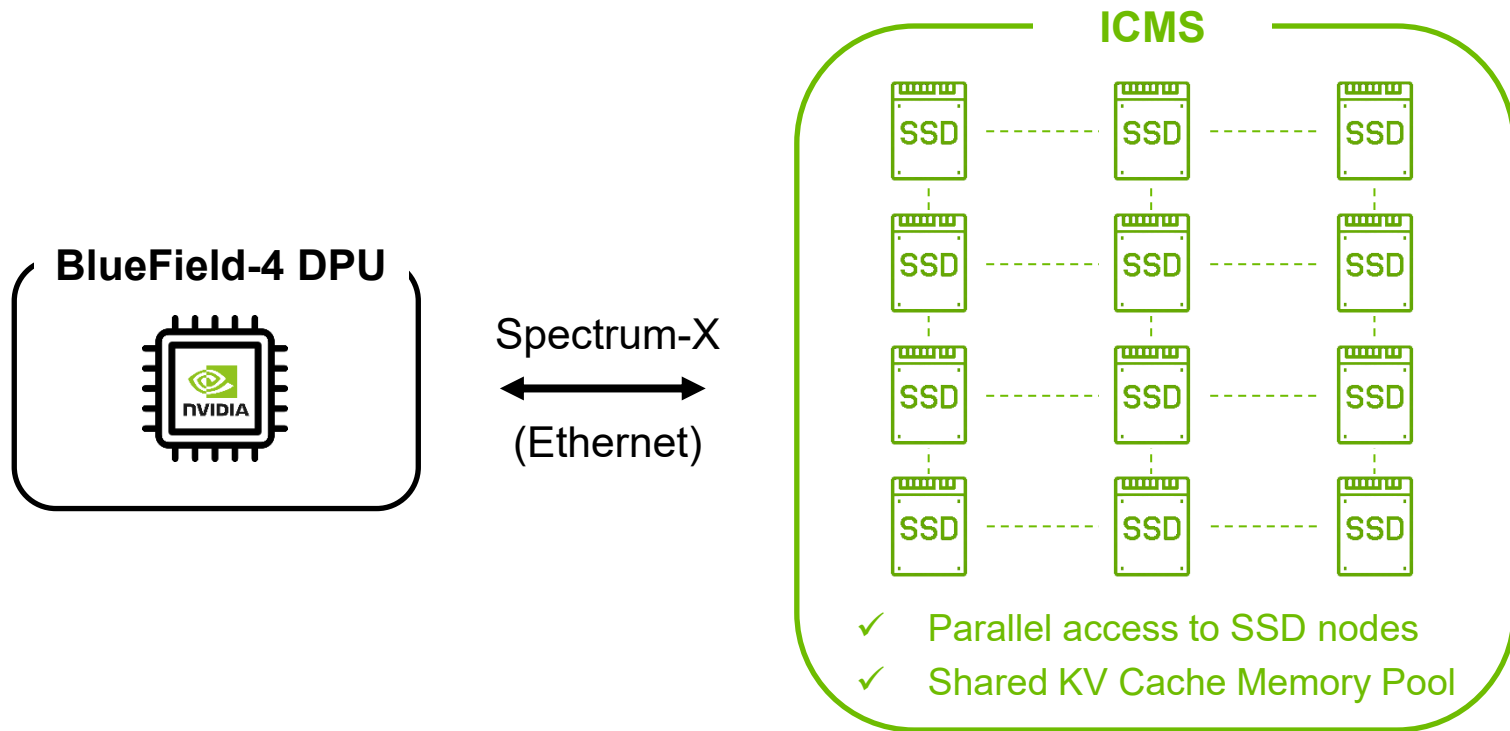
NVIDIA BlueField-4 DPU-Centric Data Plane Architecture



< BlueField-4 DPU-Centric Data Plane Architecture >

- BlueField-4 DPU는 데이터센터에서 발생하는 데이터 이동과 관련 처리를 전담하는 프로세서로, CPU와 GPU가 연산에 집중할 수 있도록 역할을 분리함.
- 네트워크, 스토리지, 보안과 같은 인프라 기능을 CPU 대신 DPU에서 분리하여 처리함으로써, 전체 시스템 자원을 AI 연산에 보다 효율적으로 사용하도록 함.
- LLM 추론 과정에서 빈번하게 발생하는 KV cache의 이동과 접근을 CPU 개입 없이 처리함으로써, 지연 시간 증가와 불필요한 부하를 줄임.
- 이를 통해 KV cache 이동으로 인한 네트워크 병목을 완화하고, 대규모 AI 추론 환경에서도 안정적인 성능을 유지함.

ICMS: Rack-Scale AI-Native Inference Context Memory Pool

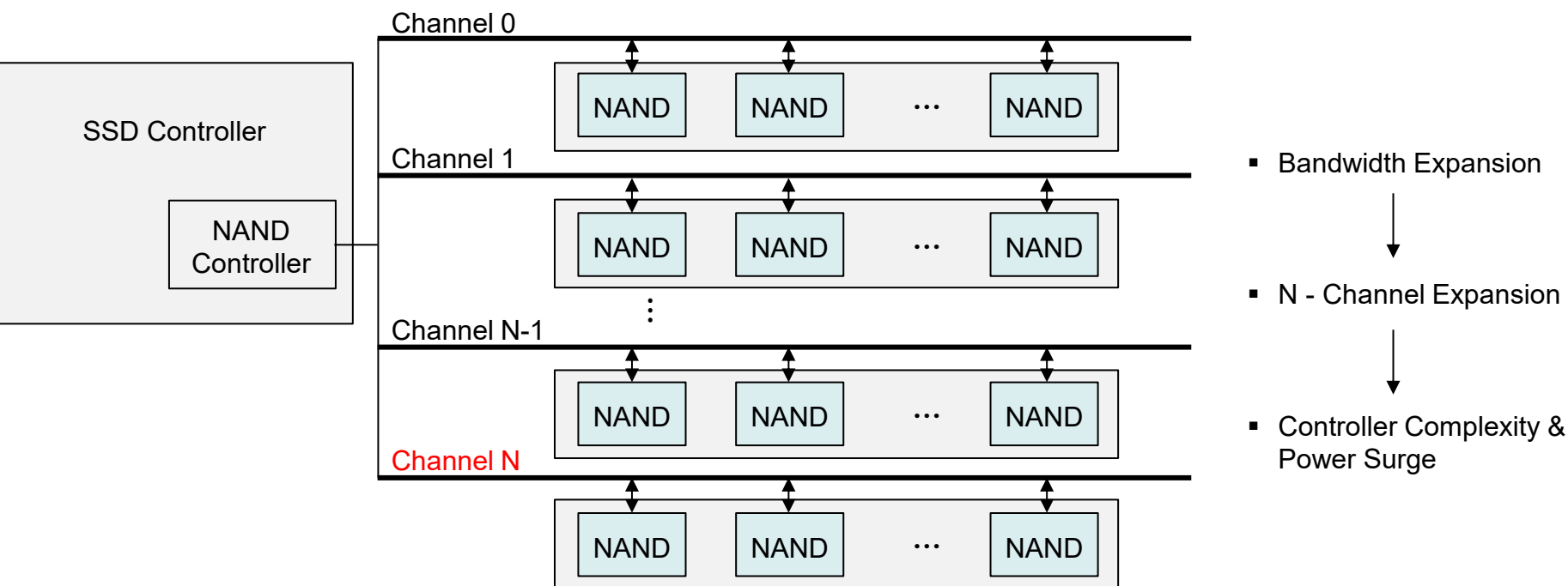


< ICMS Architecture for Shared KV Cache Memory >

- ICMS (Inference Context Memory Storage)는 LLM 추론 과정에서 생성되는 KV cache를 GPU-HBM 외부에서 저장·공유하기 위해 설계된 AI-native inference context memory storage platform임.
- GPU-HBM에 국한되던 inference context를 랙 단위의 대용량 공유 메모리 풀로 확장함으로써, long-context 처리와 대규모 동시 추론을 가능하게 함.
- ICMS를 통한 inference context memory 확장과 BlueField-4 DPU의 KV cache 관리 기능을 결합함으로써, GPU-HBM 용량 한계로 인해 발생하던 잦은 KV cache 이동을 줄임으로써, 네트워크 병목 문제를 효과적으로 완화함.
- 구성에 따라 GPU당 수 TB 규모(예: ~16 TB)의 KV cache 용량을 제공할 수 있으며, 랙 단위에서는 수백 TB급 (예: ~150 TB) inference context 메모리 풀로 확장 가능함.



Structural Limitations of Parallel SSD Based ICMS : Channel Expansion Overhead



< Structural Trade-off: Channel Expansion Overhead >

- 높은 대역폭을 확보하기 위해 채널 수를 물리적으로 확장하는 방식이 존재하지만, 명확한 하드웨어적 제약이 존재함.
- 다수의 채널을 수용하기 위해 I/O pin 수와 제어 로직이 늘어나며, 필연적으로 SSD controller의 die 면적이 증가하는 비효율이 발생함.
- 또한, 채널 수에 비례하여 전력 소모가 급증하며, 이는 고집적 시스템에서의 발열 이슈와 전력 효율 저하를 야기함.

→ 물리적 공간 및 전력 제약으로 인해, 무조건적인 채널 증가는 어려움.

HBM-HBF-SDD-ICMS Architecture Roadmap

** 1 HBM : 24 GB (1.5 GB per stack) / 1 HBF : 512 GB (32 GB per stack)

** 1 GDDR : 4 GB / 1 HBF socket (SSD) : 48 ~ 96 TB

* Bandwidth enhancement with D2D interfaces, 2 times multiplication at every generation

	Hybrid GDDR-HBF HB1 (2028)	HBF-GPU- HBM HB2 (2030)	GPU-HBM- HBF HB3 (2032)	GPU-HBM- HBF w/ SSD HB4 (2034)	GPU-HBM- HBF-Double SSD HB4 (2034)	GPU-HBM- HBF w/ICMS HB5 (2036)	HBM6 * MCC, CPU and GPU on Base die
# of GDDR	12	-	-	-	-	-	-
# of HBM	-	4	8	8	8	8	8
# of HBF	8	4	8	8+192	8 + 192	8 + 192	8 + 192
# of SSD	-	-	-	12	24	24	24
Total GDDR capacity	48 GB	-	-	-	-	-	-
Total HBM capacity	-	96 GB	192 GB	192 GB	192 GB	192 GB	192 GB
Total HBF capacity	4 TB	2 TB	4 TB	4 TB	4 TB	4 TB	4 TB
Total SSD capacity	-	-	-	48 TB	96 TB	96 TB With ICMS	96 TB With ICMS

Part5: HBF Workloads Analysis and Proof of Advantages

AI 워크로드 분석: 더 스마트한 AI 하드웨어 설계를 위한 가이드

AI 워크로드 분석은 AI 프로그램이 실제로 어떤 연산을 어떤 패턴으로 수행하는지 정량적으로 파악하는 과정입니다.
이 분석 결과를 바탕으로 AI 시스템의 연산 구조, 메모리 용량, 대역폭 등 하드웨어 설계를 최적화합니다.

AI 시스템 설계를 위한 3대 핵심 분석 요소

연산 특성 분석

코어 구조, 가속기(GPU/TPU) 종류를 결정합니다.
(예: 연산 밀도, 병렬성)



메모리·I/O 특성 분석

DRAM/HBM 용량, 캐시 등 메모리 계층 구조를 설계합니다.
(예: 접근 패턴, 대역폭)



시간·패턴 특성 분석

QoS, 스케줄링, 전력·열 설계 기준을 수립합니다.
(예: 구간별 부하 변화)



AI 작업 단계별 주요 요구사항



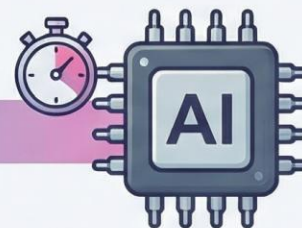
1. 데이터 준비 (ETL)

대용량 데이터 처리로 스토리지 I/O와 메모리 대역폭이 핵심입니다.



2. 모델 학습

고밀도 GPU 연산과 고대역폭 메모리(HBM)가 중요합니다.



3. 추론

낮은 지연시간과 처리량이 중요하며, 캐시 효율성이 성능을 좌우합니다.

WL analysis : case1, 80B model, INT4, 384K tokens

Model & Workload Parameters

Model Size (Billion Params)

80

Precision

INT4 (0.5 bytes)



Input Token Length

384000

Hardware Specifications

HBM Capacity (GB)

192

HBM Bandwidth (GB/s)

8000

HBF Capacity (GB)

4096

HBF Bandwidth (GB/s)

8000

HBF Latency (microseconds)

1

WL analysis : case1, , 80B model, INT4, 384K tokens

Workload Analysis

Weights Memory **40.00** GB

KV Cache Memory **375.00** GB

Total Memory Required **415.00** GB

Performance Prediction

Time to First Token (TTFT) **43,891** ms

Throughput (TPS) **19.25** tok/s

Effective Bandwidth **7,988** GB/s

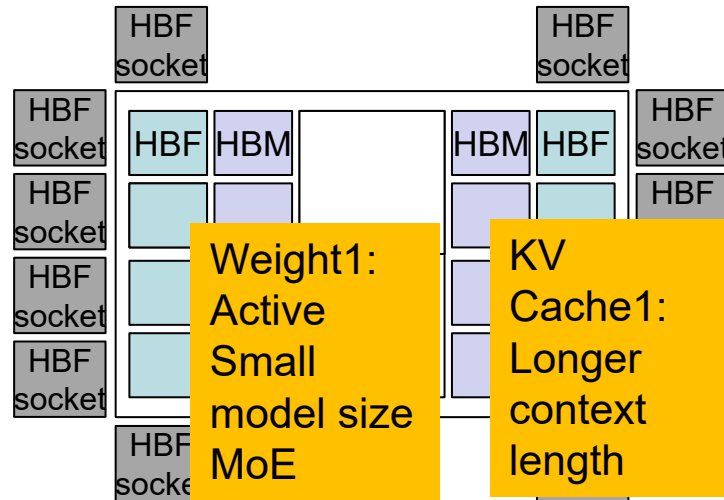
Memory Allocation (HBM vs HBF) Offloaded: 53.7% to HBF



● HBM ● HBF

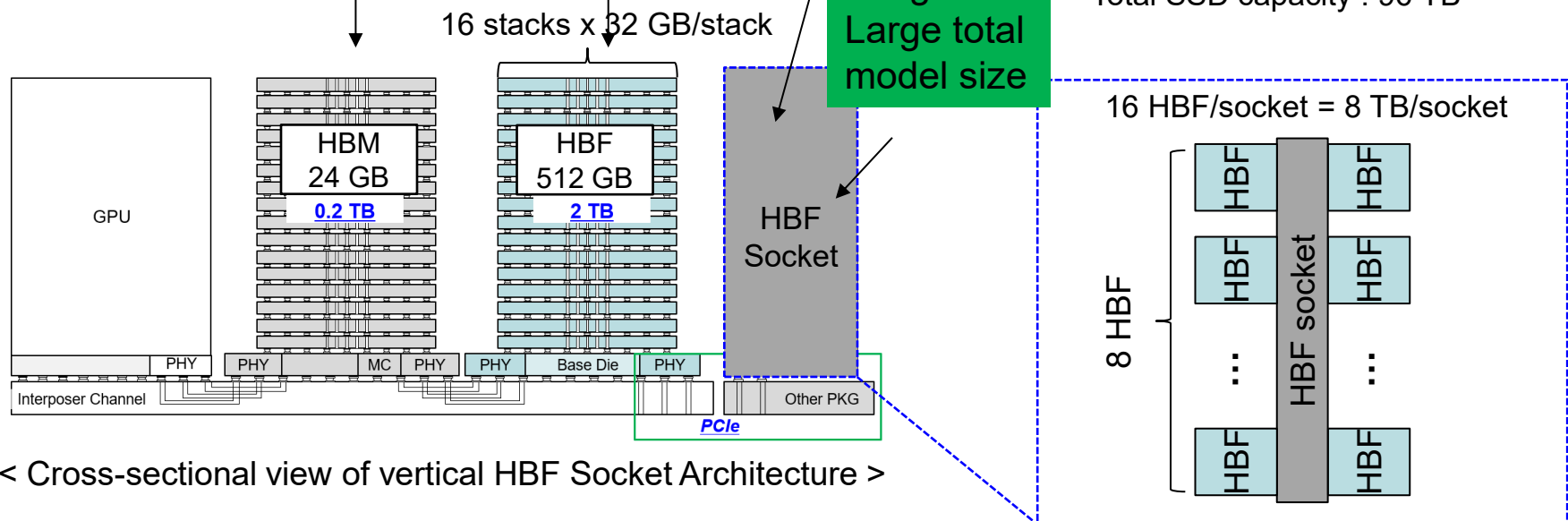
HBF for AI Accelerator – Vertical HBF Socket Architecture

**HBF4
Architecture**

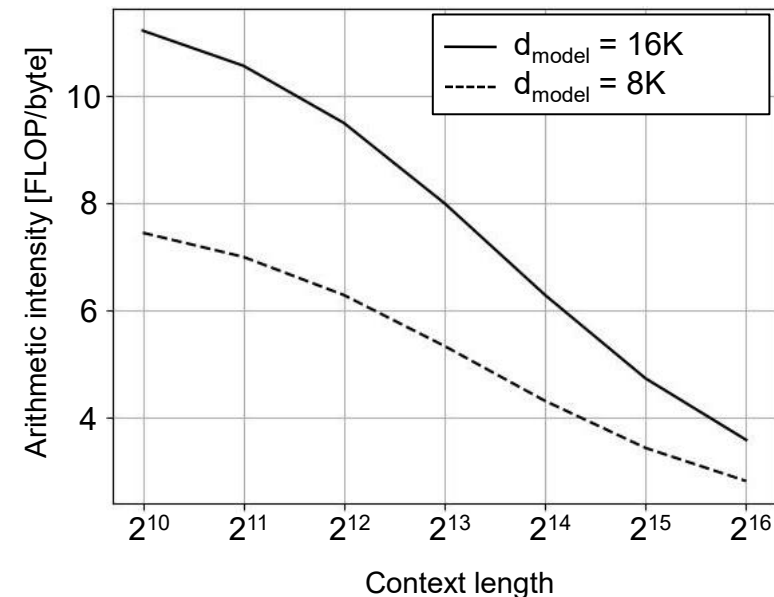


- # of HBM : 8
- # of HBF : 8
- # of socket : 12
- Total # of memory : 28
- one HBM capacity : 24 GB
- Total HBM capacity : 192 GB
- one HBF capacity : 512 GB
- Total HBF capacity : 4 TB
- One socket capacity : 8 TB
- Total SSD capacity : 96 TB

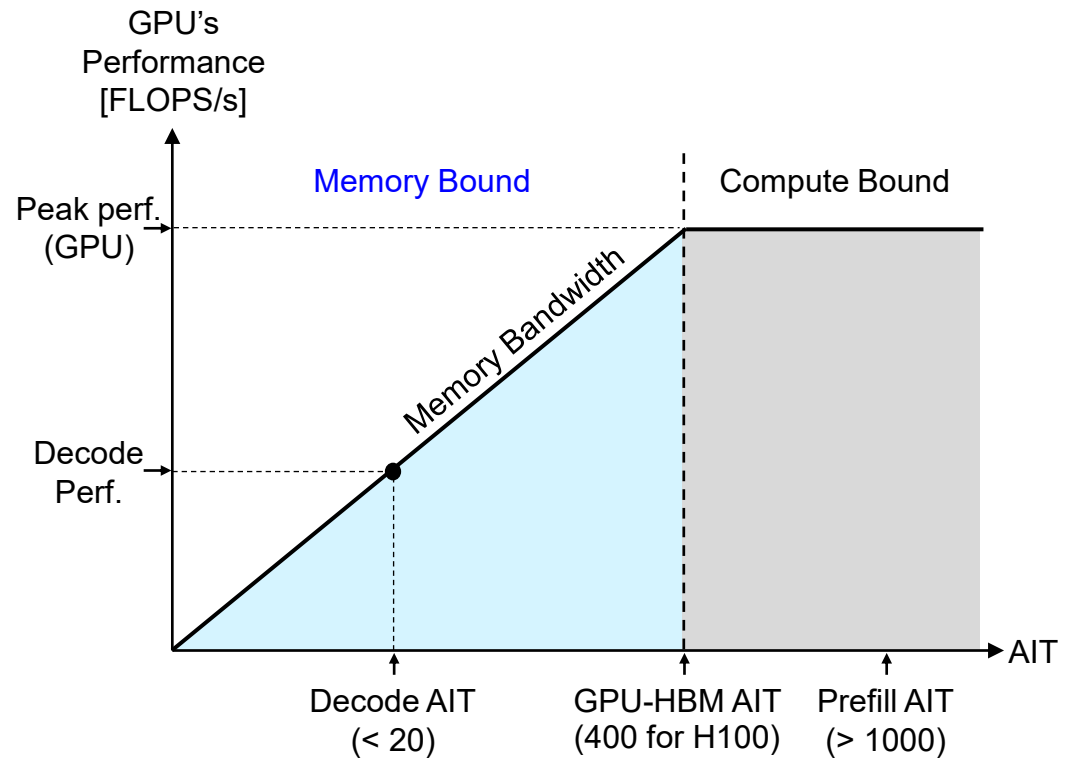
< Top-view of the vertical HBF Socket Architecture >



< Cross-sectional view of vertical HBF Socket Architecture >



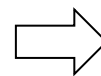
< AIT for decode process vs context length >



< Roofline model for analysis of GPU-HBM AI inference >

$$\text{Arithmetic Intensity (AIT)} = \frac{\text{Computation [FLOP]}}{\text{Memory Traffic [Byte]}}$$

$$\text{GPU - HBM AIT} = \frac{\text{GPU's Peak Performance}}{\text{HBM's Memory Bandwidth}}$$



Decode AIT << GPU-HBM AIT:
→ **Memory Bound**

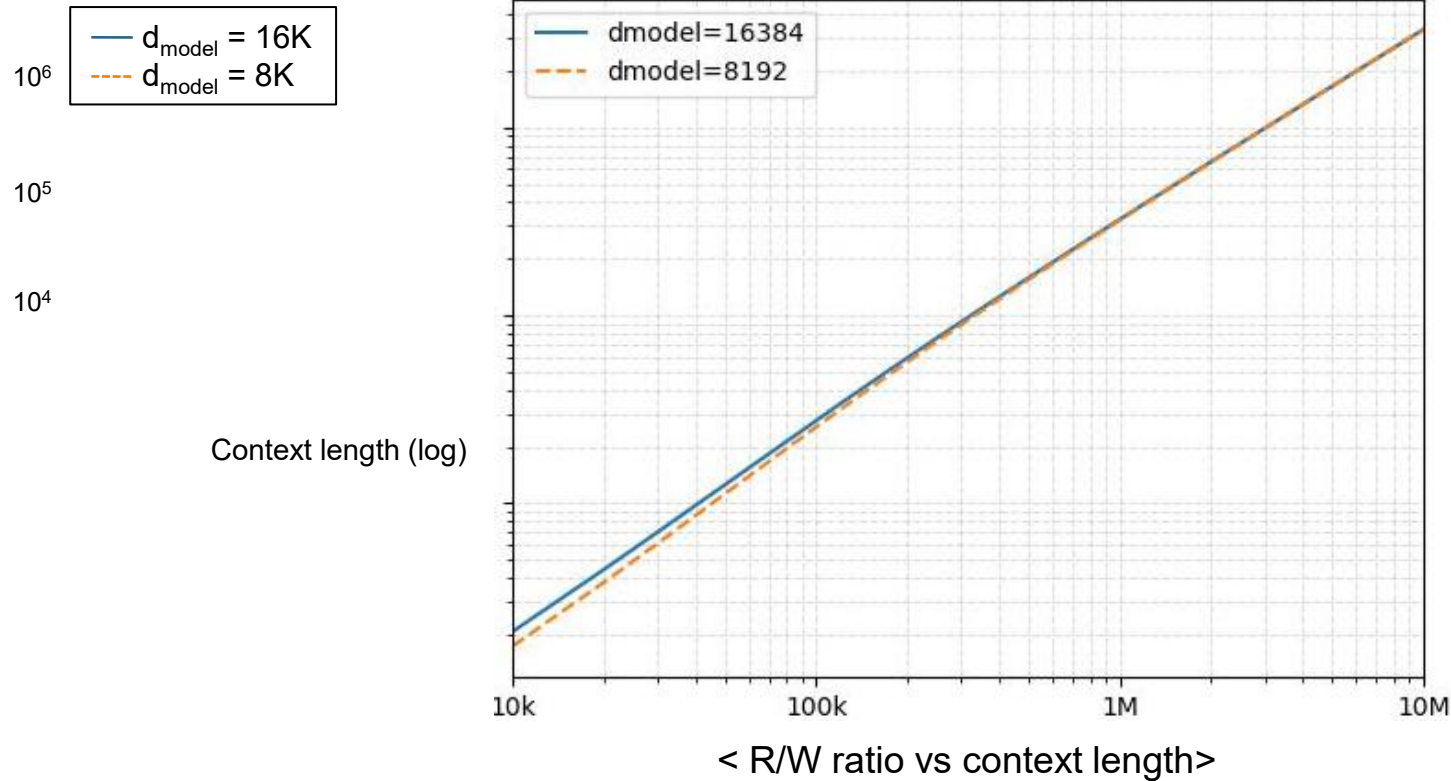
Decode Process is the main target for proposed architecture.

Memory Op.	Prefill	Decode
Read per layer	$4d^2 + (s^2 + 4sd) * B$	$\{4d^2 + 2dN(s+N/2)\} * B$
Write per layer	$(s^2 + 4sd) * B$	$\{2dN + 2d(s+N/2)\} * B$

d = model dimension
s = input length
N = generated length
B = batch size

< LLM Inference Memory Read/Write Operation >

Workload: Llama3-400B (d = 16384)

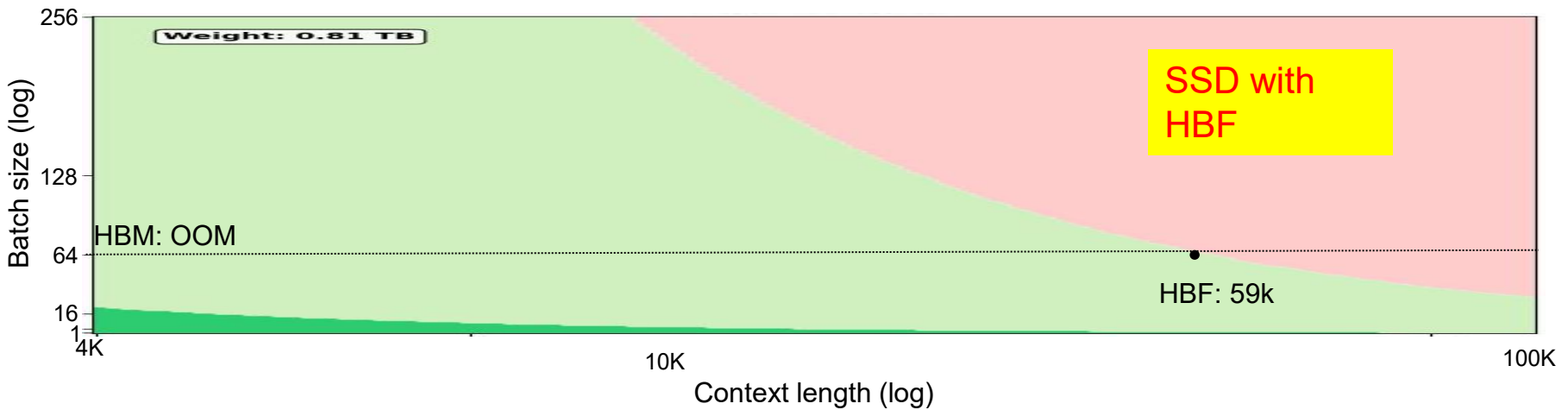


Feasible Region for AI Workload Using HBM vs HBF

d_{ff} = FFN dimension, n_L = # of layer, E = # of expert

		Equation	Llama3-400B ($N_{max} = 128k$)	Llama4-109B (17B active, $N_{max} = 10M$)	Llama4-400B (17B active, $N_{max} = 1M$)
Capacity for N_{max}	Attention	$4d_m^2 n_L$	270 GB	8 GB	8 GB
	KV Cache	$4d_m(s+N)n_L B$	2.16B TB	16.4B TB	1.64B TB
	FFN	$2d_m d_{ff} n_L E$	540 GB	208 GB	805 GB

Llama3



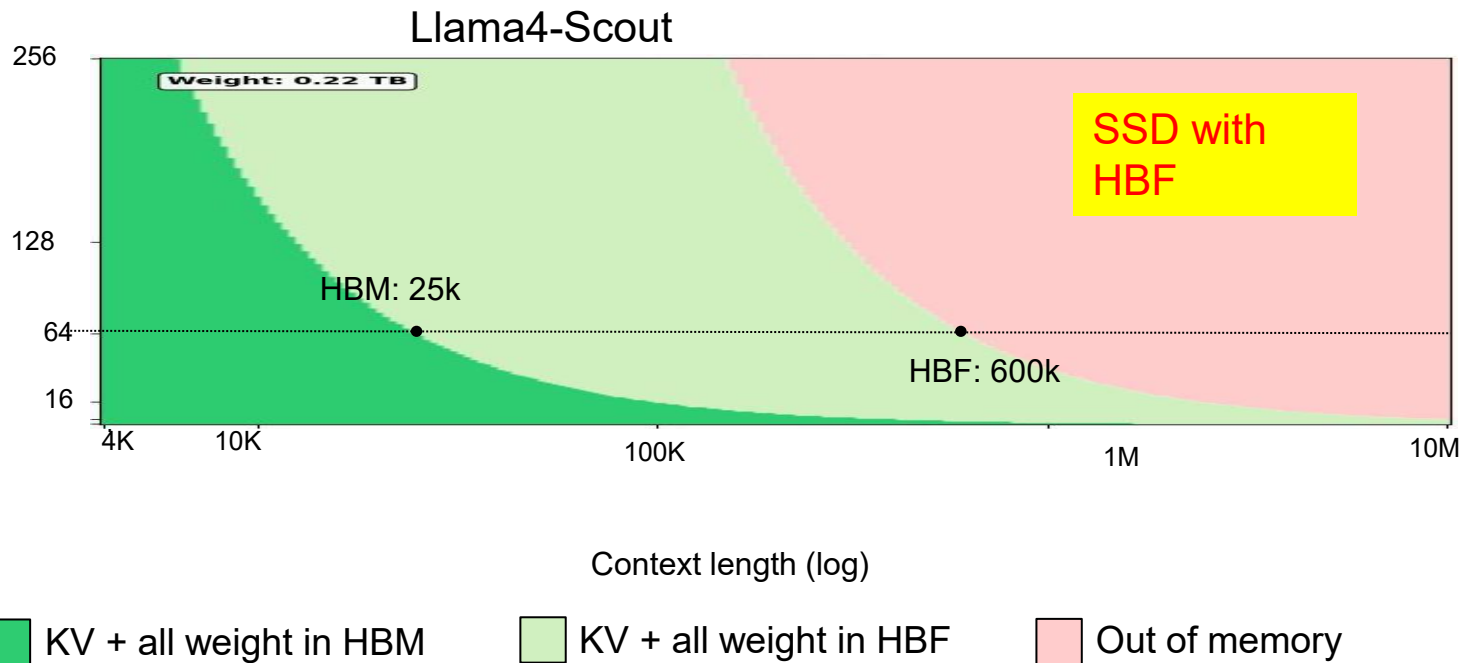
KV + all weight in HBM
 KV + all weight in HBF
 Out of memory

< Capacity heatmap for various workload using HBM and HBF >

Feasible Region for AI Workload Using HBM vs HBF

d_{ff} = FFN dimension, n_L = # of layer, E = # of expert

		Equation	Llama3-400B ($N_{max} = 128k$)	Llama4-109B (17B active, $N_{max} = 10M$)	Llama4-400B (17B active, $N_{max} = 1M$)
Capacity for N_{max}	Attention	$4d_m^2 n_L$	270 GB	8 GB	8 GB
	KV Cache	$4d_m(s+N)n_L B$	2.16B TB	16.4B TB	1.64B TB
	FFN	$2d_m d_{ff} n_L E$	540 GB	208 GB	805 GB

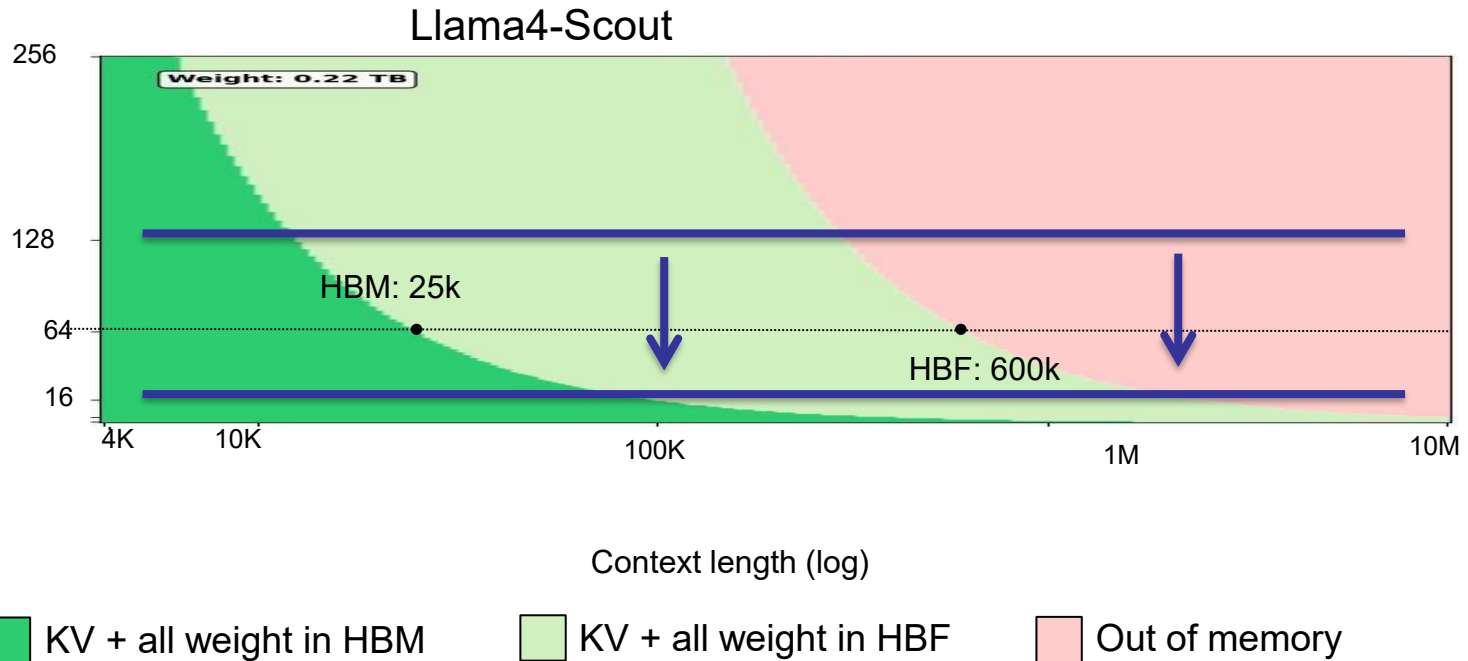


< Capacity heatmap for various workload using HBM and HBF >

Solution1: Reducing the batch size

d_{ff} = FFN dimension, n_L = # of layer, E = # of expert

		Equation	Llama3-400B ($N_{max} = 128k$)	Llama4-109B (17B active, $N_{max} = 10M$)	Llama4-400B (17B active, $N_{max} = 1M$)
Capacity for N_{max}	Attention	$4d_m^2 n_L$	270 GB	8 GB	8 GB
	KV Cache	$4d_m(s+N)n_L B$	2.16B TB	16.4B TB	1.64B TB
	FFN	$2d_m d_{ff} n_L E$	540 GB	208 GB	805 GB

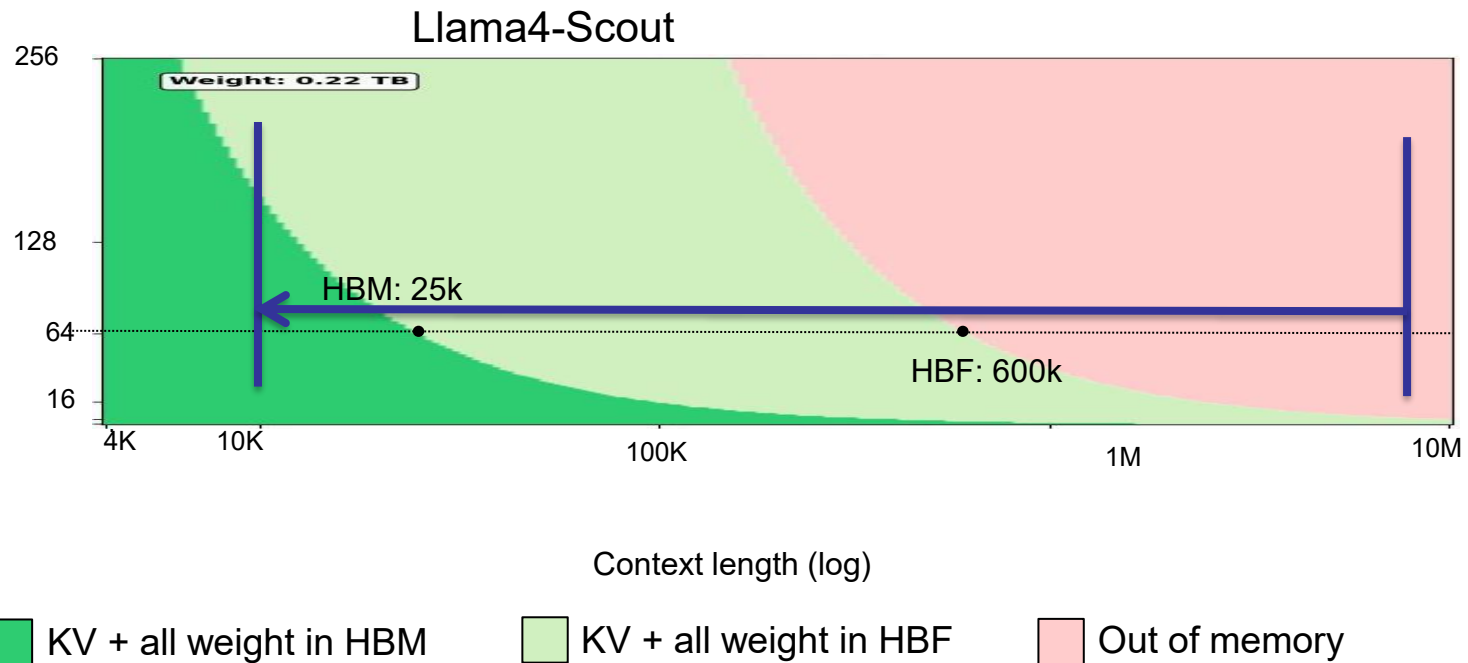


< Capacity heatmap for various workload using HBM and HBF >

Solution2: Reducing the context length

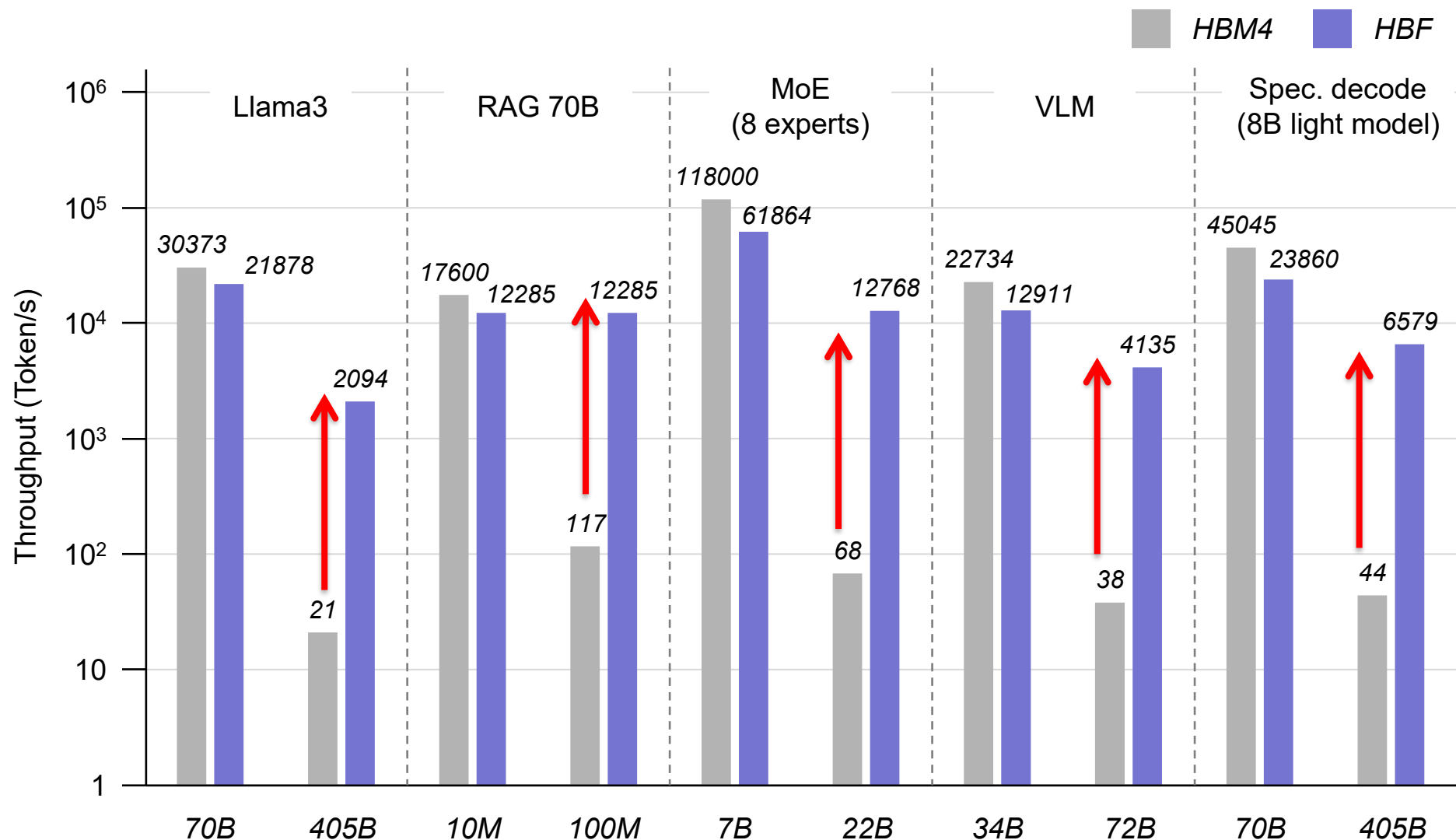
d_{ff} = FFN dimension, n_L = # of layer, E = # of expert

		Equation	Llama3-400B ($N_{max} = 128k$)	Llama4-109B (17B active, $N_{max} = 10M$)	Llama4-400B (17B active, $N_{max} = 1M$)
Capacity for N_{max}	Attention	$4d_m^2 n_L$	270 GB	8 GB	8 GB
	KV Cache	$4d_m(s+N)n_L B$	2.16B TB	16.4B TB	1.64B TB
	FFN	$2d_m d_{ff} n_L E$	540 GB	208 GB	805 GB



< Capacity heatmap for various workload using HBM and HBF >

[1/3] : Throughput in various Workloads



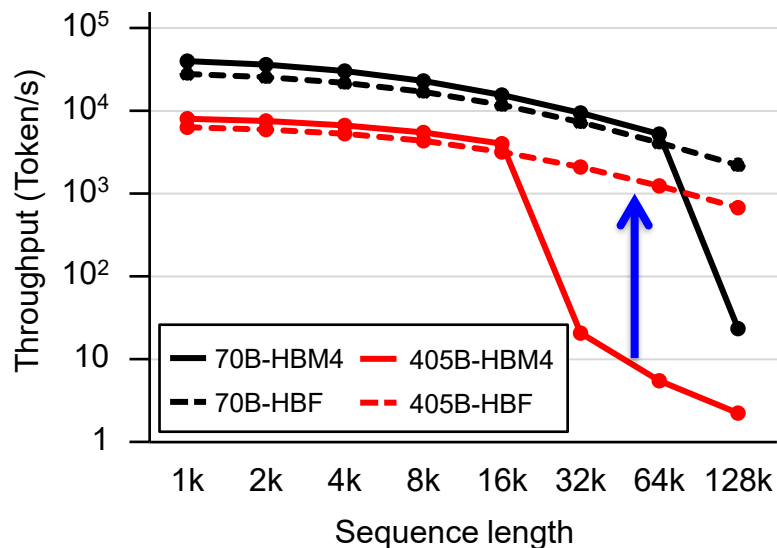
< LLM workloads and hardware specification table >

Analytical Model based Computing Performance in Decode Stage

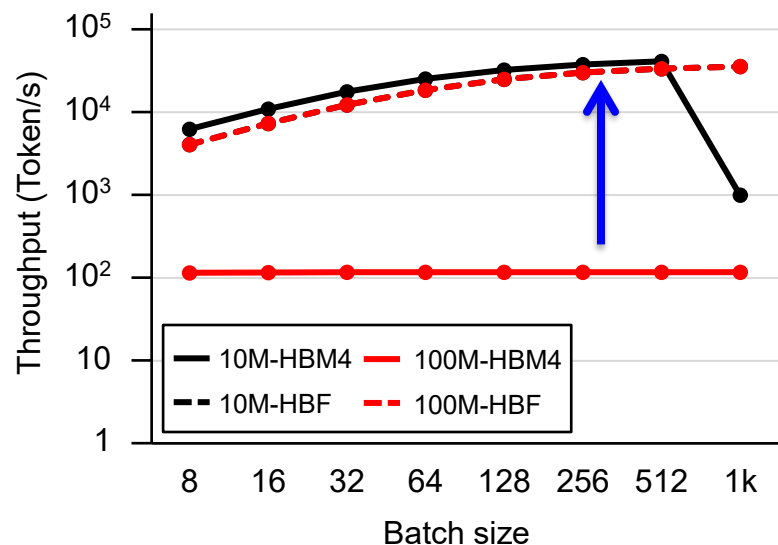
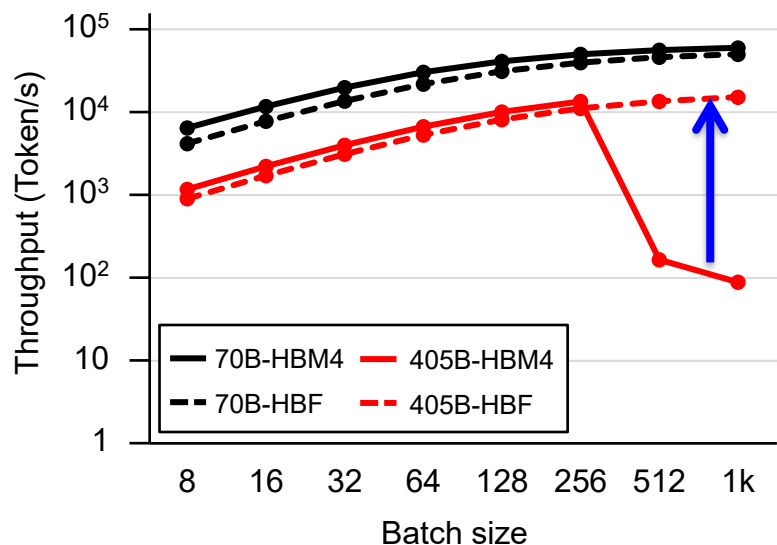
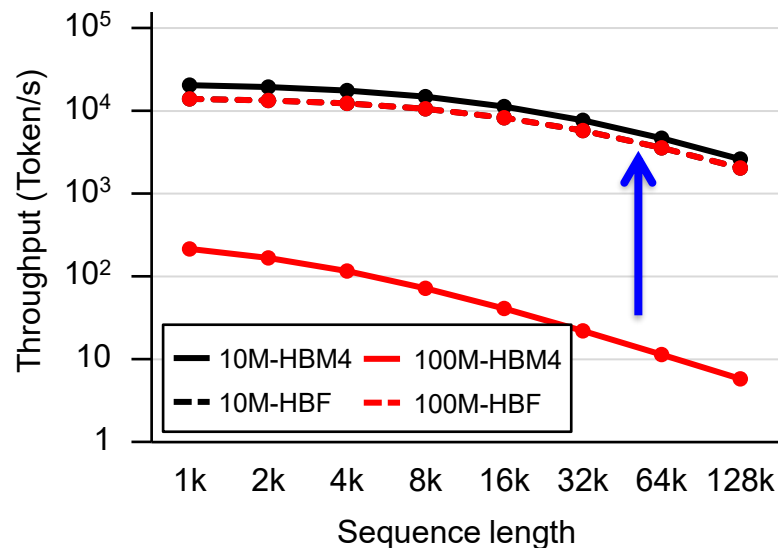
[2/3] : Throughput Sweet spot Analysis

HBF1
Architecture

Llama3 70/405B



RAG 70B (10M/100M)

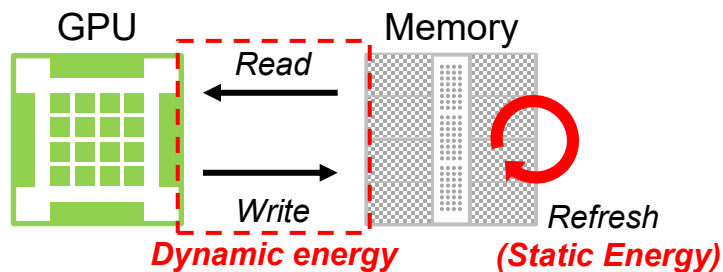


< Optimal sweet spot analysis for HBF >

Analytical Model based Computing Performance in Decode Sta

[3/3] : Energy Consumption Analysis

**HBF1
Architecture**



	HBM4	HBF
Read energy	4.0 pJ/bit	20.0 pJ/bit
Write energy	4.0 pJ/bit	100.0 pJ/bit
Refresh power	0.3 W/GB	0 (no refresh)

$$E_{total} = E_{dynamic} + E_{static} \quad \text{Token/J} = N_{tokens}/E_{total}$$

$$\begin{cases} E_{dynamic} = (B_{read} \cdot e_{read} + B_{write} \cdot e_{write}) \times N_{step} \times N_{GPU} \\ E_{static} = P_{static} \times T \times N_{GPU} \end{cases}$$

B_{read} : read bytes per step

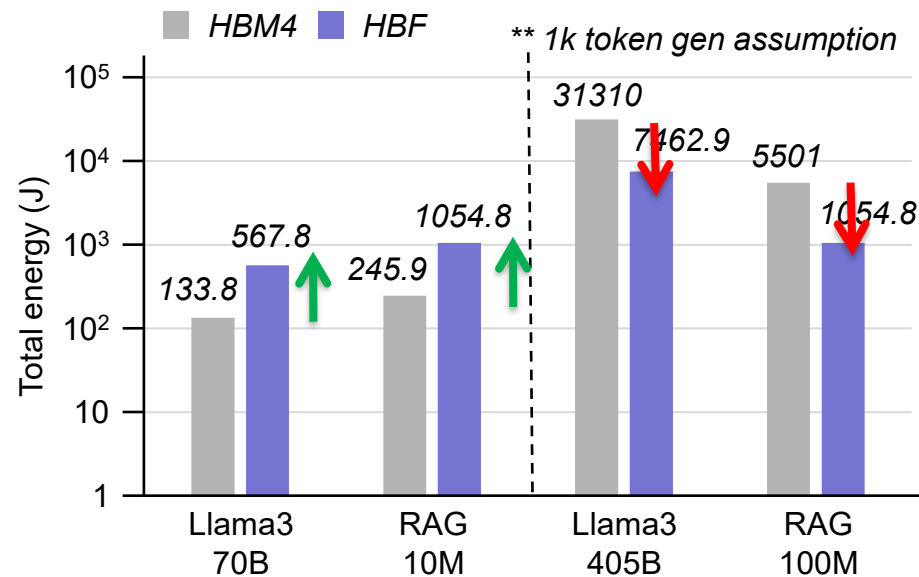
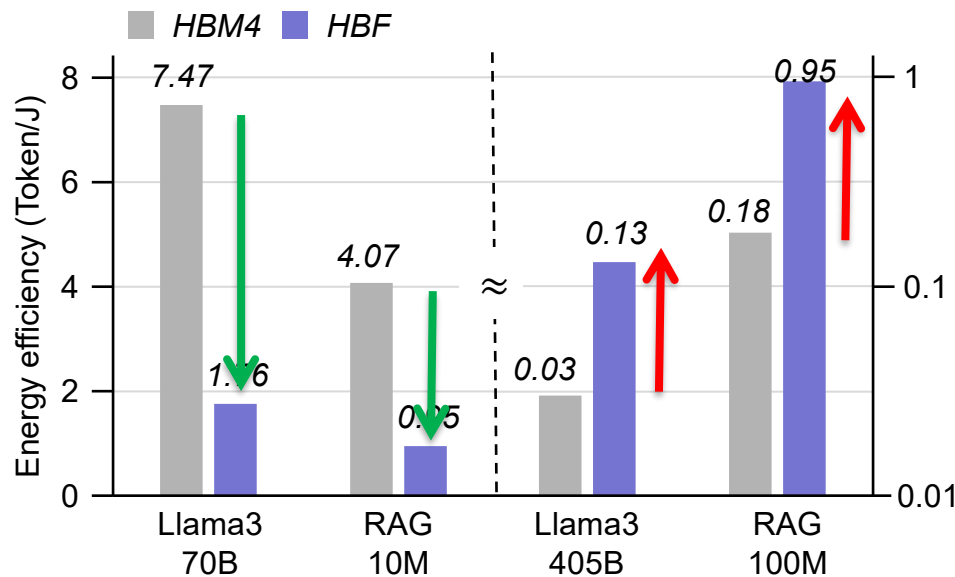
B_{write} : write bytes per step

e_{read} : read energy

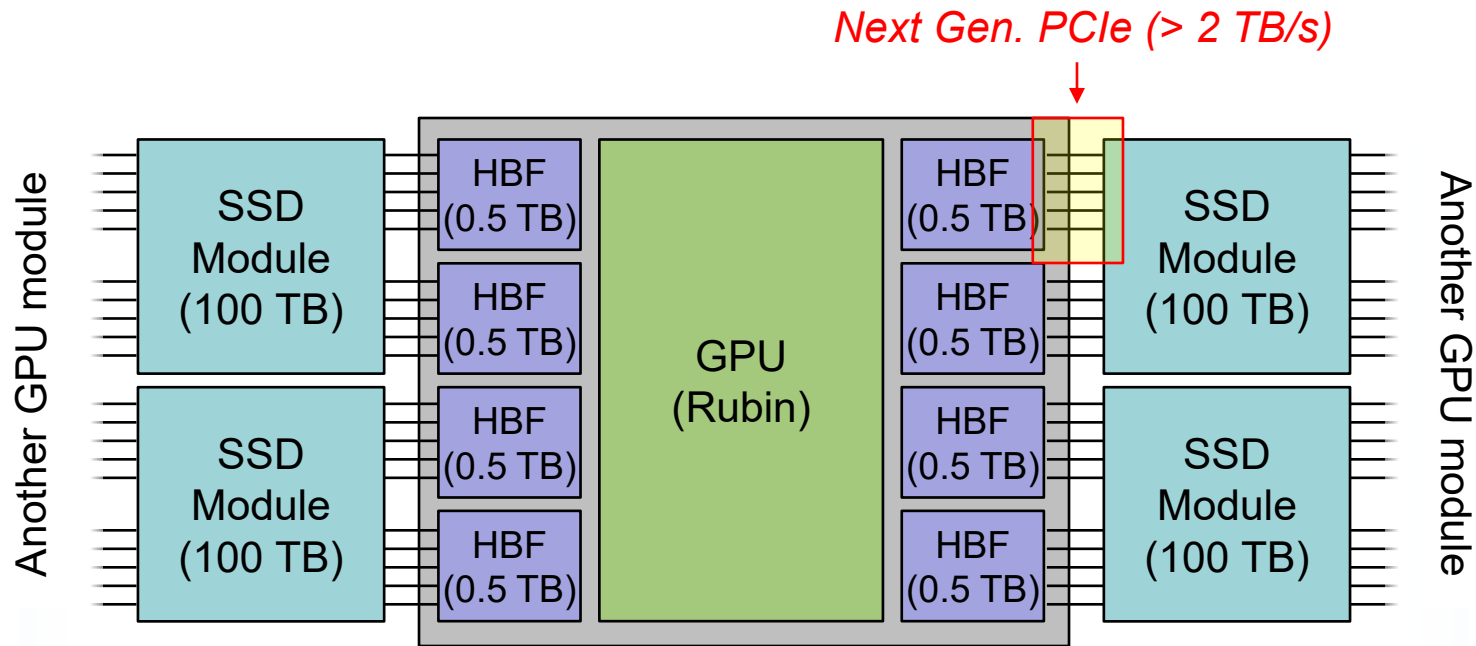
e_{write} : write energy

P_{static} : static power per GPU

$T : N_{tokens}/TPOT$



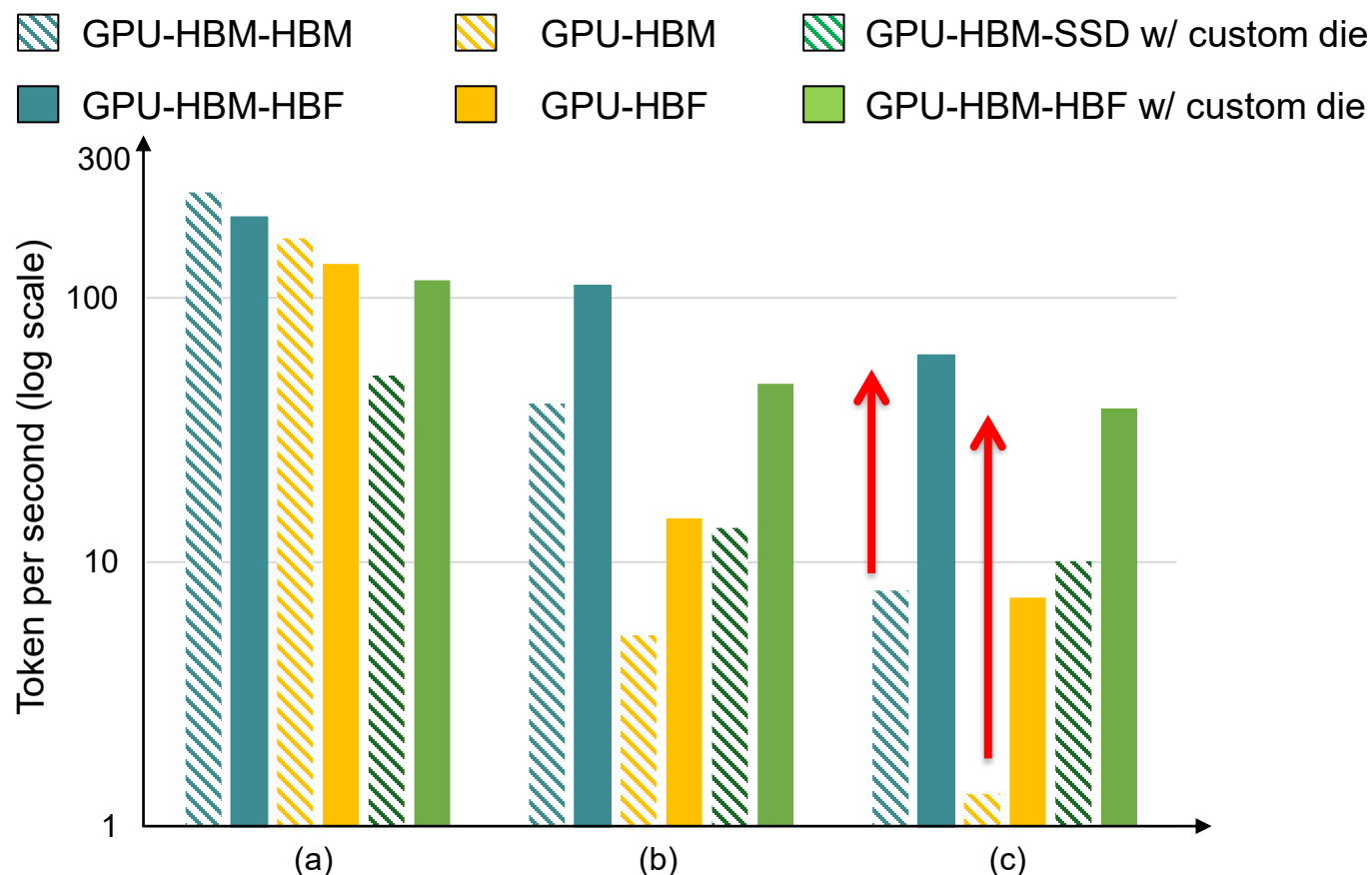
< Workloads energy consumption analysis >



< GPU-HBF-SSD architecture top-view >

- As context length increases significantly, large KV caches are needed to generate longer answers.
- Therefore, SSD modules can be directly connected to GPU module with next generation PCIe interconnect.
- NVIDIA reinforced the need for SSD-based KV cache storage by announcing ICMS at CES 2026 → **SSD** can be a great solution for large memory capacity.

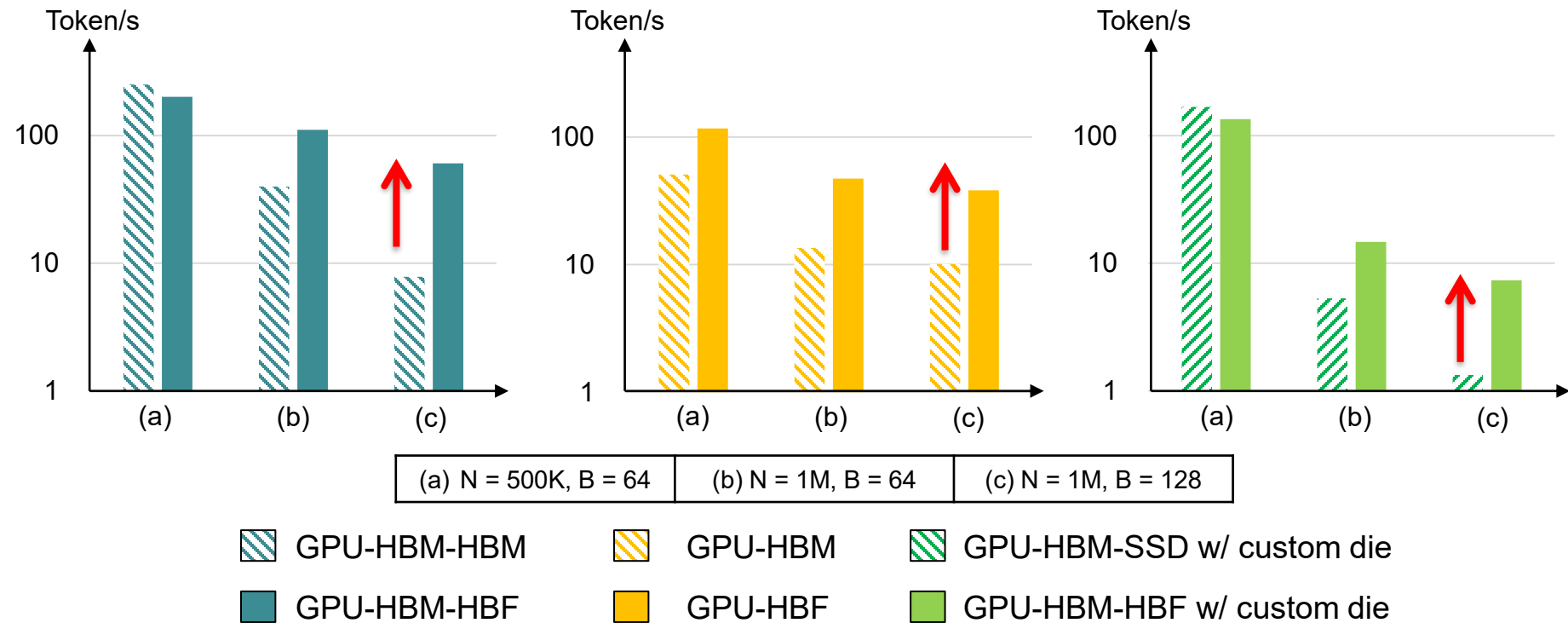
Throughput Analysis of Different HBF Architectures (One Graph)



< Token/s analysis for different HBF architecture with its respective conventional architecture

	GPU-HBM-HBM	GPU-HBM-HBF	GPU-HBM	GPU-HBF	GPU-HBM-SSD w/ custom die	GPU-HBM-HBF w/ custom die
(a) N = 500K, B = 64	250.6	201.6	168.1	134.4	50.8	116.6
(b) N = 1M, B = 64	39.8	111.1	5.3	14.7	13.5	47.3
(c) N = 1M, B = 128	7.8	60.6	1.3	7.3	10.1	38.2

Throughput Analysis of Different HBF Architectures (Separate Graph)

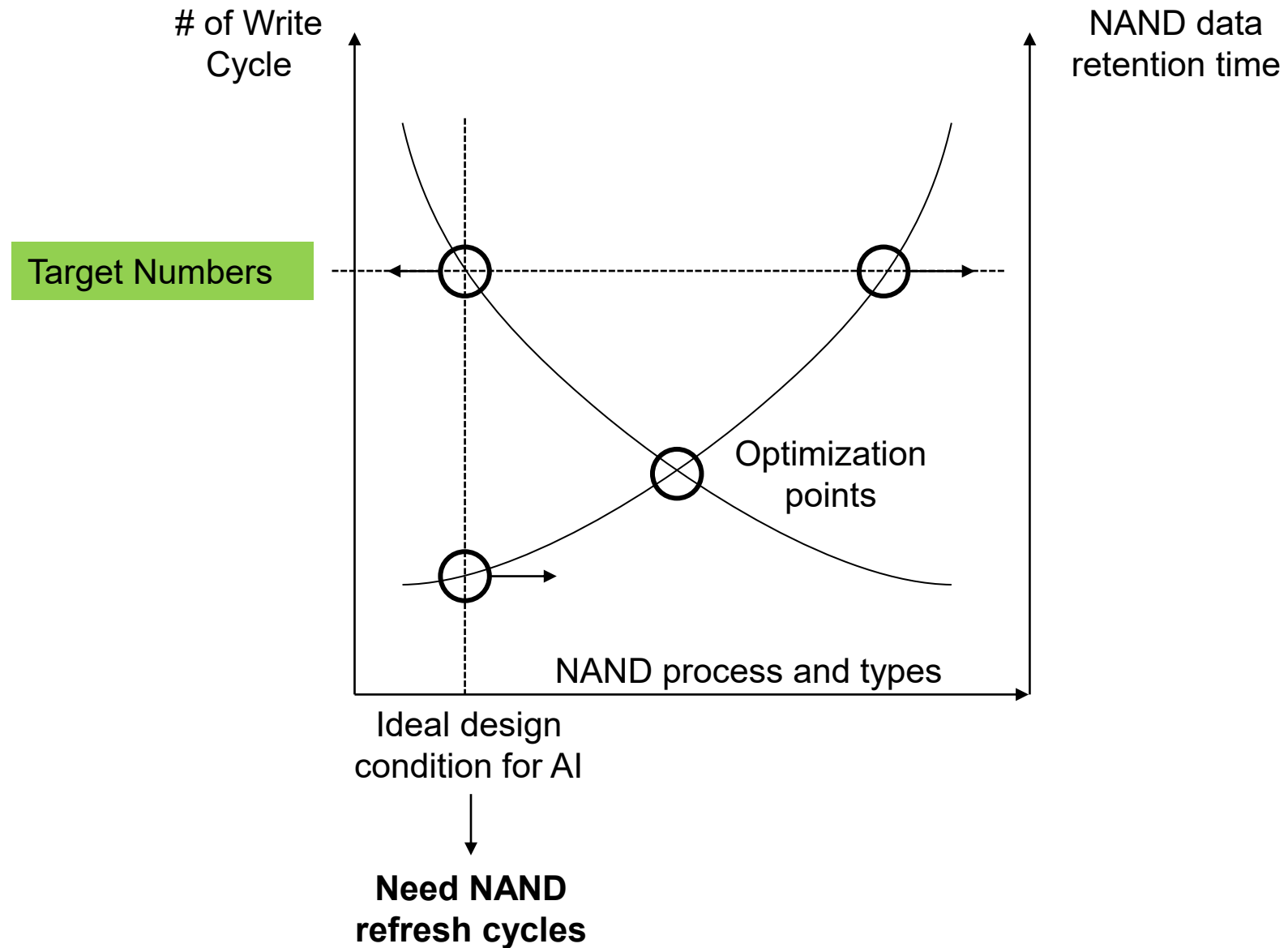


< Token/s analysis for different HBF architecture with its respective conventional architecture

- Three different HBF architectures are compared with its respective conventional architecture.
 - GPU-HBM-HBF is compared with GPU-HBM-HBM, which has the same area.
 - GPU-HBF is compared with GPU-HBM.
 - Custom die architecture is compared with an architecture that connects to SSD (PCIe), and HBF.
- Since HBM has better performance as long as capacity fits, HBF cannot leverage the benefit of low length workload.
- However, as context length or batch size increases, the benefit of HBF is clearly depicted.

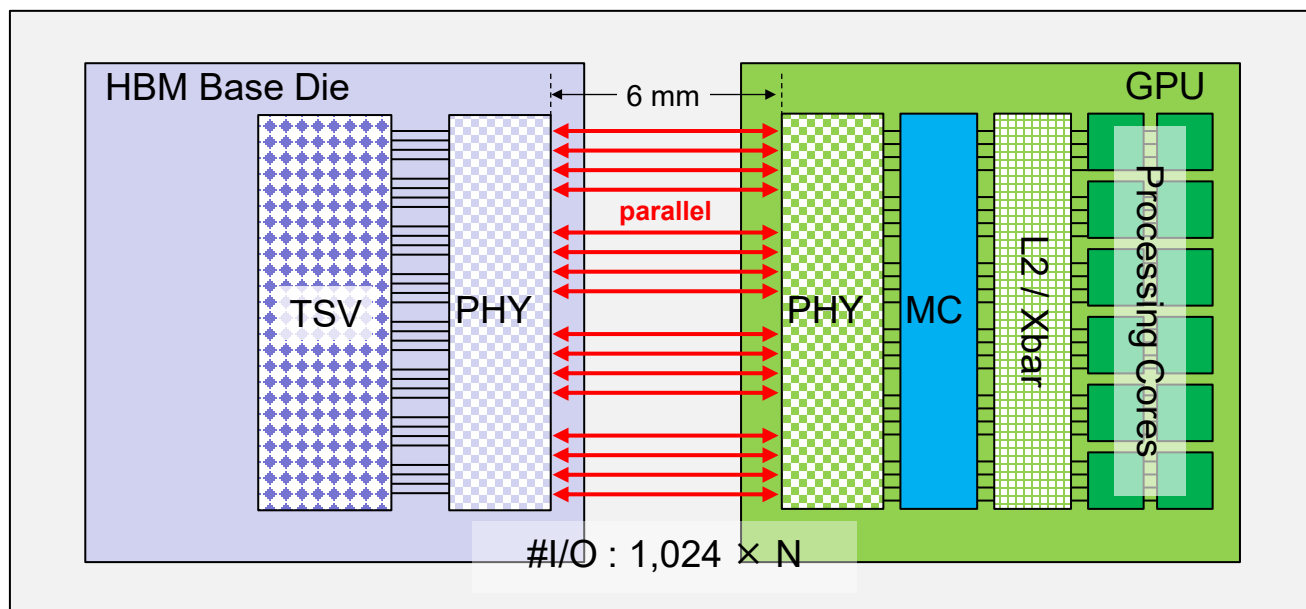
Part6: Write Endurance and Signal Integrity Designs

HBF: Durability of Writing, NAND Flash



- ✓ TSV and Bonding processes
- ✓ Signal, Power, Thermal Integrity
- ✓ Needed from the market In 3 years
- ✓ Multi-partner projects (NAND Flash, Packaging, Foundry, Academia)
- ✓ Need strong collaboration and partnership
- ✓ Investment

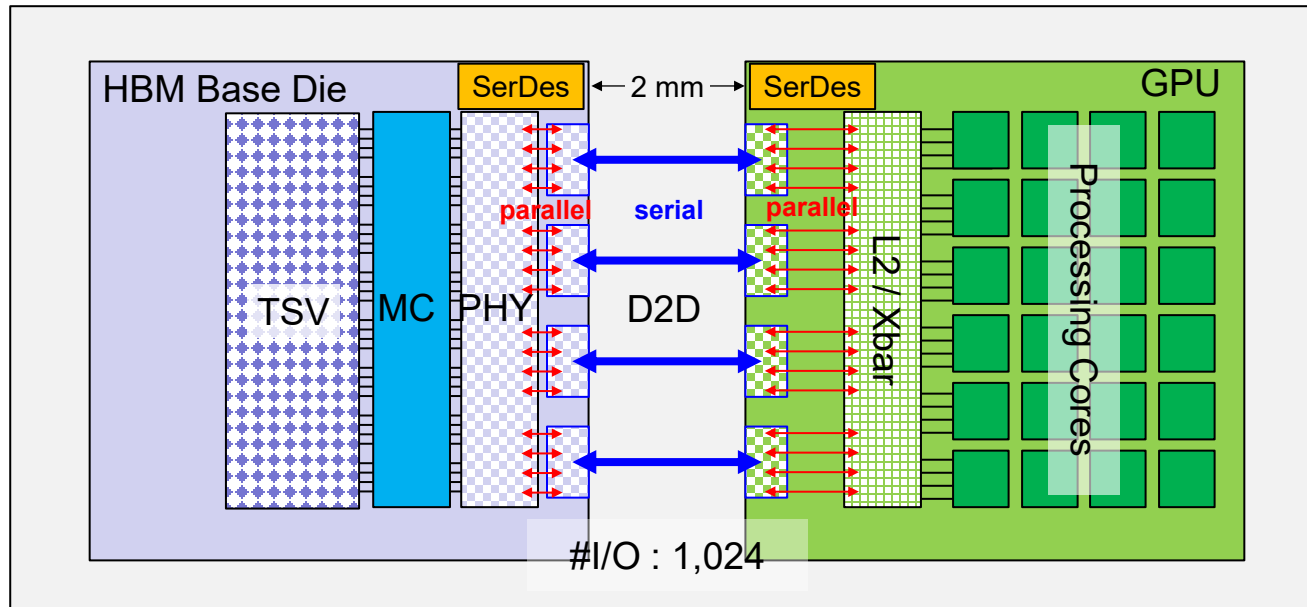
Type I. Direct Parallel GPU-HBM-HBF Interconnect



< GPU-HBM Type I : Direct Parallel Interconnect >

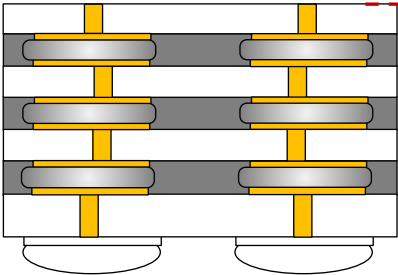
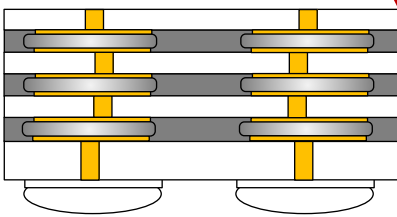
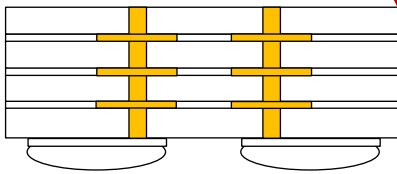
Advantages	Disadvantages
1. Increase Bandwidth 2. Increase # of I/Os 3. Minimized Latency	1. Narrow width/space/pitch $\rightarrow R \uparrow$ 2. Difficulty in pad/ball/bonding routing 3. Multi-layer metal RDL in interposer 4. # of Layer $\uparrow \rightarrow$ Interposer cost $\uparrow$ 5. Longer line length 6. Limited Data Speed (~ 64 Gbps)

Type II. D2D Interconnect between GPU-HBM-HBF



< GPU-HBM Type II : D2D Interconnect >

Advantages	Disadvantages
1. Interconnect Congestion ↓ 2. Ease of pad/bump/bonding routing 3. Short length (< 2mm) 4. Memory controller at HBM base die (Logic IP)	1. Latency at Serdes (serial ↔ parallel) 2. High data rate (100 ~ 200 Gbps) 3. High I/O power consumption 4. Increased I/O area at die

Bump/ Process	u-Bump (MUF)	u-Bump (*A-MUF)	Cu pad (Hybrid Bonding)
Structure			
Diameter	25 ~ 30 μm	20 ~ 15 μm	< 10 μm
Gap height	10~15 μm	5~10 μm	<5 μm
Stack Density	4~16Hi	4~24Hi	4~24Hi
Interconnection	MR / MUF	MR / A-MUF	Hybrid bonding
Risk	Target thickness not achieved	Feasible to overcome But, SI/ TI Risk ↑	Low technology maturity

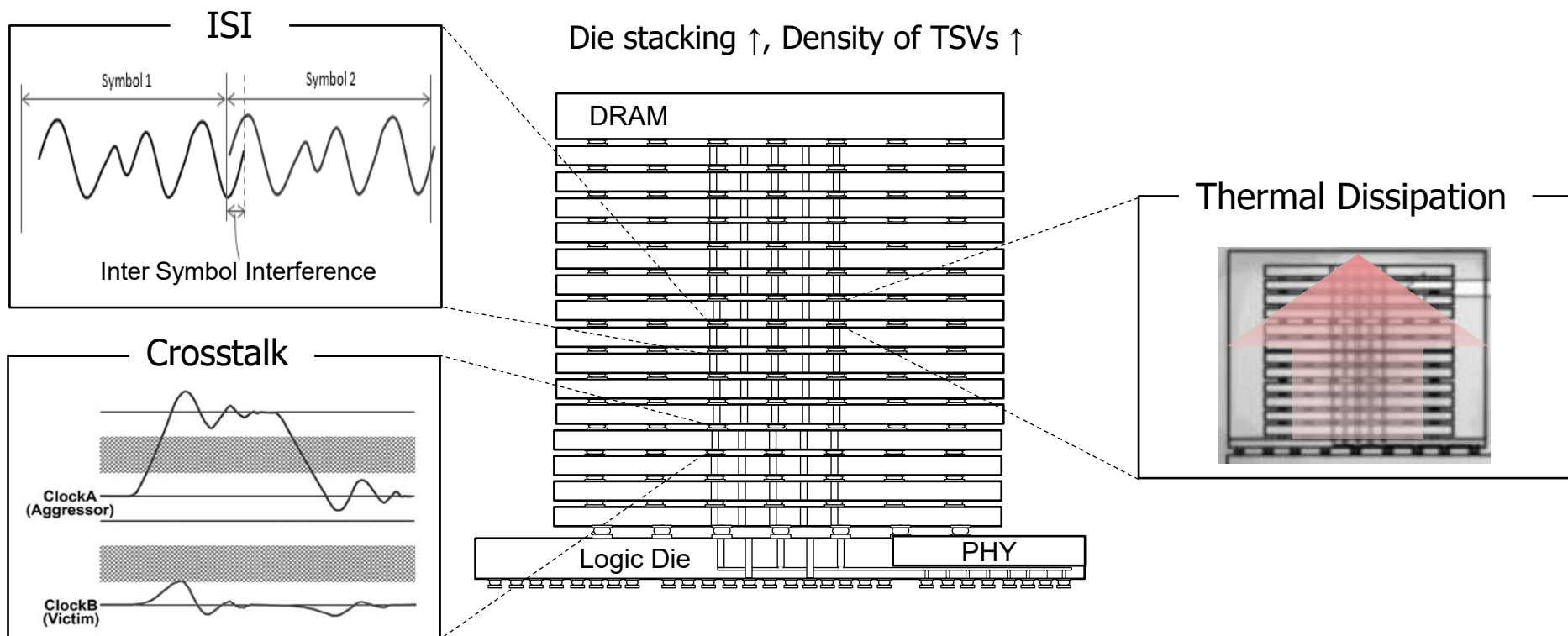
Target for HBM4

< Comparison of Bump Structures and Processes for High-Stack HBM/HBF Applications >

- The Advanced MUF process aims to achieve the target thickness required for post HBM4 while simultaneously improving the technological maturity of hybrid bonding.
- However, increased gap height and cumulative bump count compared to hybrid bonding lead to higher TI/ SI risks.

*Advanced MUF

Signal/ Thermal Integrity Issues at the Super High Stack TSV in HBM



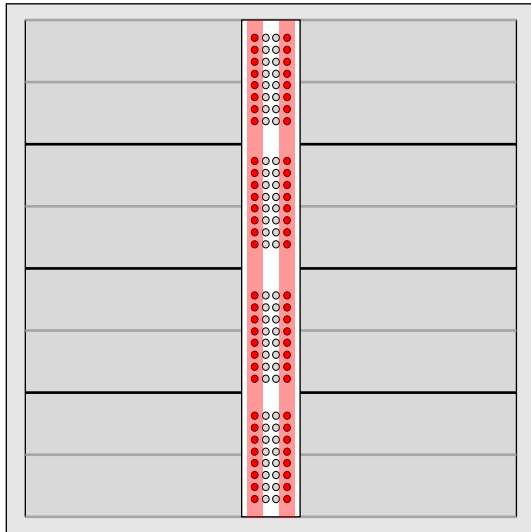
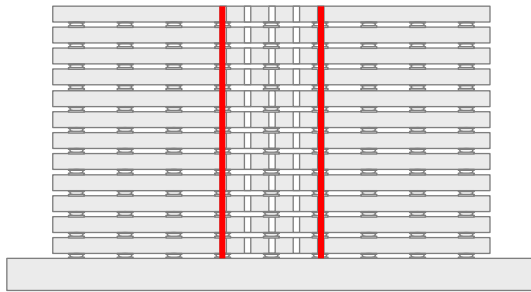
< Signal/ Thermal Integrity issues at the Super High Stack TSV in ≥ 16 Hi-HBM >

- As HBM stacking height and Bump/ TSV density increase, higher power and tighter pitch lead to severe crosstalk, compounded ISI-related signal degradation and thermal dissipation.
- In ultra-high stacks beyond 16-Hi, these SI and TI challenges become critical factors that must be addressed from the earliest design stages.

Next-Generation HBM Roadmap : Distributed Power + Thermal HBM TSV Array Placement

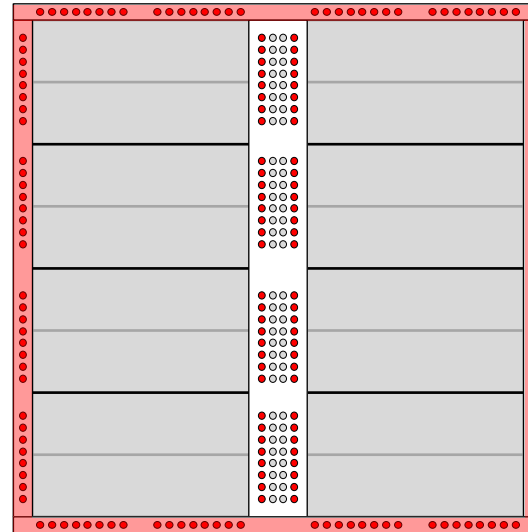
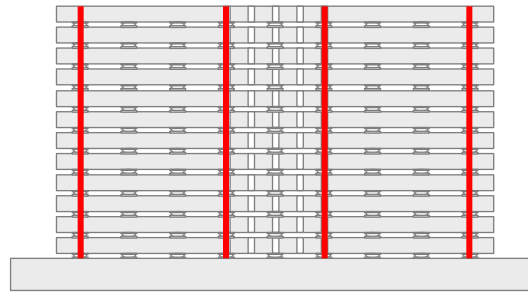
**HBM5
Architecture**

HBM3



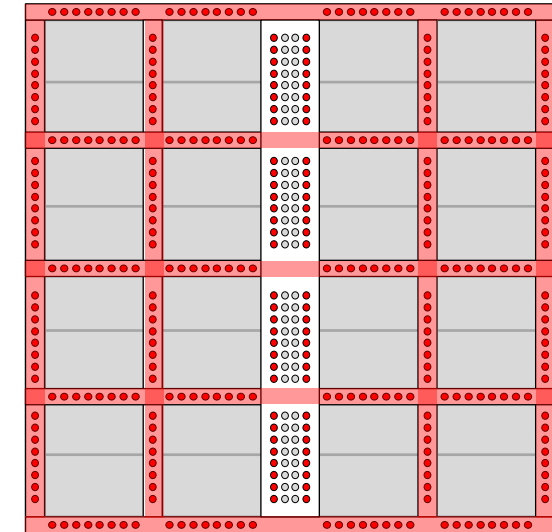
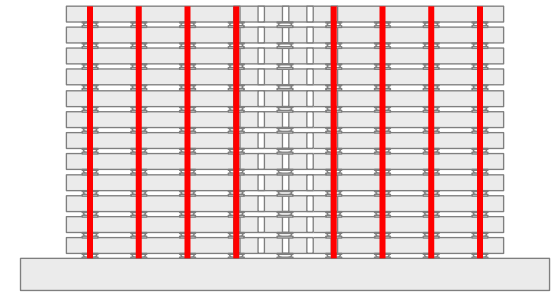
Center Array
P/G + Thermal TSV

HBM4



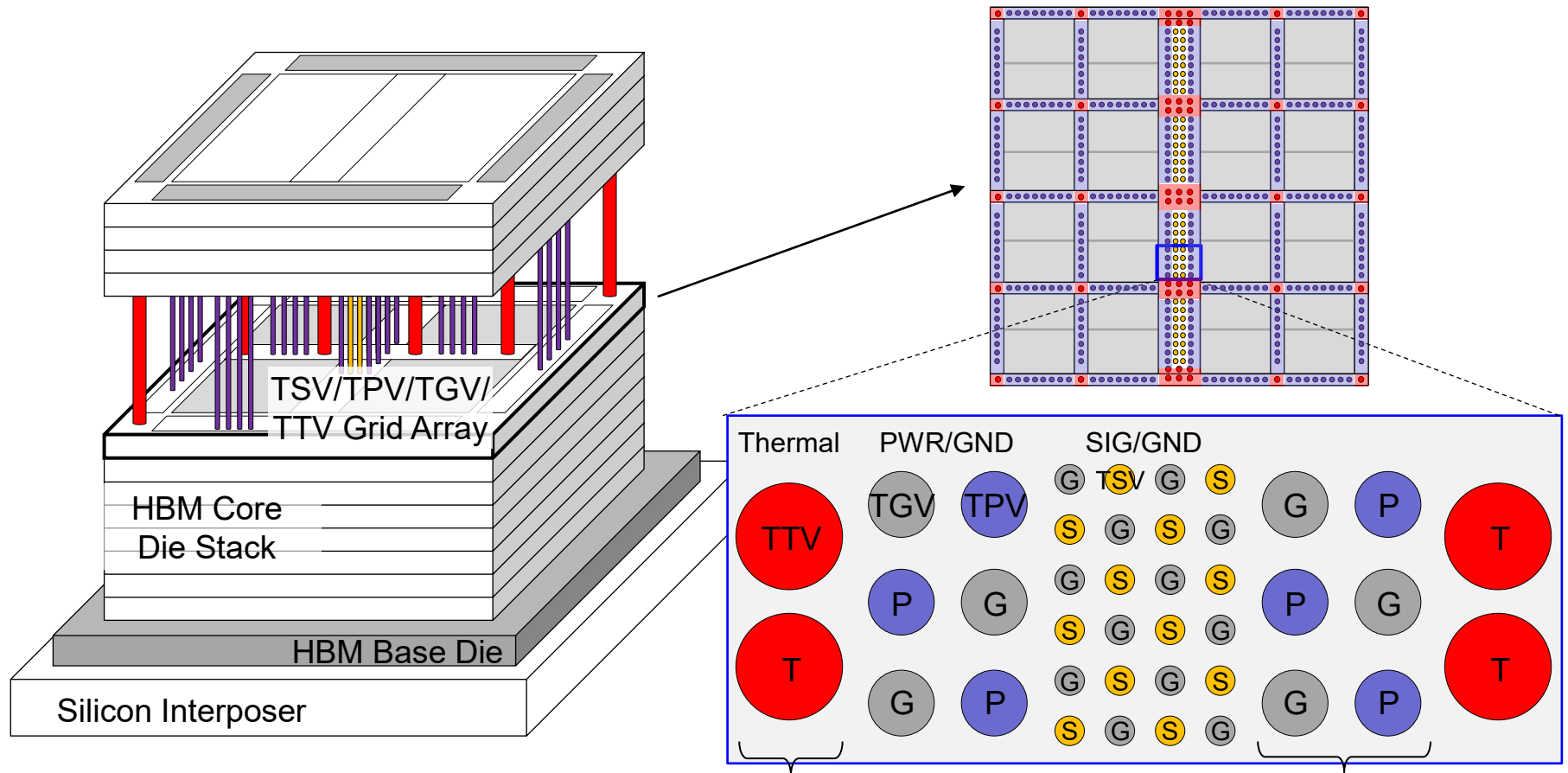
Center Array + Ring
P/G + Thermal TSV

HBM5 ~ HBM8



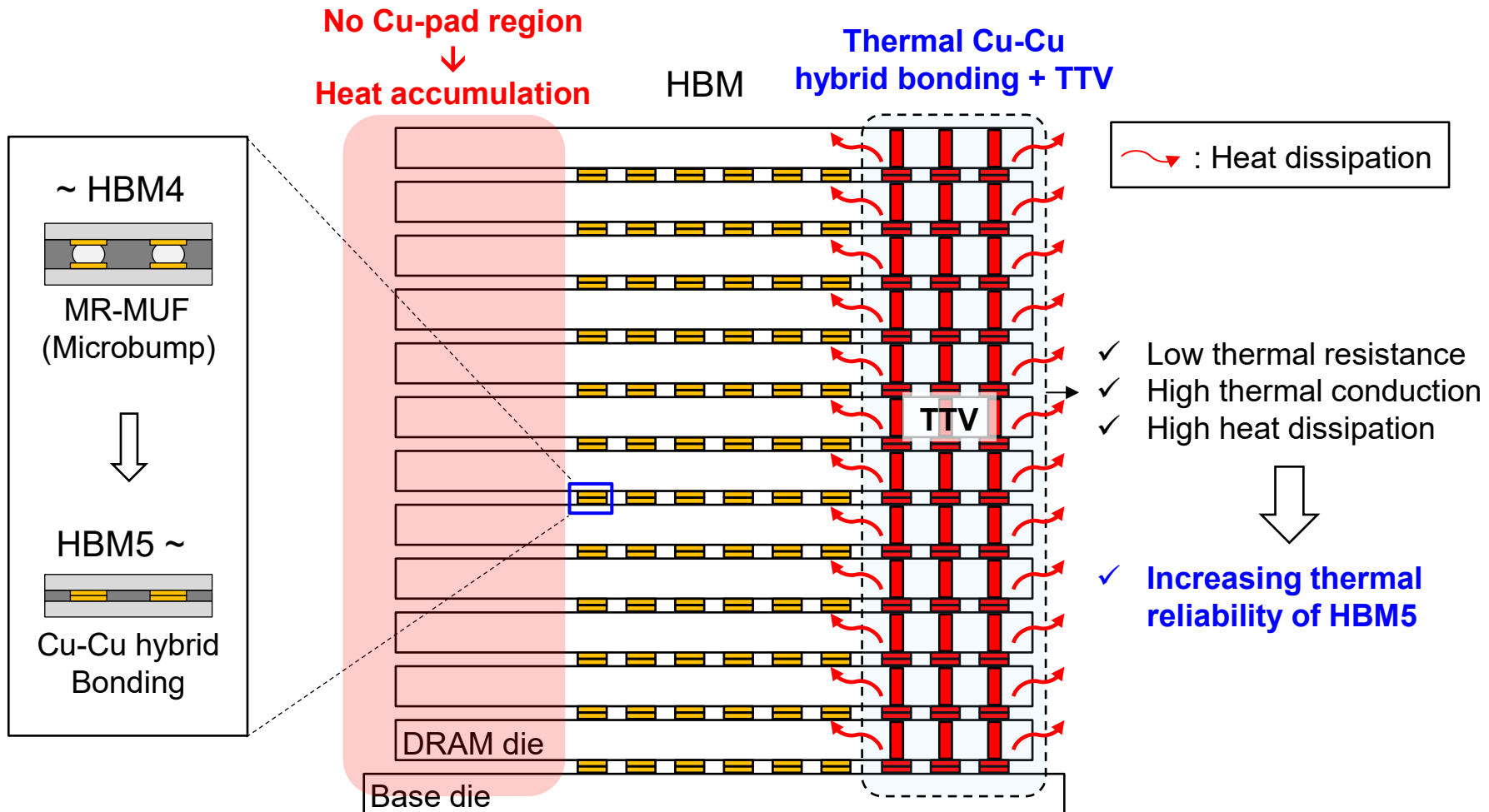
Distributed Grid Array
P/G + Thermal TSV

Distributed S/P/G/T Via Grid Array at HBM5



- ✓ Low thermal resistance
- ✓ High thermal conduction
- ✓ Low inductance
- ✓ Low electrical resistance

< Asymmetric Distributed TSV/TPV/PGV/TTV Grid Array Design for HBM5 >

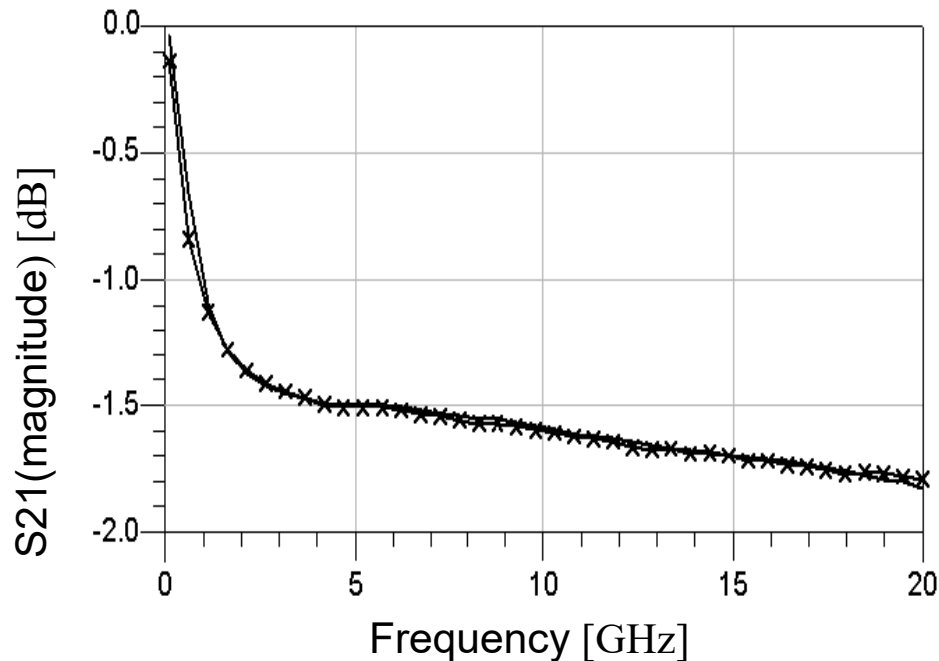


< Thermal Cu-Cu Hybrid Bonding with TTVs for Thermal Reliability in HBM5 >

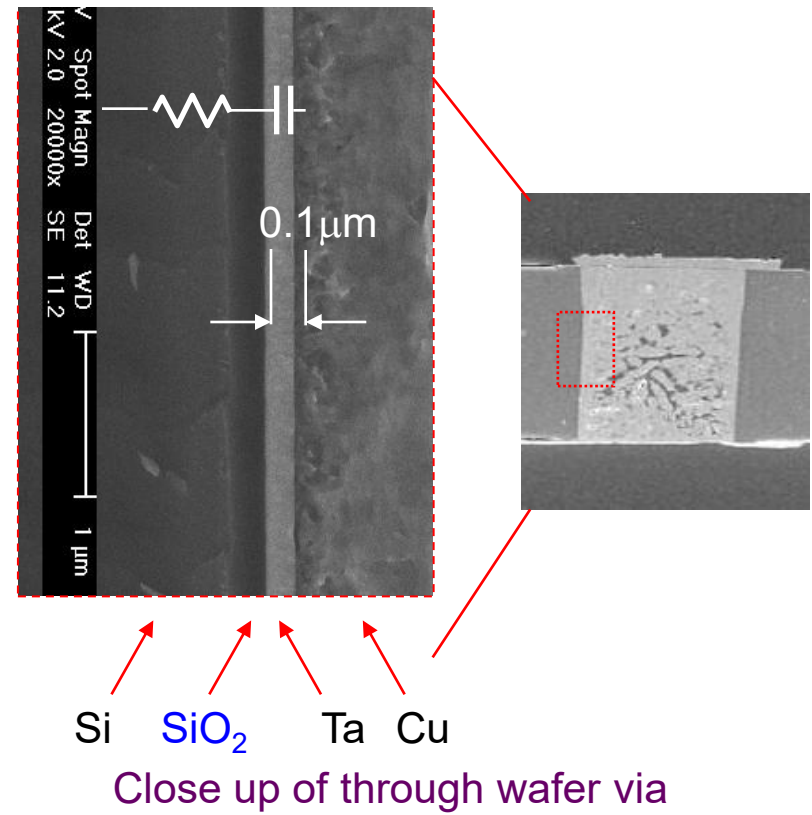
- Thermal Cu-Cu hybrid bondings are implemented on no Cu-pad (no bump) region of HBM except S/P/G Cu-pad.
- Thermal Cu-Cu hybrid bondings are interconnected to TTVs to maximize heat dissipation of HBM.

High-frequency Channel Loss in TSV

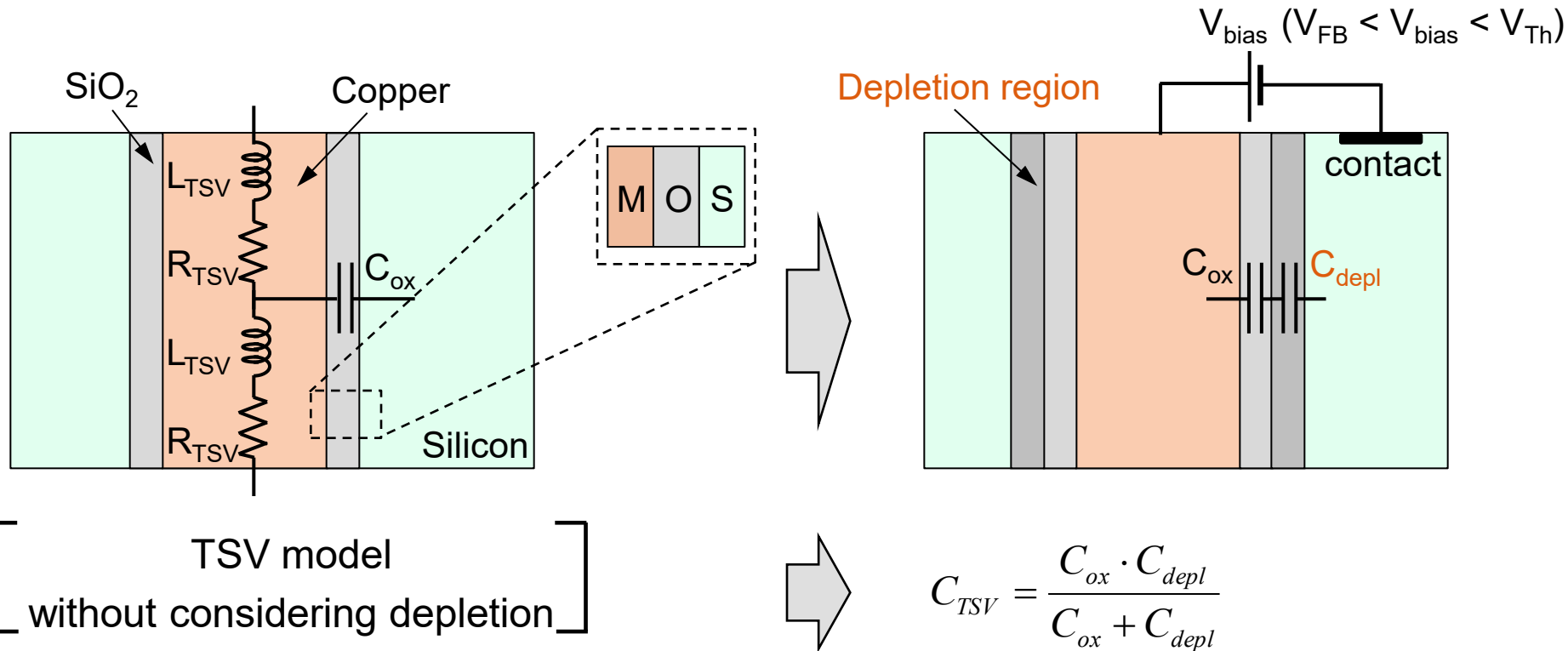
- Significant high-frequency signal loss occur at Signal Transmission Through TSV
- The signal loss through TSV is caused by substrate leakage and coupling



Magnitude of S21



TSV Depletion Phenomenon Depending on DC Bias Voltage

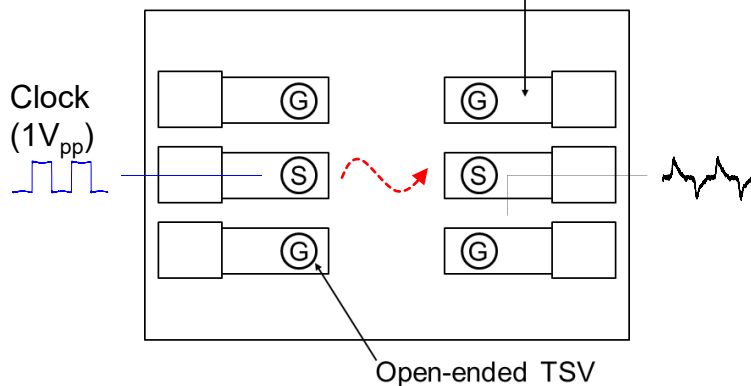


- TSV has MOS structure → depletion occurs depending on TSV bias voltage
- TSV capacitance decreases if depletion region is generated

Measured Coupled Voltage of a TSV Channel depending on Signaling

<GSG-type>

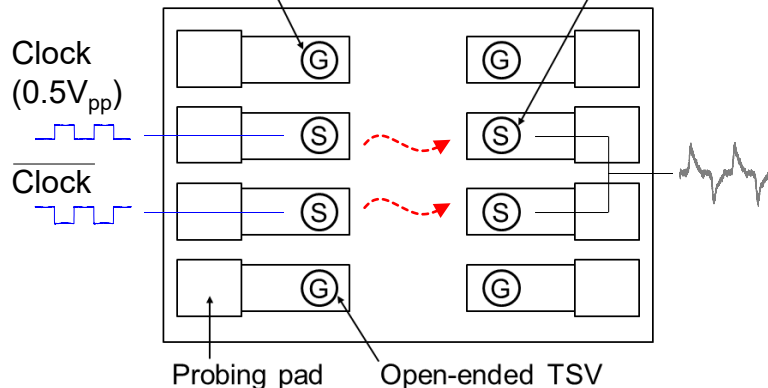
Metal line on the top die



<GSSG-type>

Ground TSV

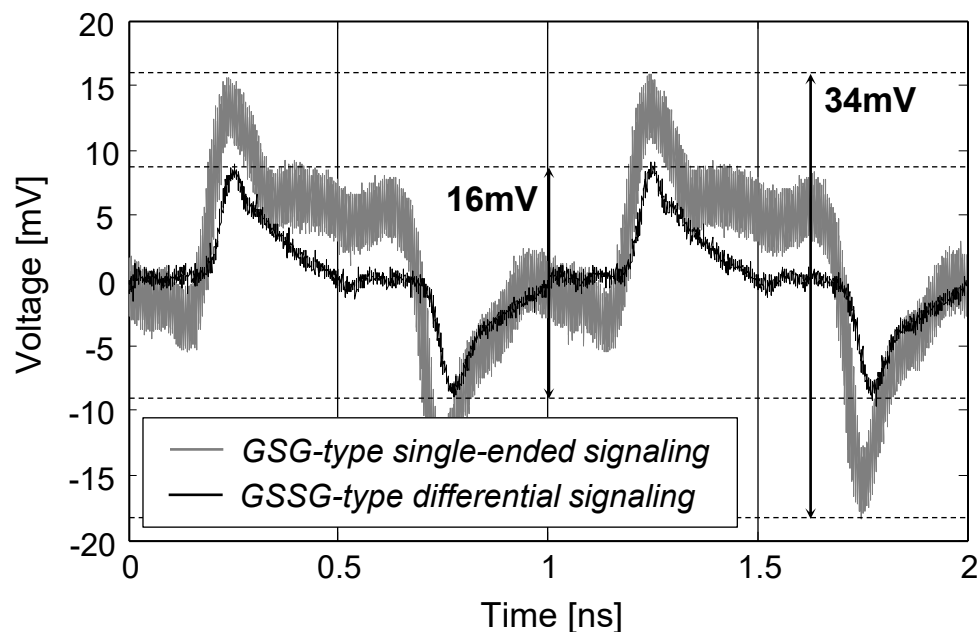
Signal TSV



Injected signal : 1GHz clock signal
Using pulse-pattern generator (PPG)
Digital cscilloscope : TDS8000B

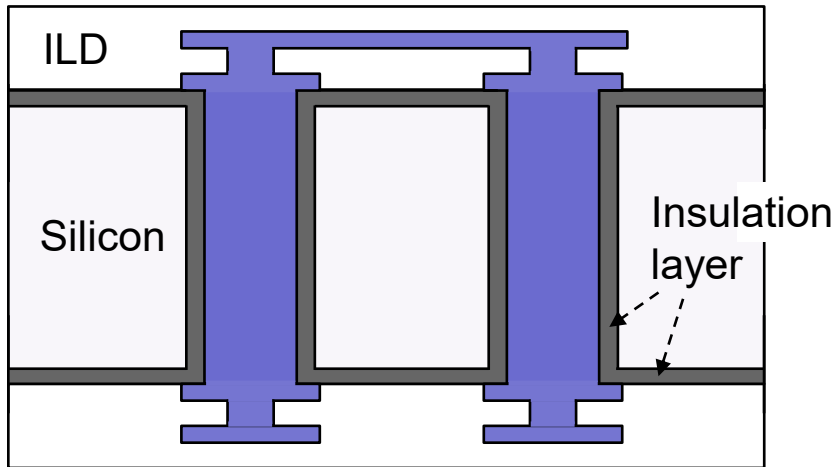
< Coupled Noise voltage >

Single-ended (GSG-type) vs. Differential (GSSG-type)

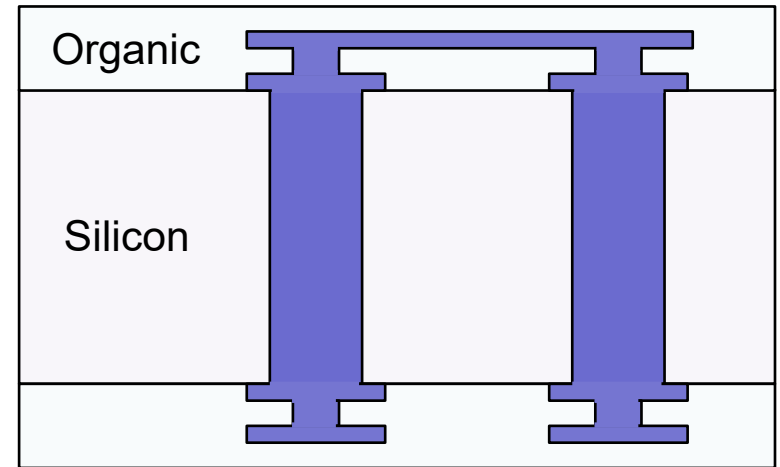


- Even with the larger insertion loss, GSSG-type differential signal TSV has better noise immunity than GSG-type single-ended signal TSV

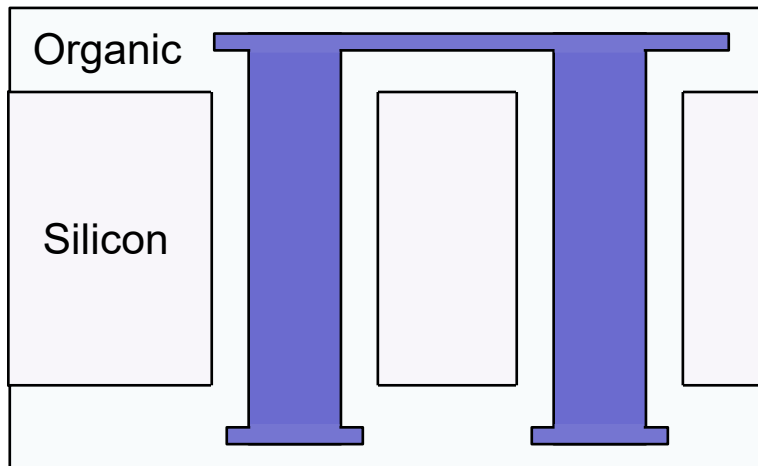
Various Structures of TSV on Interposer



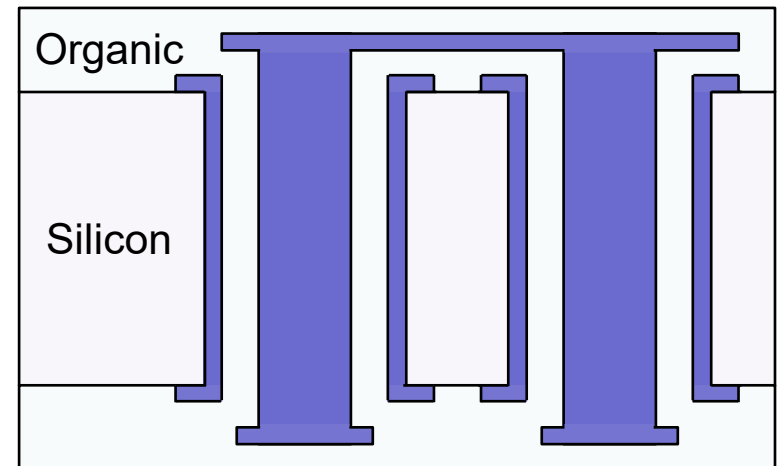
< Normal TSV >



< Through Silicon Line Via (TSLV) >

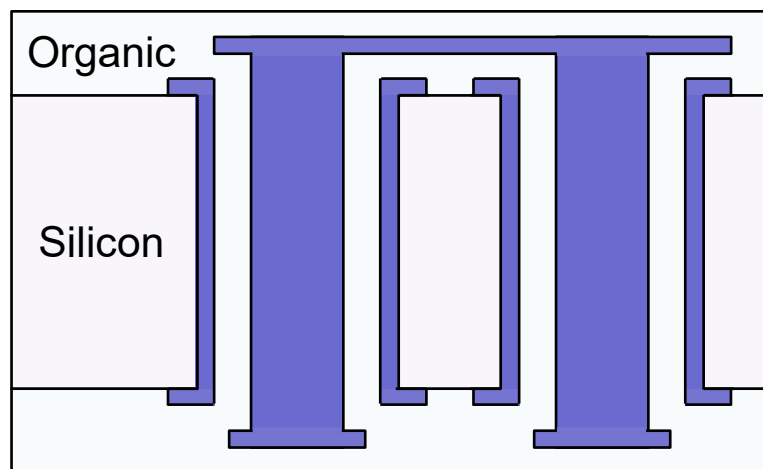


< Through Organic Line Via (TOLV) >

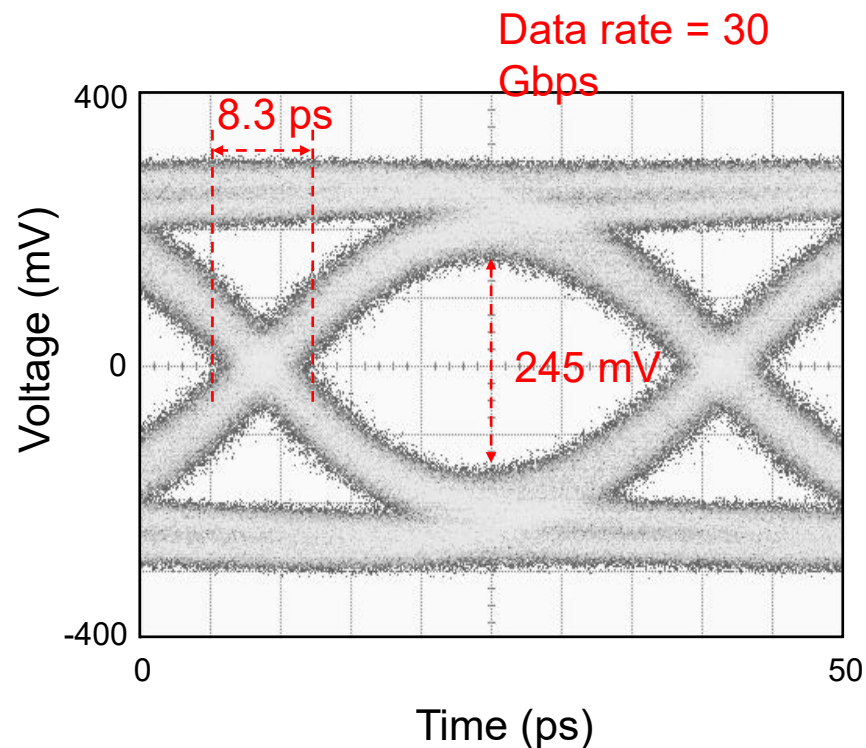


< Coaxial Organic Line Via (COLV) >

Coaxial TSV/TGV on Interposer

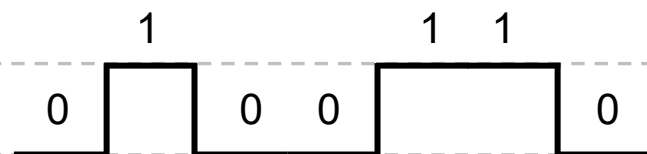
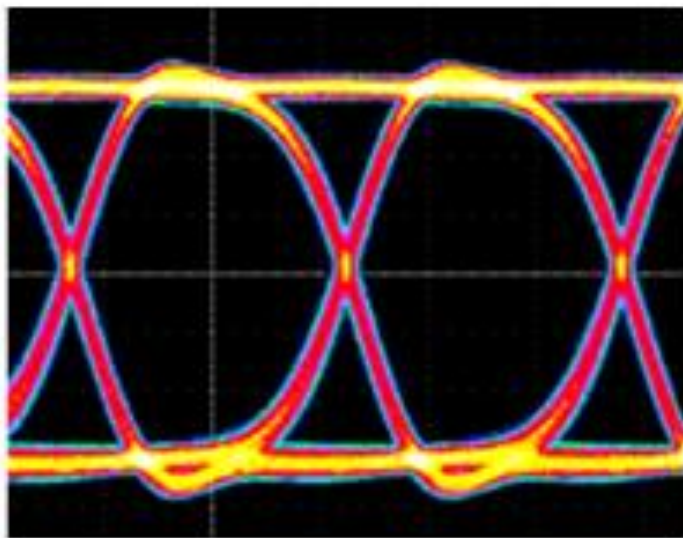


< Coaxial Organic Line Via (COLV) >



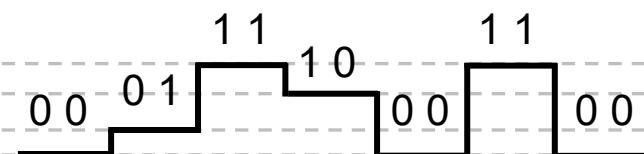
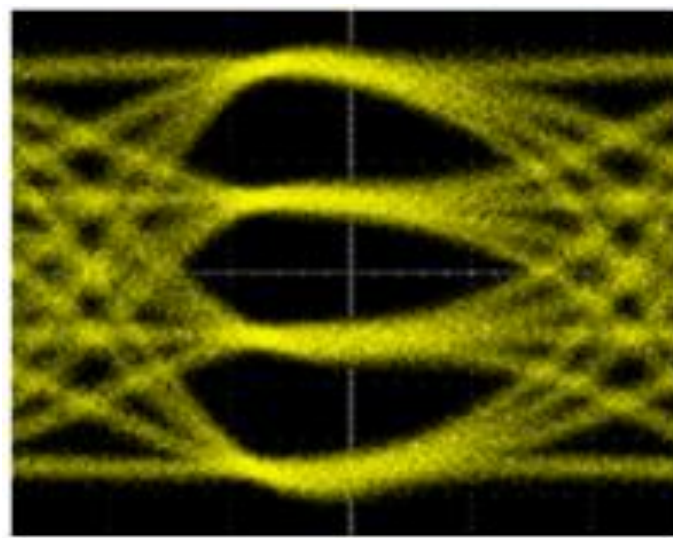
< Eye-diagram of Coaxial TSV channel >

Comparison between PAM2-NRZ and PAM4 Signaling



(2 levels- 0 or 1)

<PAM2-NRZ>

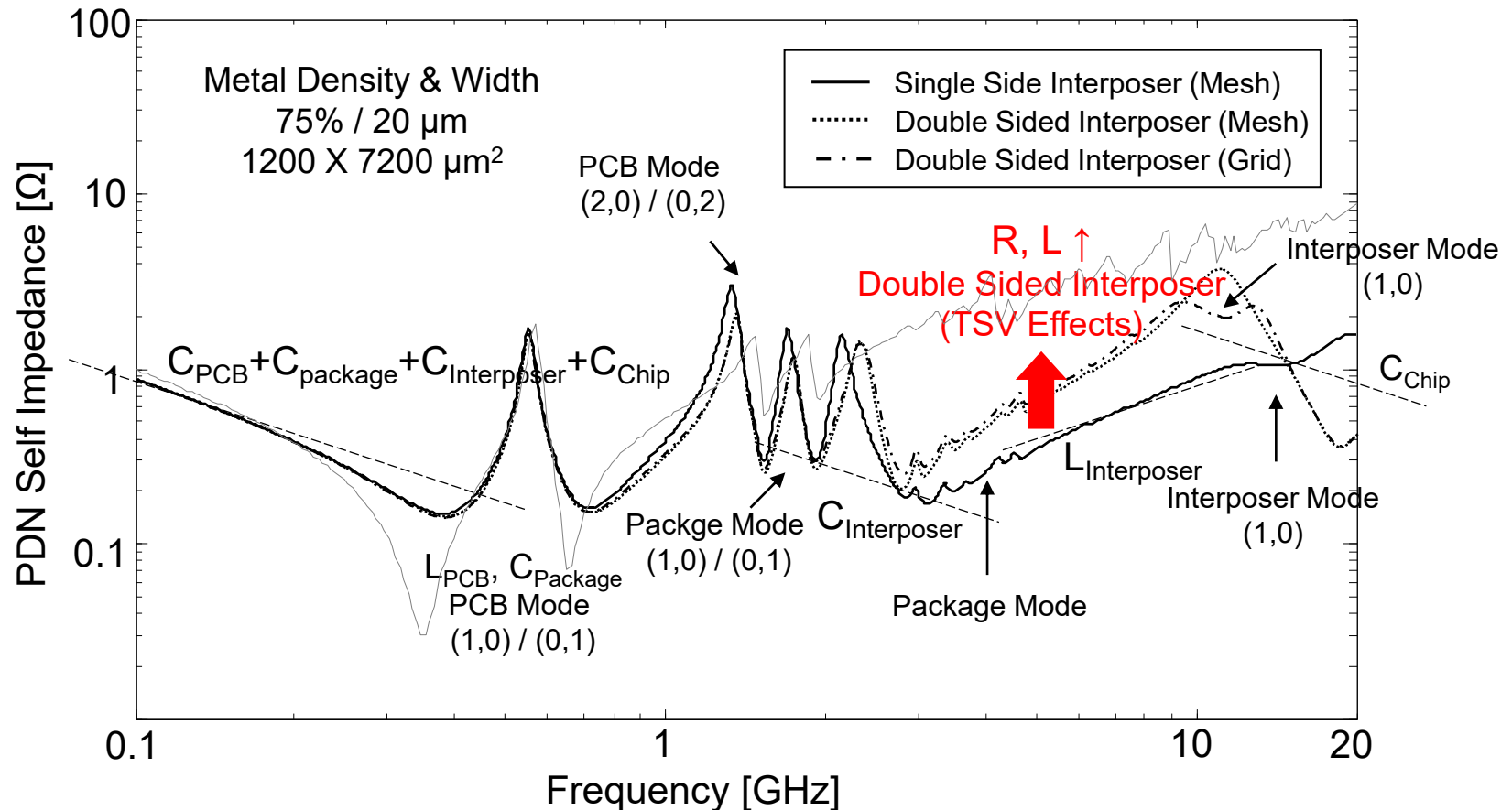


(4 levels - 00 or 01 or 10 or 11)

<PAM4>

- By programming two bits per single signal, PAM4 allows twice much data transmission than conventional PAM2-NRZ at the same rate.

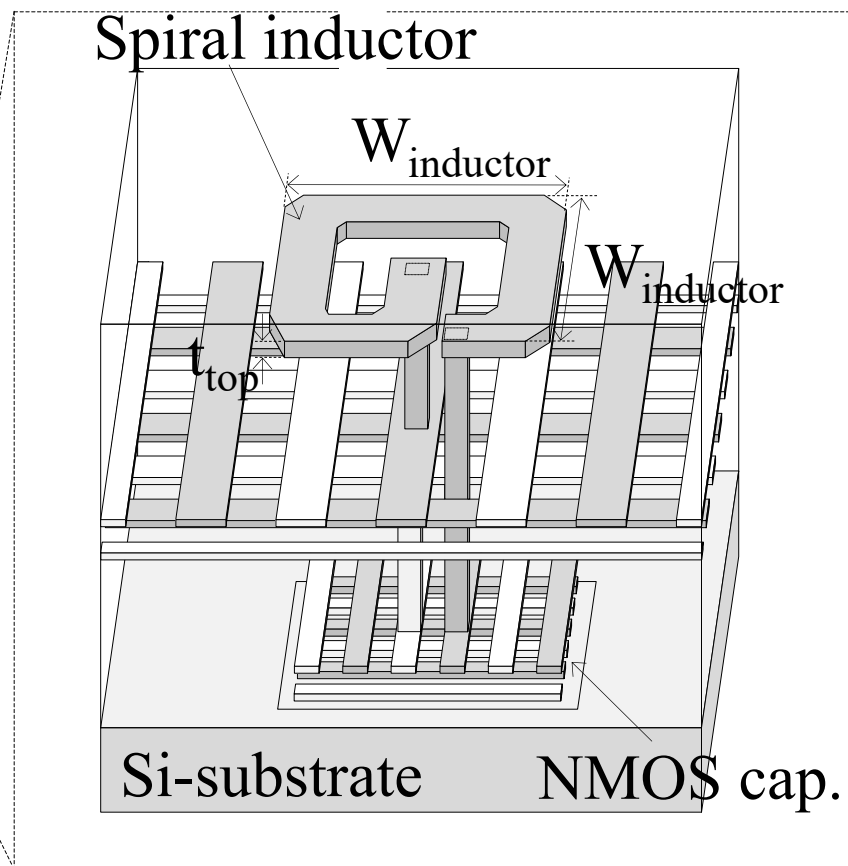
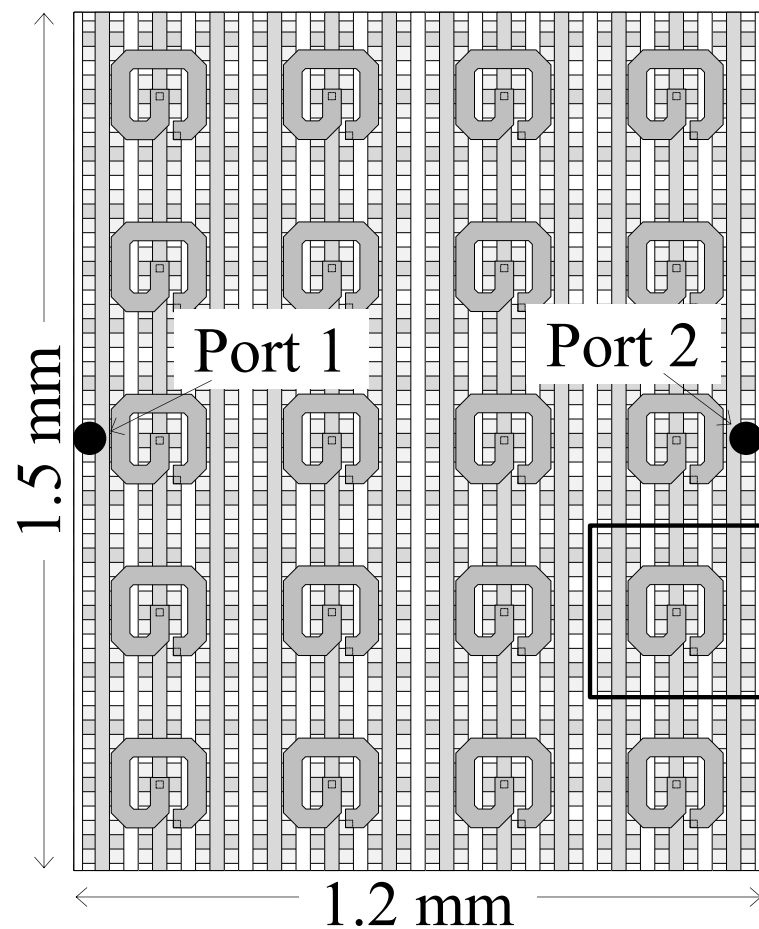
Simulated Hierarchical PDN Impedance Considering a Single- and Double-sided HBM Interposer



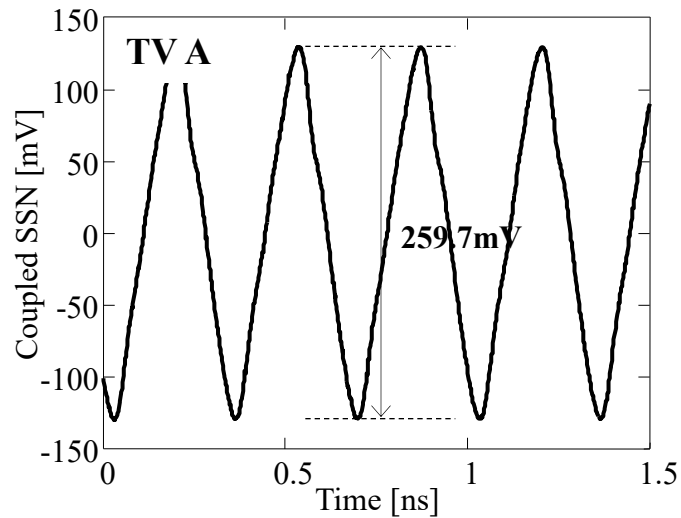
< Hierarchical PDN impedance considering a single- and double-sided HBM interposer >

- Due to the TSV effects of a double-sided HBM interposer, its hierarchical PDN impedance is relatively higher than that of a single-sided HBM interposer.

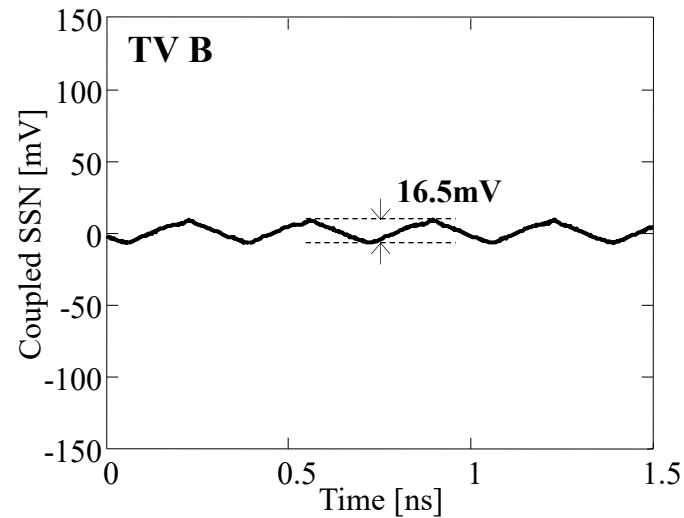
On-chip CMOS Active EBG



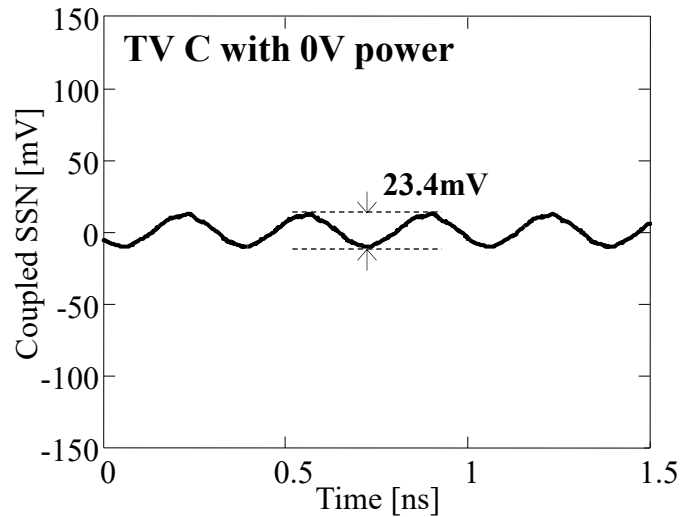
Measured SSN Waveforms with a 3-GHz Clock Noise Input



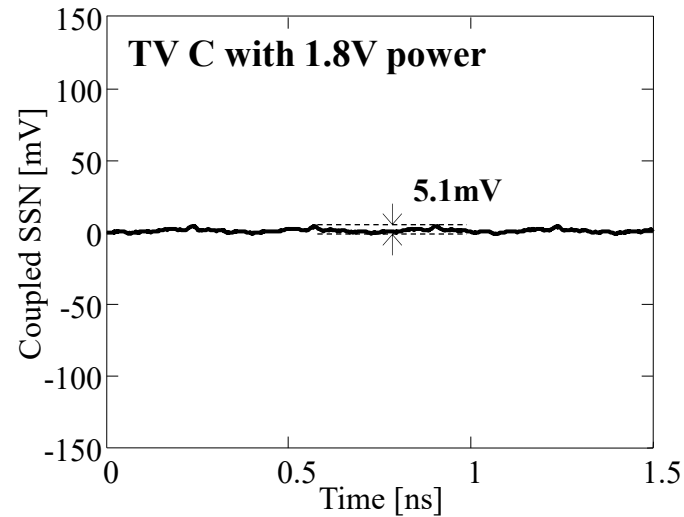
(a)



(b)



(c)



(d)

Interconnection Future(10): Design Innovations: 5 Is

Design and Analysis for High-performance System

Co-Simul. Co-Design

System on Chip
(On-Chip Level)

System in Package
(On-Package Level)

PCB, Cable, Module
(System Level)



Signal **I**ntegrity
(**SI**)

Power
Integrity
(**PI**)

Thermal
Integrity
(**TI**)

5I
for High-
Performance
System

Electromagnetic
Interference
(**EMI**)

Artificial
Intelligence
(**AI**)



Improvement of System Integrity, Performance, Reliability, Cost, ...

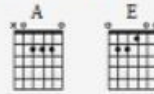
Part7: Conclusions



AI Factory: Infrastructure

AI 대량 생산, 대량 소비, 집중 생산, GPU, 전력, 냉각수

클레멘타인



♩ = 100

Music score for 'Clementine' in G major (one sharp) and 2/4 time. The tempo is marked as ♩ = 100. The lyrics are in Korean, and the melody is accompanied by guitar chords (A and E) indicated above the notes.

넙 고 넙 은 바 닷 가 예 오 막 살 이 집 한 채 고 기
 잡 는 아 버 지 와 철 모 르

A NEW AND MAGNIFICENT CLIPPER FOR SAN FRANCISCO.
MERCHANTS' EXPRESS LINE OF CLIPPER SHIPS!
 Loading none but First-Class Vessels and Regularly Dispatching the greatest number.
THE SPLENDID NEW OUT-AND-OUT CLIPPER SHIP

CALIFORNIA
 HENRY BARBER, Commander, AT PIER 13 EAST RIVER.

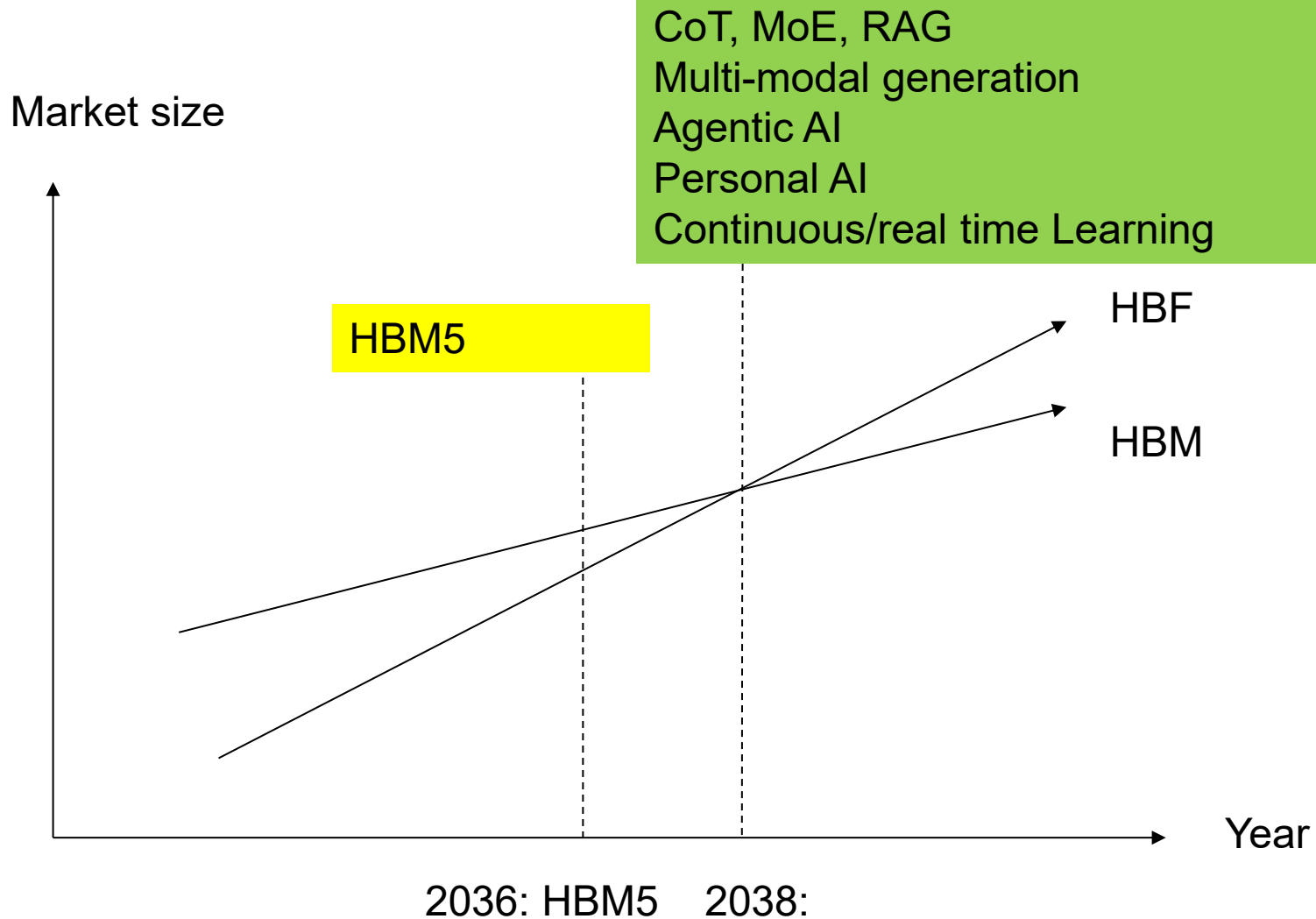
This elegant Clipper Ship was built expressly for this trade by Samuel Hall, Esq., of East Boston, the builder of the celebrated Clippers "SURPRISE," "GAMBOCK," "JOHN GILPIN," and others. She will fully equal them in speed! Unusually prompt dispatch and a very quick trip may be relied upon. Engagements should be completed at once.

Agents in San Francisco,
 Messrs. DE WITT KITTLE & CO.

RANDOLPH M. COOLEY, 88 Wall Street, Tontine Building.

NEUBITT & CO., PRINTERS.





Thank You!

HBM